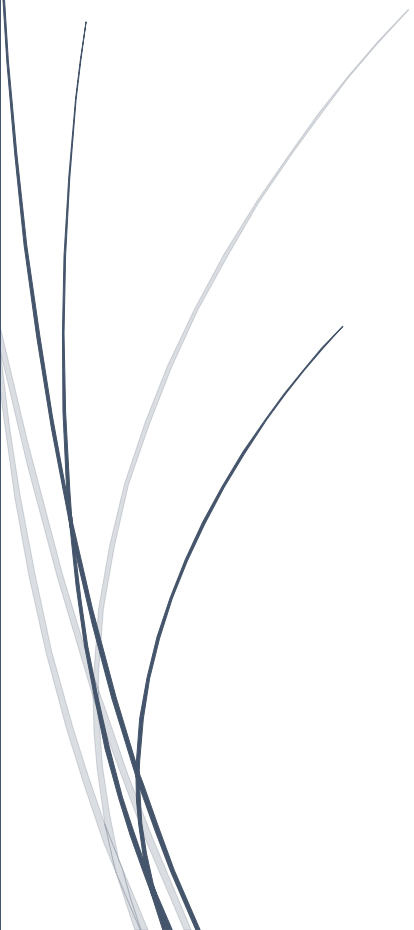




ENGLISH FOR SCIENCE

Part 1

V-LANG-650

2026-2027

BAB2 Mathematics

Nicolas DEVILEZ
UNIVERSITÉ DE MONS

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Authors: Ingrid Van Lancker, Nicolas Devilez and Sophie Delmotte ©

Essential websites

<u>OXFORD ENGLISH ADVANCED DICTIONARY</u>	<u>LINGUEE</u>	<u>HOWJSAY</u>	<u>SKELL</u>
<i>Dictionary & pronunciation: general)</i>	<i>Specialised dictionary with real-life examples</i>	<i>Pronunciation: scientific & specialised words</i>	<i>Collocations</i>
			

Contact: nicolas.devilez@umons.ac.be

Office: Plaine de Nimy, Building 5, 1st floor, 104

COURSE GUIDE & STARTER

I – OFFICIAL COURSE SYLLABUS

Course Title: ENGLISH FOR SCIENCE

Credits (UE): 4 ECTS

ECTS code: US-B2-SCMATH-001-M - English for Science

Prerequisite: None

Aim: B1

Teaching methods: 48h in-class activities + Moodle activities

Learning material:

- Student coursebook
- Grammar book
- Moodle activities

Modes of study: Attendance, active participation, oral and written exercises

Learning outcomes:

At the end of this course, students will be able to

- use basic English grammar structures in context;
- extract information from science-oriented English texts;
- communicate in English, in writing and orally, on science-related topics;
- work autonomously on audio documents (B1 level);
- build vocabulary linked to science;
- put into practice the basic rules of English pronunciation;
- create science-oriented works linked to their field of study.

Course Content and Methodology

English for Science, part 1 aims to teach you basic scientific (and more specifically mathematical) concepts and vocabulary in English and offers a thorough revision of English grammar.

The course consists of a mix of theoretical and practical activities (exercises and project) and Moodle activities.

Term 1 course content is divided into six units: (1) *Defining, Describing & Qualifying*, (2) *Compound Forms and Discipline-Specific Terminology*, (3) *Past and Perfect Tenses*, (4) *Quantification, Measurement and Frequency*, (5) *Future Tenses* and (6) *Linking Your Ideas*.

Term 2 course content is divided into six units: (1) *Presenting Skills*, (2) *Using Visual Aids in a Presentation*, (3) *Impersonal Forms*, (4) *Hypothesising*, (5) *Modality: Degrees of Truth/Reality* and (6) *Comparing*.

Important course **test dates** and **deadlines** are indicated with a  sign and in **red** below.

➤ Placement test

Term 1 begins with a **placement test** (eLAO test). If you score at level A1 or A2, we invite you to attend tutoring sessions ("cours inter-facultaires").

 **Deadline:** complete the eLAO test by **16th September 2026 – 11.59 p.m.** (see Moodle)

- Grammar

A **comprehensive revision of English grammar** is offered, ensuring that students have a solid foundation in sentence structure, verb tenses, and syntactic accuracy.

- Vocabulary

The course also focuses on **learning vocabulary common to all scientific fields and to mathematics**, emphasizing academic and scientific phraseology that is essential for clear and precise communication both in interdisciplinary and specific research contexts.

- Oral Skills

To support effective interaction, the course includes an **introduction to pronunciation and to oral communication**, helping students articulate complex ideas clearly and confidently in oral contexts.

- Written Skills

Students develop their writing skills through small **science-oriented assignments and projects**, which encourage the practical use of language in real-world scientific scenarios, fostering both creativity and academic rigour.

- Reading and Listening Skills

A key component of the course involves **document analysis**, where learners examine a variety of materials such as books, popular science articles, scientific papers, educational videos, and audio recordings. This not only enhances comprehension but also exposes students to different styles of scientific communication.

- Homework (Moodle)

Each unit contains individual and study circle homework and in-class tasks. The **homework activities** (mainly on Moodle but some activities might also be in the coursebook) **should be completed before the next class to ensure logical progression between units**. Certain recap grammar exercises such as 'Test Yourself!' can be completed later if necessary to revise for the exams.

- Self-study (Moodle)

Through the self-study activities, students engage in **individual work**, allowing them to progress at their own pace while developing autonomy and critical thinking skills.

 **Deadlines:**

- Term 1: All self-study activities are available from 14th September and need to be completed by **6th December 2026 – 11.59 p.m.**
- Term 2: All self-study activities are available from 1st February and need to be completed by **2nd May 2027 – 11.59 p.m.**

The full completion of Moodle activities (Homework + Self-study — see completion progress on Moodle) is required to obtain a **bonus** for each term (see 'III – course assessment modalities' below).

II – COURSE ORGANISATION & ETHICS

1. Groups, schedule & classrooms

The information about groups, the schedule and classrooms are available on Moodle.

2. Participation

All work, including homework, Moodle activities and active class participation (not just attendance) is **highly encouraged**. Practice makes perfect!

3. Contacting the teacher

Address your teacher only in English, inside and outside class.

Ask all course related questions on the Questions forum in English (Front Page section).

Do not send emails unless your question or message contains strictly confidential information. Emails written in French will not receive an answer.

Asking questions on the forum serves several purposes. First, you will inform your colleagues of any relevant matters that affect them as well, such as your absences. Second, you help me save time and contribute to the needs of others by asking on the forum, as others are likely to have the same question. Lastly, you will get a faster reply via the forum, as I check my forum messages before my e-mail. This is my method to prioritise students' communication over others in my workflow.

4. Plagiarism/AI awareness

The University of Mons has zero tolerance for **fraud/plagiarism**. It considers the use of **AI** (ChatGPT, Claude AI, Perplexity, etc.) as academic dishonesty or fraud if not cited accordingly (for short excerpts, in the same way as you would do it for any other work). Make sure to properly cite AND paraphrase any work you submit. Please read the documentation on the use of AI at the University of Mons ([RGE](#) Article 35 and [Annexe 0](#) Article 34)

The citation standard in this course is APA7: <https://apastyle.apa.org/>

III – COURSE ASSESSMENT MODALITIES

TERM 1: 30 points out of 100 (English for Science final grade)

- Vocabulary test: 50/100
- Grammar test: 50/100
- Term 1 Moodle activities: 5/100 bonus if 100% completed

One grammar and one vocabulary exemption test will take place during term 1. If you get 12/20 or above in grammar or vocabulary, you will get an exemption for that part. If you get less than 12/20 for either of these parts, you must take the corresponding exam during the session. You are required to attend **at least 60%** of the lessons (= **7/11** sessions) in order to be allowed to take the grammar and vocabulary exemption tests, regardless of the reason for the absence.

 The **Term 1 exemption tests** will take place on **14th December 2026**.

TERM 2: 70 points out of 100 (English for Science final grade)

- Vocabulary test: 15/100
- Grammar test: 25/100
- Final oral presentation about the project: 60/100
- Term 2 Moodle activities: 5/100 bonus if 100% completed

One grammar and one vocabulary exemption test will take place during term 2. If you get 12/20 or above in grammar or vocabulary, you will get an exemption for that part. If you get less than 12/20 for either of these parts, you must take the corresponding exam during the session. You are required to attend **at least 60%** of the lessons (= **6/10** sessions) in order to be allowed to take the grammar and vocabulary exemption tests, regardless of the reason for the absence.

 The **Term 2 exemption tests** will take place on **10th May 2027**.

TERM 3: 100 points out of 100 (English for Science final grade)

Term 1 retakes

- Vocabulary: 50 points
- Grammar: 50 points

Term 2 retakes

- Vocabulary test: 15 points
- Grammar test: 25 points
- Final oral presentation about the project: 60 points

IV – PROJECT

1. Project overview

What?

You will work **individually** on a project throughout the year and present it during the oral exam.

The project must be specific enough and be created from this list of resources:

- **One pairing (= two related chapters)** from the following **books**:
 - *How Not to Be Wrong: The Power of Mathematical Thinking* (J. Ellenberg)
 - *Taming the infinite: The Story of Mathematics from the First Numbers to Chaos Theory* (Ian Stewart)
 - *17 Equations that Changed the World* (Ian Stewart)
 - *Things to Make and Do in the Fourth Dimension* (Matt Parker)
- **One scientific paper** related to the topic explained in the chapters chosen above.
- **One popular science article** that illustrates one concept explained in one of the documents above. Use [Quanta Magazine](#) preferably.
- **One audiovisual document** that illustrates one concept explained in one of the documents above. Pay attention to the reliability of your source.
- **Optional**: any **extra documentation** deemed useful by the student.

On top of sharing information related to your field of study, the goal of this project is to **put into practice vocabulary and grammar** studied throughout the year.

Part 1 (Term 1)

Your project:

1. must contain clearly identifiable notions/concepts (that can be thoroughly defined) and offer nuances.
2. must have a specific purpose and involve one or several 'processes' (e.g., why was it created and how does it work?).
3. must contain sizeable/measurable (potentially statistical) data (with figures such as tables, graphs, etc.).
4. must be presented with a clear structure, stating its origin and effect, among other elements.

Part 2 (Term 2)

5. must include and clearly state its hypotheses, proofs and interpretations.
6. must fit the academic style criteria. Students must respect the formal, impersonal style required in written analytical scientific documents, while understanding that an oral presentation must be given in a less formal/more personal style than written documents.
7. must include comparisons of concepts (show how similar/different your project is from other similar projects and/or show the similarities/differences within the project).

2. How to start?

The **choice of documents (= 'reference list')** is on a **first-come, first-served basis**. This reference list must be submitted (and potentially regularly updated based on feedback) on **a shared Excel spreadsheet (see Moodle)**. If two or more students have chosen the same document, the time of submission will be used to determine who gets to work on it. If it cannot be determined, it will be decided by a coin toss.

 **Deadlines** (may be adapted for pedagogical reasons)

2nd Oct. 2026 11.59 p.m.	Selection of the two book chapters (click here to confirm your choice)
19th Oct. 2026 8.15 a.m.	Pitch of one book chapter Short explanation of its content (in small groups – formative exercise)
16th Nov. 2026 11.59 p.m.	Selection and presentation of the scientific paper - Assessment in class by your peers of the suitability of your paper: bring a paper version of your paper in class (2 pages per sheet, double-sided) - Short explanation of its content (in small groups – formative exercise)
30th Nov. 2026 11.59 p.m.	Selection and presentation of the popular science article - Assessment in class by your peers of the suitability of your document: bring a paper version of your article in class (2 pages per sheet, double-sided) - Short explanation of its content (in small groups – formative exercise)
22nd Feb. 2027 10.30 a.m.	Presentation of one book chapter Short explanation of its content (in small groups – formative exercise)
22nd Mar. 2027 11.59 p.m.	Selection and presentation of the audiovisual document - Assessment in class by your peers of the suitability of your document - Short explanation of its content (in small groups – formative exercise)
11th Apr. 2027 11.59 p.m.	Final Project checkpoint Submission of your final reference list. <i>No late submission will be considered. 2.5/50 will be deducted per document that does not meet the criteria.</i> Final validation in class (12th April). No further changes after this validation. - First full 8-min presentation of all documents (in small groups – formative exercise)
Two days before the oral exam	Submit your final PowerPoint presentation (use PowerPoint template + submit on final assignment) on Moodle.

3. Analysis table documents

Before inserting information from the two **book chapters** into the main body of your PowerPoint , analyse each chapter with the "**Book Analysis Table**" document.

Before inserting information from the **scientific paper** and the **popular science article** into the main body of your PowerPoint , analyse these two sources with the "**Article Analysis Table**" document. This analysis table is based on the [IMRad format](#). It will help you synthesise the information efficiently and therefore extract valuable information to create synthetic slides.

Before inserting information from the **audiovisual document** into the main body of your PowerPoint , analyse it with the "**Audiovisual Analysis Table**" document.

These three documents are available on Moodle in the Project section.

4. Project presentation and PowerPoint instructions

In class:

- Short presentations and group discussions throughout the year to prepare you for the final presentation
 - use the template provided in the Project section
 - stick to the same PowerPoint for the whole year, adding information as you proceed
 - share the file before the first presentation in September

Oral presentation (exam):

You need to prepare a PowerPoint presentation.

Duration: actual presentation: 8 minutes + 2 minutes Q&A per person. Be prepared to present all documents but note that you will likely only have time to present the gist. **Stick to timing!**

Content:

There will be four parts to your presentation (the information inside each section can be laid out in a personalised manner for your final presentation, as long as each information item is included) + Q/A session + feedback.

- a. **Overview/ Rationale** (one minute max): What is your project about? Why did you choose this topic/these topics? Why is it interesting?
- b. **Main body of the presentation: 2 book chapters + one audiovisual document + one scientific paper and one popular science article** (minimum 500 words) + optional extra documentation to understand the general context of the project:

The main body of the presentation must include all the information gathered through the year via the in-class activities:

- *Defining, describing and qualifying* – general/easy concept description (with qualifiers)
- *Terminology: compound forms* – two key complex concepts found in your documents.
- *Purpose & process* – origin and use of concept (can be same concepts as above or new ones)
- *Measuring & quantifying* – general data

- *Frequency (+ qualifying)* – graph description: more specific data
- Give an overview of the *link* between the documents.
- *Hypothesising* (conditionals and modals): explain at least two hypotheses/proofs found in your documents.
- *Comparing*: compare two concepts (one from your project and another found in literature, on the web, etc. **or** two concepts within your project).

You can reorganise the information as you see fit to correspond to your project logic.

c. **Take home message/conclusion:** what should we remember from your presentation (one recap slide - leave it on during Q/A)

d. **References**

- Cite all your sources in APA7 (reference section in the template) on each slide and at the end of the PowerPoint . This is the most reliable way of citing sources on a PowerPoint .
- The relevant use of each source in your project is left for you to decide depending on your needs. You must be capable of explaining why you chose a specific reference to support your claims and why it is a reliable source, according to academic standards. If you hesitate about the reliability of a source, please ask.

e. **Question/Answer session:**

(non-exhaustive list)

- Which document was the most difficult to understand? Why?
- What is your favourite document and why? How is it different from the others? What does it talk about?
- What was the most difficult task while doing your research and why?
- How can you apply this knowledge in daily life?
- What did you want to transmit to other people talking about this subject?
- Did you learn more about it? If yes, what did you learn?
- Would you like to continue to work on that subject later? If yes, which Master's specialisation course(s) are you looking forward to taking?
- If you had to choose another subject, what would you choose?
- What do you think about the future of your topic?

f. - feedback about the presentation (by the audience - other students)
- language feedback (by the teacher)

Visual aid:

- Make sure to include many figures (graphs, diagrams, etc.) to illustrate your presentation. Otherwise, stick to KEY WORDS.
- NO reading. You may have cards with keywords to help but strictly NO text reading.

V – COURSE MATERIALS & RESOURCES

Compulsory course materials

- Coursebooks (1 per term)
- Oxford English Grammar Course (intermediate - with answers). 2 versions exist:
 - . the older version: ISBN: 978-0-19-442082-2
 - . the new version: ISBN: 978-0-19-441487-6 (5% discount at Scientia for UMONS students)

All students must have access to both these texts during class, either in the form of a textbook, a printed copy, or an electronic copy. **Either the current edition or the previous edition may be used.** Classes will usually consist of some review of these materials, as well as listening practice, writing exercises, quizzes, and discussion.

Additional course materials

Quizlet (to practise most of the basic compulsory vocabulary required in this course)

Procedure to join Quizlet

1. Go to the [Quizlet website](#)
 2. Join Quizlet
Sign up
Fill in
 - username: one word only, no spaces
 - email address
 - password
 - tick 'I agree'
 3. Go to your email inbox and confirm
 4. Come back to this page and click on
 - a. [Quizlet FS class term 1](#)
 - b. [Quizlet FS class term 2](#)
 5. Settings
 - Choose profile picture
 - Language: choose English UK
 - Email alerts: untick all
- [English resource Bank](#) : Learn online; study and practise speaking English in Mons or abroad; watch English programmes; use academic and business glossaries...
 - [Grammar Bank](#): Learn and practise grammar online (intranet)
 - Blattes, S., Jans, V., & Upjohn, J. (2013). Minimum competence in Scientific English. EDP Sciences. <https://doi.org/10.1051/978-2-7598-0989-9>
 - Upjohn, J., Hay, J., & Colle, P. (2006). Minimum competence in Medical English. EDP Sciences.
 - Solovey, N.V., Letunovskaya I.V. (2016). English for students of mathematics: study guide.
 - Nekovar, J. (2011). Mathematical English (a brief summary). Université Paris 6.

Other useful resources

1. Pronunciation tools

- **Howjsay:** Use [Howjsay](#) to listen to the pronunciation of any word, including specific, scientific, and technical terms. This tool is particularly useful for mastering difficult pronunciations.
- **Contextual Listening ([YouGlish](#)/YouTube):** Use YouGlish to hear how specific words are used in different contexts and accents on YouTube. This can help you understand the nuances of pronunciation and usage. If you cannot find the word with this tool, simply type it on YouTube and check reliable videos from conferences, etc. (note: **avoid** at all costs AI pronunciation videos such as this [one](#))
- **Dictionaries:** Refer to dictionaries such as [Merriam-Webster](#) and [Cambridge](#) to check the phonetic transcription of words. This will help you understand the correct pronunciation and improve your phonetic skills.

2. Dictionaries

[Wordlink](#). This fantastic tool can link most webpages word-by-word to online dictionaries. Copy and paste a URL into the box above, specify the language, and click 'Go' to display a copy of the page with all words linked to dictionary lookup:

. **monolingual** (target language: to English): Oxford Learner's Dictionary or WordReference

. **bilingual** (target language: to French): WordReference, Linguee, Larousse

- [Merriam Webster](#) (monolingual with pronunciation)
- [Macmillandictionary.com](#)
- [Wordreference](#) (bilingual)
- [The Free Dictionary](#) (bilingual)
- [Oxford Online Dictionary](#) (monolingual with pronunciation)
- [Linguee](#) (translation engine)
- [Tradooit](#) (bilingual concordancer)
- [Cambridge Online Dictionary](#) (monolingual, bilingual, speciality areas, ...)
- [Collins Online Dictionary](#) (monolingual)
- [Just The Word](#) (monolingual concordancer)

3. Magazines and Press ([popular science articles](#)) [SOURCES FOR YOUR PROJECT]

Before looking up information on any website, check if the University of Mons has paid access through the [UMONS resources](#) (especially Press Reader - look under 'Bases de données'). The University of Mons has institutional access to all these magazines:

- New Scientist
- Scientific American
- Science News
- BBC Science Focus
- Cosmos
- Smithsonian

Free resources online (non-exhaustive list)

- [Quanta Magazine](#)
- [ScienceDaily](#)
- [Phys.org](#)
- [Science Mag](#)
- [Live Science](#)
- [EurekAlert](#)
- [E-science](#)
- [BBC Science and Environment](#)
- [ABC Science](#)
- [NASA Science](#)
- [Science News for Students](#)

4. Scientific resources ([journals](#)) [SOURCES FOR YOUR PROJECT]

The University of Mons has an [institutional access](#) to many journals. Identify a journal from your field of study. To do so, you can use this [website](#) – start by selecting the subject area (e.g. Chemistry) and subject category if you wish (e.g., Organic chemistry), select a journal and find a scientific paper that corresponds to your project topic.

Chemistry

Organic Chemistry

All regions / countries

All types

2023

☐ Only Open Access Journals
 ☐ Only SciELO Journals
 ☐ Only WoS Journals
 ?

Display journals with at least 0

Citable Docs. (3years)

Apply

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1 - 50 of 195

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	Title	Type	↓ SJR	H index	Total Docs. (2023)	Total Docs. (3years)	Total Refs. (2023)	Total Cites (3years)	Citable Docs. (3years)	Cites / Doc. (2years)	Ref. / Doc. (2023)	%Female (2023)
1	Progress in Polymer Science	journal	6.211 Q1	331	50	158	14572	4512	154	26.51	291.44	37.38
2	Nature Synthesis	journal	4.759 Q1	25	221	179	6882	1465	136	8.18	31.14	27.05
3	Chem Catalysis	journal	3.455 Q1	29	273	438	12894	2864	254	6.54	47.23	31.14

Note that **you should not choose a review paper**, which is a different type of scientific publication (slightly more complex and especially longer than a conventional research paper). If you do not know the difference between a research paper and a review paper, go to this [page](#).

5. General and scientific audiovisual resources

General

- [Breaking News English](#)
- [BBC Learning English](#)
- [Voice of America - Learn American English with VOA Learning English](#)
- [English Central](#)

Scientific (specifically mathematics in **bold**)

- [TED-Ed](#) (short videos on various subjects including science, 2 to 7 mins with or without subtitles)
- [Domain of Science](#)
- [Kurzgesagt](#) (short videos on various subjects including science, 2 to 7 mins with or without subtitles)
- [Ted Talks](#)
- [Inside Science](#)
- [Science in Action](#)
- [Science Friday](#)
- [The Open University](#)
- [Science](#)
- [The Life Scientific](#)
- [PBS Space time](#)
- [Numberphile](#)
- [3Blue1Brown](#)
- [Mathologer](#)
- [Michael Penn](#)
- [The Bright Side of Mathematics](#)
- [Another Roof](#)
- Many more: [Aleph 0](#), [Dr Trefor Bazett](#), [MIT OpenCourseWare](#), [Eddie Woo](#), [Stand-up Maths](#), [Combo Class](#)

6. APA style (referencing guidelines)

- Reference list: <https://apastyle.apa.org/instructional-aids/creating-reference-list.pdf>
- In-text citations: *In-text citations*. (n.d.). <https://apastyle.apa.org>. <https://apastyle.apa.org/style-grammar-guidelines/citations>
- Free APA7 citation generator: *APA7 Citation Generator: CiteFast automatically formats citations in APA 7th Edition*. (n.d.). Cite Fast. <https://www.citefast.com/?s=APA7#> [Webpage](#)

7. Proofreading, paraphrasing tools & improving your English with AI

- [DeepL write: AI-powered writing companion](#)
- [Writefull](#)
- [LanguageTool](#)
- [Online AI Editor & Grammar Checker - Wordvice AI](#)
- [Lengusa | Write better English!](#)
- [Ludwig • Find your English sentence](#)
- [Grammarly](#)

VI – HOW TO IMPROVE YOUR ENGLISH - DISCUSSION

In groups of 3-4:

1. Discuss ways to improve your language skills (Speaking, Writing, Listening, Reading and Vocabulary), and write down your ideas below:
2. Make a contract with yourself: decide on 3 changes in your life to learn English more actively and write them down below:
 -
 -
 -

VII – 10 RESOURCES TO IMPROVE YOUR LISTENING SKILLS & SPOKEN ENGLISH IMMERSIVELY

1. **Travelling/living abroad!**: this is based on the idea that the active use of a foreign language in a variety of real-life circumstances and contexts is the best way to learn it. This method has several advantages. It accelerates learning and helps improve oral expression, promotes cognitive growth, improves memory and problem-solving ability. It can also help increase cultural understanding and empathy, helping students have a greater awareness and respect for other cultures. Read [this experience](#) to learn further on the topic.
2. **AI Applications**: To enhance your English language skills independently, consider using various AI applications. Some of these apps, such as Pronounce and ELSA Speak, offer limited free plans. For a comprehensive list of AI pronunciation tools, you can visit this [website](#).
3. **Original Version with English Subtitles**: Make a conscious effort to watch your favourite series and movies in their original version with English subtitles. This method is effective because it allows you to hear the natural flow of the language while simultaneously reading the text. [Link](#) to read further on the topic.
4. **English-Speaking TV Channels**: Watching English-speaking TV channels can also be beneficial. Many of these channels have excellent apps and YouTube channels, such as the BBC News App available on [Google Play](#) and the [Apple Store](#), and the [BBC News YouTube channel](#).
5. **YouTube Videos**: You can also watch virtually any video in English on YouTube, including entertaining ones that make learning English enjoyable.

6. **Music in English:** Listening to music in English is another great way to improve your skills. For active listening practice, you can use platforms such as [LyricsTraining](#).
7. **Video Games:** Playing video games in English, especially massively multiplayer online games (MMOs) like Fortnite, World of Warcraft (WoW), and League of Legends (LoL), can be particularly helpful. Games with rich storylines, such as Zelda, Horizon, Grand Theft Auto (GTA), and The Last of Us, also provide immersive language experiences. For additional resources, you can explore the YouTube channel [Real English for Gamers](#).
8. **Podcasts:** Listen to podcasts on Spotify, Deezer, and Apple Music. Many podcasts come with transcripts, which can aid in understanding and learning new vocabulary.
9. **Language Exchange/Events:** Attend English events or find language exchange opportunities in Mons through platforms like My Language Exchange. These events provide a great way to practice speaking with native speakers and other learners. *The Bookshop* in Mons offers such [events](#) free of charge. A colleague also organises events through her Facebook group [English Evenings](#).
10. **Museums:** Visit museums that offer audio guides in English. This can help you improve your listening skills while learning about various subjects.

VIII – 17 TIPS TO IMPROVE YOUR PRONUNCIATION

1. You should be aware of the **movement of your lips, tongue, and throat**, and it's helpful to check your progress in the mirror. You might begin with this [video](#).
2. You can **record yourself using your phone's voice texting feature**, such as WhatsApp's own conversation.
3. **Practising shadowing with transcripts** from TED Talks, audiobooks, podcasts, videos, songs, and other sources can be beneficial. Shadowing is a technique where you listen to a native speaker and try to mimic their speech as closely as possible, including their intonation, rhythm, and pronunciation. You can also create your own transcripts with the free online text-to-speech AI tool [Luvvoice](#) in order to practise your pronunciation (which is very useful to prepare for a presentation). Combine it with Reading Coach (see 5.).
4. **Imitating common phrases and words** used in casual conversations, like "oh I donno" or "hi whatcha doing?" can be a good start. You might begin with this [video](#).
5. **Reading aloud** regularly helps you strengthen your mouth muscles. You might begin with [Reading Coach](#).

How to use it: select 'add your own passage' and paste the text you want to practise reading. Then, allow access to your microphone for recording. Read out loud, making sure to articulate and speak loudly enough and slowly. Practise individual words to improve your pronunciation.
6. **Emphasising key phrases** can improve your clarity and expression.
7. Focusing on **word stress** enhances your pronunciation.
8. **Comparing different sounds and words**, especially [minimal pairs](#), can refine your listening and speaking skills.
9. **Practising the [seven intonation patterns](#)** in your speech is extremely important. One example with this [Friends video](#).

10. **Working on contractions** can help you sound more natural in conversations. You might begin with these two videos: [affirmative contractions](#) and [negative contractions](#).
11. **Singing along to slow songs** by artists such as The Beatles or Frank Sinatra can be enjoyable, as new songs tend to be fast.
12. **Trying tongue twisters** can be a fun way to improve your pronunciation and articulation.
13. **Comparing English sounds with French** can help you understand the differences and improve your pronunciation. Have a look at these [eight points of comparison](#).
14. **Remaining consistent with British English** in terms of pronunciation, vocabulary, and grammar is important (see this [video](#)). If you're not confident with another form of English, it's best not to improvise. There are exceptions for students who have studied at SHAPE or lived in the US, Australia, etc. who might be more familiar with another form of English.
15. **Accepting feedback** from friends or teachers is crucial for improvement. Remember, it's nothing personal!
16. **Practising regularly** is key because practice makes perfect.
17. [EngVid pronunciation exercises](#) can be a helpful resource for improving your pronunciation.

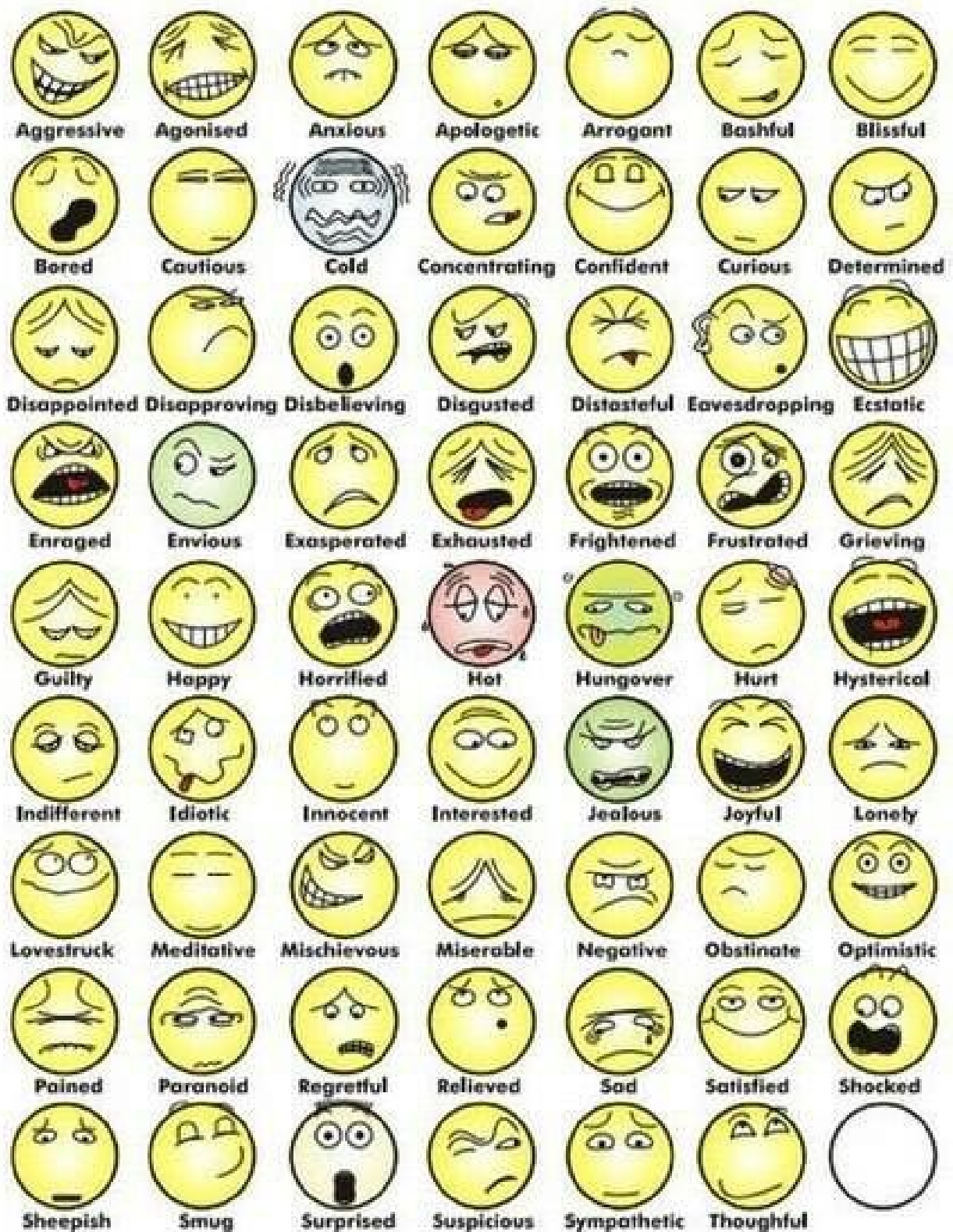
IX – INTRODUCTIONS

1. Read the questions quickly, then get up and introduce yourselves to at least 4-5 classmates.



- What is your name, how old are you & where are you from?
- Tell me something about your hobbies and interests
- Why did you choose to study mathematics at the University of Mons?
- What is interesting about mathematics for you?
- When did you know that you were a math person?
- Experience of learning English: for how many years have you been learning English?
How do you immerse yourself?
- When did you last use English? With whom? In which situation? How did you feel before, during and after communicating in English? (Emotion vocabularyⁱⁱ below to help you)
- Have you ever been to an English-speaking country? Which one?
- What do you expect from this course?

How do you feel today?



2. Plenary: judgemental Language - give examples from your personal experience.

- What is "good/bad" or "(in)correct" English like? Is that question even acceptable?

Give examples from your personal experience.

- Why was it "good" or "bad"?

You will probably find it difficult to answer this because there's an inherent problem in using judgements like "bad" or "good". What do these words mean? Why are they problematic, not just to qualify English but as a whole?

There is no such thing as good or bad English. The use must be **contextual**.

Adapt your language level to the communication situation (writing a science report ≠ writing an email, speaking at a conference ≠ talking to your friends, etc.) and brush up on your grammar and vocabulary skills.

Otherwise, **do your best and persevere! Practice makes perfect!**

X – HOMEWORK FOR NEXT CLASS

1. Do the placement test. The procedure is on Moodle.
2. Get the coursebook from the 'Presses Universitaires' if you want a paper copy (**on request this year**).
3. Get the grammar book (PDF or paper copy)
4. Grammar: Revise Present Tenses
 - Grammar book, Section 2 (Present tenses), pp. 13-16., pp. 17-21
 - AngloLink [Video](#)
5. Download on your smartphone:
 - the 'WordReference' app
 - BBC news (for short demo: <http://www.bbc.com/news/help-33449119>) - select the following topics: World - Europe - UK - France - Belgium - Science & environment. You can of course select more topics that are of interest to you.

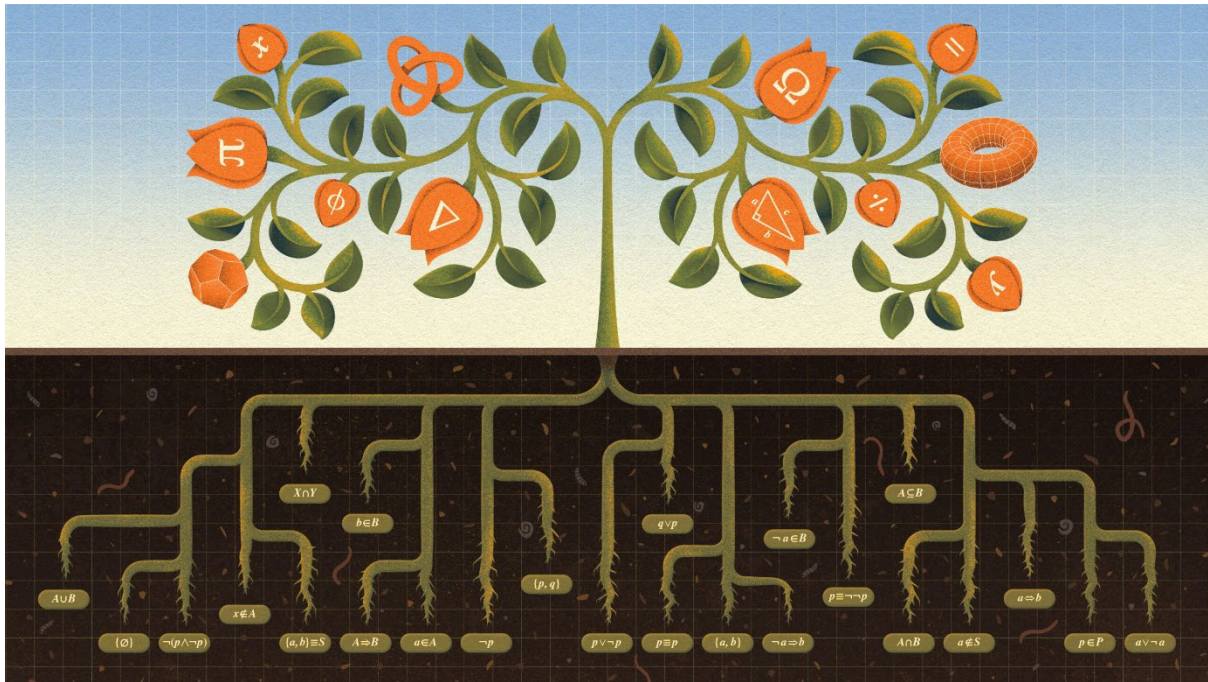
Don't forget to check the news (general and scientific) so that you can talk about one item next week.
6. Reading & research: '*Why Math's Final Axiom Proved So Controversial*'ⁱⁱⁱ. Check questions 1-3 and try to answer before reading the article, then read the article (similar to what you need to find for your popular science article) and answer the rest of the questions (4-9).
7. If you think you have time to explore the books for next week, go to Moodle and have a first look at the three books selected and think about the two chapters you would like to work on.

Why Math's Final Axiom Proved So Controversial

By Gregory Barber

April 29, 2026

Zermelo-Fraenkel set theory is so widely accepted that modern mathematicians hardly think about it. But believing in its core principles didn't come easily.



Kristina Armitage/Quanta Magazine

Introduction

[1] How do mathematicians decide that something is true? They write a proof.

[2] Often they start with proofs that already exist, building on or drawing connections between proven claims. Each of these proofs, in turn, has relied on other proofs to make its point, and so on. Proofs upon proofs. Truths upon truths. But eventually this process must come to an end. At some point, things are true simply because they are.

[3] These truths are the axioms, the ground rules. And it is tempting to stop there — to declare, as [Penelope Maddy](#), a philosopher of mathematics at the University of California, Irvine, put it, “that axioms are obvious or intuitive or conceptual truths.”

[4] After all, most mathematicians simply accept that their work relies on an axiomatic system — namely, “Zermelo-Fraenkel set theory with the axiom of choice,” or ZFC — if they bother to acknowledge the axioms at all. ZFC is a list of 10 basic principles that together form the foundation on which nearly all of modern mathematics is built.

[5] But a closer inspection reveals a more unsettled, human process of establishing truth. “Any honest, clear-eyed examination of how the axioms of ZFC came to be adopted would have to acknowledge that a wide range of mathematical considerations went into these decisions,” Maddy said.

[6] That process, which began over a century ago, is still very much in progress.

Paradoxes and Doubt

[7] The late 1800s were a time of paradoxes and doubt, the result of mathematicians beginning to search for cohesive ideas about what rules the mathematical universe obeyed. There were axiomatic systems out there, but they tended to be for specific areas of math: Euclid's postulates for geometry; various schemes for standardizing arithmetic. But how did they all fit together? Could all of math be derived from one common set of rules?

[8] Mathematicians found a potential solution — and more doubts — in the work of Georg Cantor.

[9] At the time, Cantor was studying the real numbers — that is, all the numbers that appear on the number line — and what they could say about the nature of infinity. He had found that there were [more real numbers than whole numbers](#), giving rise to the profound realization that not all infinities are the same size.



Public Domain

The German mathematician Ernst Zermelo came up with the axiom of choice, the most controversial of the 10 basic principles that today underlie all of mathematics.

[10] To make this comparison, Cantor had used a seemingly simple tool: the set. A set is a collection of objects, or elements. It might be a collection of numbers, like the real numbers, or a collection of shapes, or even a collection of other sets. Over time, it became clear that complex and disparate mathematical ideas — nearly all of them — could be represented with these same elementary entities. As a result, the set emerged as a potential tool for ironing out any inconsistencies between different areas of math.

[11] But early set theory lacked canonical rules. It was possible to define sets with any property, which led to exactly the sorts of paradoxes that were bothering mathematicians at the time. Consider, for example, the set of all sets that are not members of themselves. Does this set contain itself? Whether you answer yes or no, you get a contradiction now known as Russell's paradox.

[12] As mathematicians obsessed over these dilemmas, ZFC emerged out of a struggle with another idea of Cantor's.

[13] In 1883, Cantor introduced his "well-ordering principle." He claimed that it should be possible to arrange any set so that all of its (non-empty) subsets would have a smallest element. For finite sets, this is intuitive. You can always put the lesser items first. But for infinite sets, it is less obvious. Take the set of integers $\{\dots, -2, -1, 0, 1, 2, \dots\}$. The negative numbers form a subset, but they also get lower and lower for eternity. It seems as if there can be no least element.

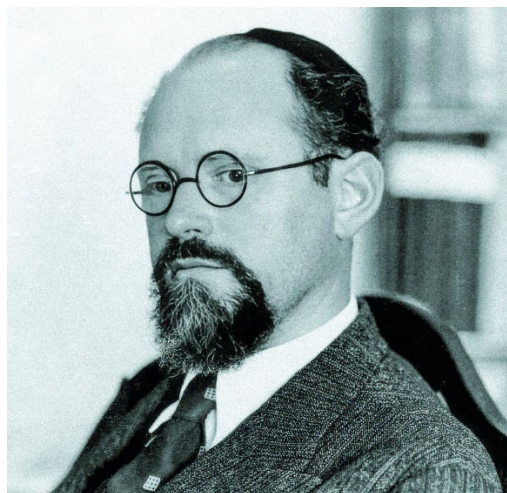
[14] But what if you arrange the original set of integers like this: $\{0, -1, 1, -2, 2, \dots\}$? Now you can say that the smallest element is the one that comes first in any subset. In this way, -1 becomes the smallest element of the subset of negative numbers.

[15] Cantor's "law" was that this should be possible for all sets, even if you can't explicitly construct the proper ordering. It was one way of arguing that infinite sets behave like finite ones.

[16] In 1904, the German mathematician Ernst Zermelo proved it. He did so by showing that Cantor's law was equivalent to a principle he had developed while exploring the properties of sets. This principle, the so-called axiom of choice, said that if you start with multiple (or even infinitely many) non-empty sets, you can choose one element from each of those sets to create a new set.

[17] Zermelo developed his other axioms to prove this equivalence. "He was just listing all the assumptions that he needed to get the proof through," said [Joan Bagaria](#), a set theorist at the University of Barcelona. That list included the basic idea that there is such a thing as a set, which is defined by its elements. Other axioms dealt with the formation of sets from other sets, or with the existence of infinite sets.

[18] Zermelo's list of axioms emerged at a time when many mathematicians, such as Abraham Fraenkel, were also tinkering with set theory's foundations. Many of them found themselves arriving at different formulations of similar ideas — and some new ones, too, that resolved problems arising from newer theories having to do with larger forms of infinity. In 1930, Zermelo released a "final" list that included revisions to his own axioms as well as a handful of additions — but not, at first, the axiom of choice. Mathematicians were more hesitant to include it, because unlike the other axioms, it defined sets without giving an explicit way to construct them.



Abraham Fraenkel was one of the founders of modern set theory, adding several key rules to its core list of axioms to help get rid of various paradoxes.

The David B. Keidan Collection of Digital Images from the Central Zionist Archives

[19] Zermelo was pleased that his list of principles, known as ZF, appeared to cleanse the set-theoretic universe of many major paradoxes such as Russell's. But he lamented that he was unable to prove that his axiomatic system was "consistent" — that it didn't yield contradictions.

[20] He need not have worried. Just a few years after the arrival of ZF, Kurt Gödel showed that no axiomatic system capable of basic arithmetic can be used to prove its own consistency. Moreover, any consistent system must also be incomplete, meaning that there are true mathematical statements that cannot be proved using the system's axioms.

[21] In fact, in the 1960s, the Stanford mathematician Paul Cohen proved that the axiom of choice is "independent" of the other axioms — that is, under the rules of ZF, the axiom of choice cannot be proved true or false.

[22] Once it was clear that logic could not validate the axiom of choice one way or the other, the question became: Is it useful? And it was. It makes a great deal of other math possible — especially [math related to infinite objects](#). After that, the axiom of choice gained much more widespread acceptance. “Without choice, your tools are very limited,” Bagaria said. “It’s like doing math with your hands tied behind your back.” And so the C (for “choice”) came to be affixed to the list of axioms that were originally developed to support it.

[23] The axiom of choice demonstrates the folly of believing that mathematical axioms are self-evident or obvious. An axiom can also be accepted for plenty of other reasons, as Maddy put it — such as for its power to generate interesting theorems.

[24] The ZFC axioms are often regarded as perhaps the most universal truths that humanity has managed to articulate — for while it may be possible for physicists to imagine universes in which physical laws are turned inside out, mathematical laws will remain constant.

[25] It is a paradox without resolution: The foundations of mathematics are as universal, as solid as anything humanity knows, a core part of nearly every mathematical truth. And yet they remain simply what we choose to believe.

The 10 Axioms of ZFC

1	Null Set: There exists a set that has no elements, called the empty set.	
2	Extensionality: If two sets have the same elements, they are the same set.	
3	Pairs: For any two sets, you can form a new set whose elements are those two sets.	
4	Power Set: For any set, you can form a new set that contains all its subsets.	
5	Union: If a set contains other sets, you can gather all the elements of those other sets to form a new set.	
6	Infinity: There exists an infinite set.	
7	Separation: From any set, you can form a subset of elements that satisfy a given property.	
8	Replacement: For any set, you can form a new set by applying a given operation to each of its elements.	
9	Regularity: No set can be an element of itself.	
10	Choice: Given a collection of sets, you can choose one element from each of them to form a new set.	

1. Read only the title of the text. What do you already know about the topic?
2. Does the text seem reliable? Why?
3. Do you think the text will meet your expectations? Why?
4. After reading the text quickly, write down at least 5 keywords. What do they mean?
5. After reading the text more carefully, divide it into sections, e.g. introduction, main part(s), conclusion etc. Give a content-related keyword to each section.

Section	Content-related keyword

6. Write down the main idea of each paragraph (they are **[numbered]** in the text) using *one or a few words only* for each.

Paragraphs	Main idea
1	
2	
3	
4	
5	
6	
7	
8	
9	

10	
11	
12	
13	
14	
15	
16	
17	
18	
19	
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21	
22	
23	
24	
25	

7. Combine your answers to questions 4-6 into a single short summary of one paragraph.

8. Why or how is the article important and what is/are the most important point(s) - the big idea?

9. Write down a reference to the article according to [APA referencing guidelines](#).

On the next page, combine these answers (4-7) and be prepared to **pitch** them in about 90 seconds in class (No PowerPoint needed). *Rehearse your pitch out loud* before coming to class.

What is a pitch?

The purpose of a(n) (elevator) pitch is to raise sufficient interest.

Discover what it is exactly in this [video](#)^{iv}. You can also watch a few of these 3-minute [pitches](#) from Oxford Mathematics.

To sum up:

1. A pitch shrinks your idea — values, promise, and character — into one statement. It answers **4 main questions**:
 - What is the initial **problem**? (e.g. water pollution)
 - What is your **idea**? (e.g., one specific novel cleaning technique)
 - What does it do? What **solution** does it offer? (e.g., some details about the technique)
 - What is the **value** of your idea? How does it differ from other similar solutions? Why is it interesting/exciting? What makes it unique? (e.g., explain why this technique is more cost- or time-efficient to tackle water pollution)

Guess what the characteristics of a pitch are. It should:

-
-
-
-
-
-
-

Pitch Structuring

1. Establish the problem or need
 2. Introduce the solution or offering
 3. Share the value/benefit or your idea (what makes you different)
 4. Optional conclusion/summary
- + Provide evidence or support/example

[illegible]

PITCHING IDEAS & IDENTIFYING SOURCES

I – READING & RESEARCH: WHY MATH'S FINAL AXIOM PROVED SO CONTROVERSIAL

1. Discuss: plenary: What was the aim of the activity?
2. In groups of 3-4: pitch your answers in 90 seconds and compare them. Agree on a common answer if possible.
3. Plenary: recap.
 - What are the key ideas of the text? Why do you find it important?
 - Guess what the characteristics of a pitch are. It should:
 -
 -
 -
 -
 -
 -

II – PROJECT: SOURCE SELECTION

GENERAL INFORMATION vs. CURRENT RESEARCH (SCIENTIFIC PAPER & POPULAR SCIENCE ARTICLE)

Do you understand the differences between all the sources you need to find (see **IV – PROJECT** above)? Do these short exercises on source selection.

GENERAL INFORMATION vs. CURRENT RESEARCH

Look at the following articles. Are they about general/encyclopaedic information or current research? Justify.

Title	Source	General info	Current research
From Symphony to Structure: Listening to Proteins Fold	TheScientist		
I: Molecular Mechanics and Dynamics	BIOCHEMISTRY ONLINE		
Molecular dynamics simulations and drug discovery	BMC Biology		
Scientists Say: Doppler effect	ScienceNewsExplore		
Scientists use Doppler to peer inside cells	Phys.org		
Salient Object Detection via Fusion of Multi-Visual Perception	MDPI		
Game theory	Wikipedia		
New Knot Theory Discovery Overturns Long-Held Mathematical Assumption	Scientific American		

➤ SCIENTIFIC PAPER (NOT A REVIEW/SUPERSTUDY)

1. How do you recognise a scientific paper? Look at this [example](#).

- Content:
- Length:
- Authors:

2. Fill in the following table.

Title	Source	Scientific paper? (tick)	If scientific paper, research or review (circle)	Suitable as scientific paper?
Loop Quantum Gravity– An Introduction	Proceedings of Science		Research – Review	yes - no
Loop quantum gravity theory offers glimpse beyond the event horizon	Phys.org		Research – Review	yes - no
Design of a linear motor-based magnetic levitation train prototype	International Journal of Reconfigurable and Embedded Systems (IJRES)		Research – Review	yes - no
A comprehensive review of nanoscale MOS capacitors applications in photonics	ScienceDirect		Research – Review	yes - no
Second-order nonlocal shifts of scattered wave-packets: What can be measured by Goos–Hänchen and Imbert–Fedorov effects?	ScienceDirect		Research – Review	yes - no

➤ POPULAR SCIENCE ARTICLE: NEWS/MAGAZINE ARTICLE (min. 500 words)

Title	Source	General info	Current research	Length (min 500 w.)	Suitable as current research article? Justify
Bitcoin: electricity consumption comparable to that of Poland	Polytechnique insights			1088 words	yes - no
The gambling strategy that's guaranteed to make money and why you should never use it.	Scientific American			1547 words	yes - no
Mastering the Importance of Poker Math	BETMGM			926 words	yes - no
You didn't see it coming: New research shows turbulent flows can be caused by minute triggers.	Phys.org			800 words	yes - no
Unveiling Roulette's Mathematical Secrets	Women's Resource Centre			1036 words	yes - no
We could use one of Jupiter's moons to hunt dark matter	New Scientist (find the article yourself in under 3 minutes – use UMONS library access)			_____ words	yes - no

III – FIND YOUR PROJECT TOPIC

Personal research

1. Individually, think of **a coherent project topic from one book chapter pairing (= two related book chapters)**. Stick to **one central idea**. Check out the books and the [book chapter pairing tool](#) on Moodle, browse the internet (scientific papers, Quanta Magazine, etc.) and look up **(current) research findings** in your field of study.

CENTRAL PROJECT IDEA	

Link between book chapter n°....	and book chapter n°....

Now, get up and discuss project ideas with your classmates.

2. Do not directly ask others about their project idea or immediately share yours. Use small talk (why you are personally interested in such or such topic) and find roundabout ways to direct the conversation towards the project.
3. Present your project topic idea to your classmates and ask for advice. Give advice about theirs. Avoid any negative comments.
4. Identify whether some project ideas overlap and discuss strategies/compromises to keep each idea unique.

IV – PITCHING AN IDEA

- How have you presented your ideas so far? Do you remember what a **pitch** is?

The purpose of a(n) (elevator) pitch is to raise sufficient interest.

To sum up:

A pitch shrinks your idea—values, promise, and character—into one statement. It answers **4 main questions**:

- What is the initial **problem**? (e.g. water pollution)
- What is your **idea**? (e.g., one specific novel cleaning technique)
- What does it do? What **solution** does it offer? (e.g., some details about the technique)
- What is the **value** of your idea? How does it differ from other similar solutions? Why is it interesting/exciting? What makes it unique? (e.g., explain why this technique is more cost- or time-efficient to tackle water pollution)

It should:

1. Give **one big benefit** rather than several small ones, as most people respond better to that.
2. Tell a **story**.
3. Be very brief (usually under **180 seconds**. In this case, **90 seconds** is more than enough!)
4. Remain **simple**. Leave technical and other detail for questions.
5. Include some **numbers** if you have them.
6. Be given in a **positive** and **enthusiastic** tone.

V – PITCH STRUCTURING

1. Establish the problem or need
 2. Introduce the solution or offering
 3. Share the value/benefit or your idea (what makes you different)
 4. Optional conclusion/summary
- + Provide evidence or support/example

SpaceDev Exercise: identify the pitch structure

SpaceDev was founded with the vision of producing innovative and affordable space solutions to enable humanity to explore and develop space through practical and profitable steps. To realise this vision, we are developing evolutionary building blocks – starting with affordable microsatellites and their associated subsystems, and safe affordable propulsion and launch systems. To retain and build the team necessary for these developments, SpaceDev is fostering a corporate culture that attracts and retains creative, practical, and energetic employees who are driven to “make space happen”.

VI – AUDIO: SANDER SONG'S PITCH^{vi}

1. What concrete problem does he provide to explain the theory?
2. What is the theoretical concept and the difficulty linked to this theory?
3. Explain the gist of his research. What solution does it offer specifically?
4. Why do you think it was a good/bad pitch? (structure, tone, voice, etc.)

VII – START ON YOUR PROJECT: INSTRUCTIONS

1. Download the PowerPoint template (see *Project* section on Moodle) and set up a shared POWERPOINT document for your future project.

Indicate the title of your project and copy and paste the link (click on 'paramètres du lien' , '**personnes UMONS**' and share in '**consultation possible**' access) of your shared POWERPOINT into this shared Excel sheet '[MATH - Project topics + article selection - 26-27](#)' so that all students from your class can access your file without the possibility of modifying it.

2. Determine the central idea of the project (to do so, you can try to answer the **four questions** from '**PITCHING AN IDEA**') in one or two paragraphs.
 - Please write a formal and continuous text. Do not simply "answer" the questions as if they were a questionnaire (and even less so include them in your text!).
 - Create a structured text '**PITCH STRUCTURING**'.

3. Prepare yourselves to pitch this idea orally next week under 90 seconds.
Try to **finalise the book chapter pairing choice** and give some details about your project idea.

Do not read a text! You are allowed to have key words.

Your written project idea must be different in style from your oral pitch, i.e., the former is to inform in a neutral way, the latter is to convince or sell your idea.

VIII – HOMEWORK FOR NEXT CLASS

PROJECT

1. Prepare and rehearse your project pitch (based on the book chapter pairing choice).
2. Make sure that your project idea is unique and obtain your teacher's final approval.
3. Students absent on 21st September: Project: find topic/chapter pairing + put the info in the '[MATH - Project topics + article selection - 26-27](#)' shared file.

GRAMMAR

4. Revise Questions
 - Grammar book, Section 8 (Questions), pp. 103–107. Check your answers.
 - coursebook: questions
 - question structures: Test Yourself!
 - AngloLink [Video](#)
5. Revise Passive voice*
 - Grammar book, Section 7 (Passives), pp. 93–95.
<https://www.grammarbank.com/simple-present-passive.html>

* This is a Term 2 grammar chapter, but it is already required to understand it decently well now.

VOCABULARY

6. U1: Defining, Describing & Qualifying: Preparatory Tasks (including **Show and Tell preparation activity** at the end of the lesson) (Moodle)
7. U1: Purpose & Process: Preparatory Tasks (Moodle)

TEST YOURSELF ON MOODLE

8. U1: Qualifiers: Test Yourself! – can be done later if too much work
9. U1: Suffixes: Test Yourself! – can be done later if too much work

QUIZLET

10. Create an account, discover the tool and start practising vocabulary.

UNIT 1: DEFINING, DESCRIBING & QUALIFYING



I – PRACTISE FOR YOUR PROJECT

1. Pitch your project idea to your classmates (move around in the class). Be convincing and enthusiastic about the project you are defending.
2. Take notes about at least 2-3 projects.

3. Roundtable: one student explains the project of another student and gives their impressions. Why did you find it interesting? Give some details that stood out.

II – SHOW AND TELL

1. In groups of 4-5, take turns to describe the object you have chosen and try to guess what it is by writing your answer on a small piece of paper.
2. Do not give your answer orally during the description.
3. The student who guesses the most objects wins.

III – TIME'S UP! SPEAKING PRACTICE & VOCABULARY BUILDING

1. In groups of 4-5 (2 teams within one group): take turns to describe the words you have received on cards and make the rest of your team guess them (3 rounds – 1st round: sentences; 2nd round – one word; 3rd round – mime)
2. The team that guesses the most words win the round.
3. Count the points for each round and determine the winning team.

A. Categorisation

One of the simplest ways of defining a mathematical term is by using the word “**to be**” and assigning it to a general category.

- The definition answers the implicit question: “*What (sort of thing) is X?*”

It is a number • figure • shape • expression • equation • formula • function • operation • theorem • property • proof • method • algorithm • transformation • object • instrument • tool • device

POTENTIAL MODIFIERS

It's a kind of / sort of / type of (device)

- ▶ A prime number is a **sort of integer** that has exactly two distinct divisors: 1 and itself.
- ▶ A quadratic equation is a **kind of polynomial equation** in which the highest power of the variable is 2.

B. Constituent elements

A mathematical object can be defined by listing and describing its parts or components.

- Implicit question: “*What are the basic elements of X?*”

It is made (up) of • It consists of • It is composed of • It comprises • It includes • It contains • It is formed by

POTENTIAL MODIFIERS

typically • basically • mostly • mainly

- ▶ A fraction **consists of** two integers: a numerator and a denominator separated by a fraction bar.
- ▶ **Typically**, a coordinate system **comprises** two perpendicular axes — the x-axis and the y-axis — meeting at the origin.

C. Location

Definitions can be refined by indicating the position or direction of a mathematical object.

- Implicit question: “Where is X ?”

It is situated / found / located / placed • It lies • It moves / shifts / tends / proceeds / progresses (+ preposition)

FREQUENTLY USED PREPOSITIONS

above • along • at the origin / end(point) of • at the bottom/top • behind • beneath • below • beside • between • in front of • near • next to • in the middle • at the centre of • in the interval • inside • on the axis • on the left(-hand side) / right(-hand side) of • outside • through • towards • underneath • within

For more information related to prepositions of place, see:

<https://www.grammar.cl/rules/prepositions-of-place.htm>

- ▶ The vertex of a parabola **is located at** the point where the curve changes direction.
- ▶ An asymptote is a line that the graph of a function approaches but never reaches as the variable **tends towards** infinity.

D. Use

Definitions may explain the purpose or application of a mathematical object.

- Implicit question: “What is X for?”

It is to / for (+ -ing) • It is used to / for (+ -ing) • It serves to • It allows (+ -ing) • It enables / helps to

POTENTIAL MODIFIERS

above all • commonly • essentially • frequently • mainly • typically • widely

- ▶ Integration **is essentially used to** calculate the area under a curve or between two curves.
- ▶ The determinant of a matrix **serves to** determine whether a system of equations had a unique solution.

E. Purpose

→ Purpose is concerned with questions of “**why**” or for “**what reason**” something is done.

■ Nouns

the purpose • function • use • role

- ▶ The Euclidean algorithm is used for computational **purposes**, particularly in cryptographic applications.

the aim • goal • target • objective

- ▶ The main **objective** of number theory is to study the properties and relationships of integers.

■ Verbs

it is designed to / for • is devised to / for • is intended to / for • is aimed to / at (+ ing) / for
• is responsible for • is planned to / for

- ▶ The Gaussian elimination method is designed to reduce a system of linear equations to row echelon form.
- ▶ This notation was devised for expressing the asymptotic behaviour of functions in a compact and rigorous way.

it functions (as) • operates (as)

it produces • it generates • it yields

- ▶ The Fourier transform **operates as** a tool for decomposing a function into its constituent frequencies.

■ Other expressions

in order to / so as to • so that

N.B. – “In order to / so as to” have basically the same meaning as “to”. They are merely stronger ways of expressing the idea of purpose.

“So that” is followed by a clause composed of subject + verb.

- ▶ We apply the change of variables **to** simplify the integral.
- ▶ We apply the change of variables **in order to / so as to** simplify the integral.
- ▶ We apply the change of variables **so that the integral can be simplified.**

F. Process

→ Process is concerned with questions of “**how**” or “**in what way**” something is done.

■ Adverbial and prepositional phrases

by means of • through • thanks to • via •
using

therefore • thus • thereby • hence •
it follows that

► The result has been obtained **by means of** a substitution that transforms the integral into a standard form.

► We factor out the common term, **thereby** simplifying the expression considerably.

■ Verbs

to enable • make (it) possible (for) • allow • permit

► Diagonalisation **enables** / **makes it possible for** us to compute powers of a matrix efficiently.

► The change of variable **allows** / **permits** the integral to be evaluated in closed form.

N.B. – Meaning

“**To enable** / **to make possible**” have the primary meaning of making something feasible or mathematically tractable.

“**To allow** / **to permit**” have the primary meaning of making something valid under given conditions or axioms.

However, in modern usage both are often interchangeable in mathematical writing.

The substitution $x = \sin\theta$

enabled
made it possible for
allowed
permitted

the integral to be evaluated.

N.B. – Many impersonal forms are present in this unit (see underlined verbs above). We will study them further in Term 2.

TO DESIGN VS. TO DEVISE

1. 'The car was _____ in Italy and made in Brazil.'
2. 'The company has _____ a computer programme that allows people to design their own homes.'

→ to _____: to plan and organise the structural elements of an object/project (often on paper)

→ to _____: to elaborate, invent a new system or solution involving careful thought and innovation

Note:

- 'TO' is used before a verb

e.g. 'These measures are designed TO reduce pollution.'

- 'FOR' is used before a noun (or verbal nouns, often ending in '-ing')

e.g.: 'Computers were designed FOR processing massive amounts of data.'

e.g. 'A new system was devised FOR underwater exploration.'

IN ORDER TO / SO AS TO / SO THAT

1. "I regretfully must withdraw from this campaign _____ take care of my wife who was just recently diagnosed with Parkinson's disease."

2. "I regretfully must withdraw from this campaign _____ I can take care of my wife who was just recently diagnosed with Parkinson's disease."

→ so that + ...

→ in order to/ so as to + ...

THEREBY (+ _____)

Students perform in hospitals, thereby _____ (gain/gaining) a deeper awareness of the therapeutic power of music.

G. Shape

We often make definitions by describing the shape or appearance of a geometric object.

- Implicit question: “What does X look like?”

It is circular / round / oval / curved • It is square / oblong / rectangular / triangular • It is hollow / solid / rigid / flexible • It is narrow / wide / thin / thick / flat • It is symmetric / asymmetric / regular / irregular • It is convex / concave • It is closed / open / bounded / unbounded

POTENTIAL MODIFIERS

relatively • quite • fairly • about • roughly • approximately

- ▶ A circle is a **round, closed** curve where every point is equidistant from the centre.
- ▶ A hyperbola consists of two **roughly symmetric**, open curves that each approach their asymptotes.

N.B. – One simple way of describing a shape is to use the structure: **NOUN + “-SHAPED”**.

cube-shaped • triangle-shaped • diamond-shaped • disc-shaped • cone-shaped • sphere-shaped • U-shaped • L-shaped • bell-shaped • tree-shaped

- ▶ A normal distribution curve is **bell-shaped**, symmetric about the mean.

H. Relative pronouns

In order to gain in precision, definitions often resort to the use of relative pronouns: **who • whose • (of) which • that ...**

For people: who • whose

- ▶ A mathematician is a scientist **whose** work focuses on abstract structures and logical reasoning.

For objects: which • that

- ▶ A prime is an integer greater than 1 **which / that** has no divisors other than 1 and itself.

For more information related to relative pronouns, see:

- Grammar book, Section 19 (Relatives), pp. 267–280.
- https://www.youtube.com/watch?v=IlkARyk4CTI&t=746s&ab_channel=Anglo-Link

1. Adjectives

■ Importance

important • significant • crucial • meaningful • relevant

- Euler's identity is still **relevant** to contemporary problems in complex analysis and number theory.

■ Dimension

large-scale • enormous • huge •
tremendous • widespread

≠

small-scale • tiny • negligible • minute

- The error introduced by this approximation is **negligible** for values of n greater than 1000.

■ Hierarchy

major • main • chief • leading •
primary • foremost • outstanding

≠

minor • secondary • common •
average • typical • standard

- The **primary** condition for convergence is that the sequence of partial sums remains bounded.

■ Intensity

dramatic • striking • steep • sharp • marked • notable

- There is a **sharp** increase in the value of the function as x approaches 0 from the right.

■ Aptitude and utility

appropriate • suitable • useful • reliable • efficient • powerful • rigorous

- Newton's method is a **suitable** algorithm for finding the roots of a differentiable function.

2. Adverbs

■ Essence

essentially • fundamentally • basically • on the whole

- **Fundamentally**, there is little difference between the two methods of solving this system of equations.

■ Manner

carefully • steadily • thoroughly • accurately • rigorously • properly • systematically

- The proof must be **rigorously** formulated to be accepted as valid in formal mathematics.

■ Negation

hardly (any) • scarcely (any) • barely (any)

- After simplification, **hardly any** terms remain in the expression.

■ Degree

a little • a bit • slightly

relatively • quite* • rather • fairly

very • extremely • exceedingly

- The computed value is **slightly** greater than the theoretical prediction, suggesting a rounding error.

■ Approximation

nearly • almost • roughly • more or less • virtually • to a certain extent • partly

≠

Completeness

totally • quite* • entirely • utterly • thoroughly

- The two curves are **virtually** identical over the interval $[-2, 2]$, making them interchangeable for practical purposes.

* The word **quite** has two opposite meanings in English depending on whether the following adjective is **gradable** (quite good = fairly good) or **ungradable** (quite impossible = completely impossible).
See <https://dictionary.cambridge.org/grammar/british-grammar/quite>

VI – PRACTISE

1. Describe an object. Look at the following example:



A mouse **is** a device **which** basically consists of a laser or LED mechanism in a plastic case **and/that** is used to control a cursor with one hand.

2. Construct sentences with the same pattern, selecting phrases from each box.

- A. A volcanologist
- B. A rock scientist
- C. A beaker
- D. Goggles
- E. A microscope
- F. A bar code reader



**IS /
ARE**

- _____ a geologist
- _____ a tool for holding liquid
- _____ a tool for magnifying small objects
- _____ a hand-held or stationary input device
- _____ a volcano scientist
- _____ a type of safety glasses



WHICH/THAT/WHOSE/WHO

is (mainly used) to/for

is located in

typically works for/in



**AND
THAT/WHO/
WHOSE AIM IS
TO**

_____ is used to capture and read information contained on little vertical lines

_____ job could be for instance to find iron in the Earth's bedrock

_____ determines the likelihood of an imminent eruption and studies the past eruptions of a volcano

_____ are designed to protect the eyes from spills

_____ is typically made of glass

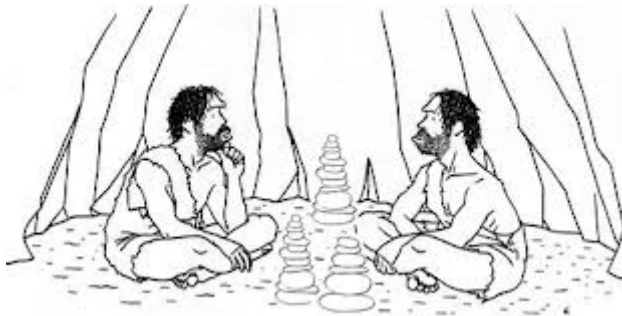
_____ consists of several lenses

3. Ever since the beginnings of mathematics, we have tried to look for devices to help us calculate. Describe these devices in pairs by using the unit vocabulary:

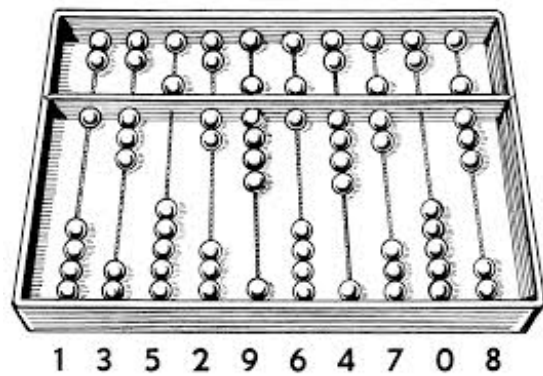
- What sort of device is it?
- What are the basic elements of this device?
- Where is it found/used?
- What is it used for? What is its purpose?
- How does it work?
- What does it look like (colour, shape)?

Look up information on the internet if you are not sure what each device is and/or consists of.

Pebbles and Sticks

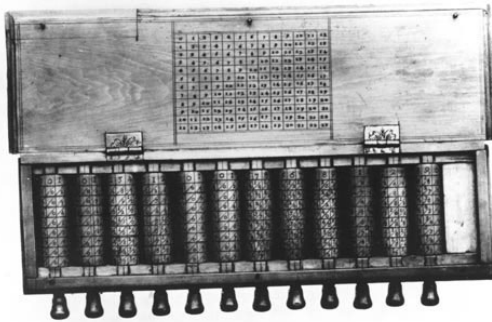


Abacus



Napier's Bones / Rods

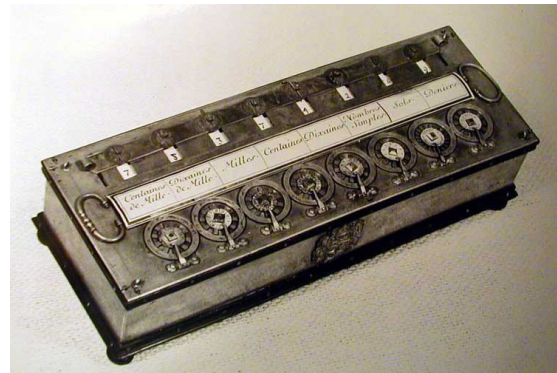
2A Napier's Rods



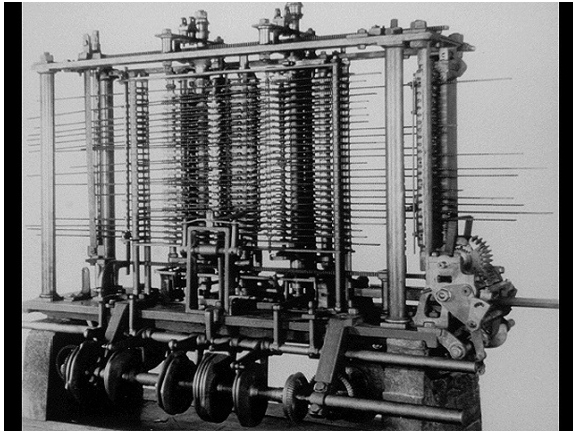
© Napier's Bones - cylindrical form

2B Napier's Rods

Pascal's Arithmetic Machine



Babbage's Differential Engine



The Arithmometer



Mathematics helped design computers and now computers help solve mathematics, and even open new fields for it: *a never-ending loop*.

A. PREPARATION

Match the words with the definitions.

spiral – to perceive – prevalent – restricted – renowned – a corkscrew

a device for removing corks from bottles	
a plane curve winding around a central point	
limited	
famous	
to notice or realise	
very common in a particular place	

B. LISTENING COMPREHENSION

➤ 1st listening

Take notes of the gist.

➤ **2nd & 3rd listening**

Are the sentences true or false? Justify your answers orally.

- | | | |
|--|------|-------|
| 1. The speaker starts the talk with two examples of helices found in nature. | True | False |
| 2. A helix is a three-dimensional object. | True | False |
| 3. The main advantage of a spiral staircase is its strength. | True | False |
| 4. DNA contains the genetic information for most, but not all, living things. | True | False |
| 5. A mirror image of a helix can appear to change direction. | True | False |
| 6. The screw is an example of a conical helix. | True | False |
| 7. Helices are much more common in manufactured objects than in natural objects. | True | False |
| 8. It is a mystery why there are so many helices in the world. | True | False |

C. VOCABULARY

Write the adjective form of the word in brackets in the gap.

- To understand this, we will use a complicated formula.
(mathematics)
- Doctors are now confident that the condition is caused by a defect. (genes)
- When bamboo is cut, you can see it is a straight pole with a hole down the middle. (cylinder)
- Two-..... space can be viewed as a projection of the physical universe onto a plane. (dimension)
- The pattern is made up of colours: white and black.
(alternate)
- The iris of the eye is composed of muscles which you can see as the brown, blue or green colour of the eye. (circle)

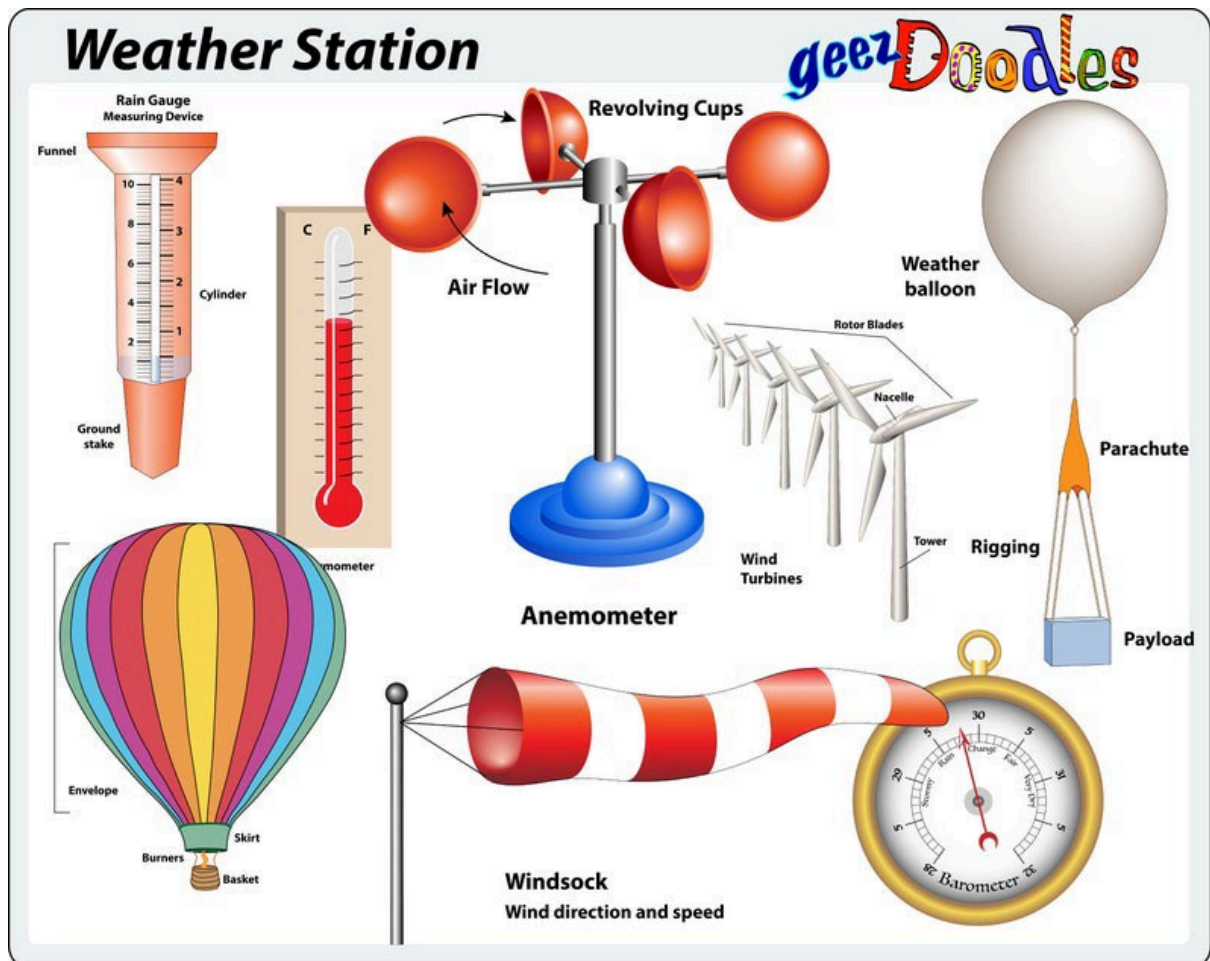
D. DISCUSSION

Do you ever watch scientific lectures or talks like this online? If yes, on which websites?

VIII – PRACTISE^x

Make comments about the use of 2 items (from pics below) using phrases such as:

*aims to - purpose - goal - designed for - provide - in order to
via - thereby - enable - by means of - make it possible*



1. Rain gauge	
2. Thermometer	
3. Anemometer	
4. Wind turbine	
5. Weather balloon	
6. Hot air balloon	
7. Windsock	
8. Barometer	

A. BRAINSTORMING

1. Give examples of both major and minor accidents.
2. Were they brought about by faulty technology or human error?
3. Was any action taken as a result of the accident?
4. Give examples of how technology can be used to increase safety.

B. LISTENING

1. Human error accounts for 18% of air accidents. True/False? Justify.
2. How often do pilots generally fall asleep while piloting?
3. What is pilot fatigue due to?
4. What does ENRI stand for?
5. What are the biological functions that ENRI can process?
6. Why is this system particularly efficient?

C. PROBLEM/SOLUTION IDENTIFICATION

In the transcript below, find the excerpts which examine the PROBLEM and the excerpts which indicate SOLUTIONS (why was it created – purpose & how is it achieved – process).

Despite the thousands of parts which are used in the construction of aeroplanes, very few air-crashes are caused by component failure. It is, in fact, human error that is responsible for 80% of air accidents. This explains why one of the primary objectives of the aircraft industry today is to devise/design ways of improving these figures. One area of particular concern is pilot fatigue. This problem is far more widespread than is commonly realised and Dr Mark Rosekind claims that "70% or more pilots say that they have fallen asleep at least once while piloting". Fatigue is particularly linked to the crossing of time zones which can severely disturb the circadian rhythm.

The Japanese Electronic Navigation Research Institute (ENRI) is currently in the process of developing a monitoring system which will be able to provide early warning of fatigue by means of voice recognition techniques.

Biological functions, such as blood pressure, heartbeat and voice, have a fractal structure which can be identified. Under stress or fatigue, these physiological manifestations undergo change and consequently, as can be seen in the figure, the pattern alters, gradually losing its structure and becoming chaotic. By using the mathematics of chaos theory, Kakuichi Shiomi, chief researcher of ENRI, has been able to design/devise software which enables these minute distinctions to be identified. The pilot's voice pattern can be matched with a control model of an alert voice. It takes no more than 10 seconds for data samples to be processed which means that fatigue can be measured in real time without interfering with the normal activity of the flight crew.

According to Shiomi, the system makes it possible to detect tiredness in test subjects 10 to 20 minutes before they become aware of it themselves.

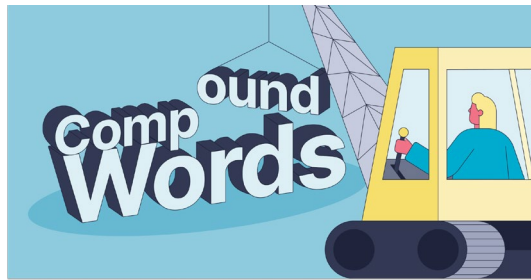
X – HOMEWORK FOR NEXT CLASS

1. - U2: Compound forms & Discipline-specific Terminology: Preparatory Tasks (Moodle)*

- Compound Forms: [Video](#)

* This grammar/vocabulary chapter is taught in Term 1, revised in Term 2 and only evaluated in Term 2, due to its great complexity for many students.

UNIT 2: COMPOUND FORMS & DISCIPLINE-SPECIFIC TERMINOLOGY



I – STARTER

1. What is a compound form? Give an example.
2. What are they used for?
3. Do you know other languages that use the same process to create new words?
4. Identify compounds in the texts below.^{xii}

Long-range weather report

A dry weekend for many areas with some sunshine in the southeast of England.

By midweek, most places will see a fair amount of rainfall, with some heavy downpours in Scotland, particularly the Highlands.

Northern Ireland and western parts of the mainland are also likely to see strong winds, with speeds up to 70 mph forecast which could cause uprooted trees and some structural damage.

However, by Thursday the wind should die down but it will remain overcast and wet at times.

The overall picture is that the weather will remain changeable, with temperatures feeling agreeably warm during the sunnier spells. This could lead to occasional thunderstorms.

Please remember to put your clocks back one hour when British Summer Time finishes at 2 a.m. on Sunday morning.



The AI breakthrough in mathematics

A brand-new substance — a groundbreaking proof assistant — which could transform the workload of mathematicians worldwide, has been developed using artificial intelligence and supercomputing.

The findings were made by a DeepMind research team working alongside Terence Tao, a Fields Medal winner and long-standing advocate of computer assisted problem solving.

Mathematicians say the large language model could potentially reduce proof verification time by up to 80% in complex multi-step calculations.

Since its development, the new software tool has been used to verify a longstanding open conjecture in number theory.

DeepMind researchers used their neural network platform to narrow down 28 million potential proof pathways to 14 promising candidates in under a week — a theorem checking process that could have taken more than three decades using traditional blackboard and notebook research methods.

Automated numerical-sampling and curve-fitting workstation

This purpose-built workstation automates your time-consuming, routine data-acquisition and curve-fitting tasks. The workstation accepts a variety of off-the-shelf input devices and sensor arrays. To operate, simply connect the workstation to your data-collection hardware and the numerical dataset to be processed, input the target error bound and convergence tolerance, and start the unit. Note: variations in sampling rate and signal-to-noise ratio can affect output accuracy.

The workstation retrieves a raw data series from the input buffer, determines the baseline offset, calculates the least-squares best-fit curve, adjusts the model parameters to minimise the residual error, and stores the fitted output to the results folder. This step-by-step sequence continues until all datasets have been processed, freeing you for other tasks.

All communications between the host computer, the data-acquisition board, the signal-processing module and the output display are sent via a built-in high-speed USB interface. Statistical information such as the baseline offset, root-mean-square error and calculated goodness-of-fit coefficient for each dataset are stored in a structured data file for further analysis or printing.

The workstation comes complete with a multi-channel data-acquisition board, a real-time signal-processing module, a high-resolution analogue-to-digital converter and a Windows-compatible laptop computer, your choice of sampling rate, a noise-reduction filter, three input-port adapters to accept a variety of sensor signals, and a 2-m cord with a standard IEC* plug.

* International Electrotechnical Commission

5. Complex compound forms

Open any research paper and you will be struck by how much information is packed into a few words. A number theorist studies **prime number distribution**; a statistician runs a **least-squares regression analysis**; a topologist worries about **fixed-point theorems** and **boundary value problems**. A single **error correction algorithm** can rest on a **finite field construction**, a **round-off error estimate**, and a **worst-case complexity bound**. These are all expressed without a single connecting word.

This is the power, and the difficulty, of the **compound noun**. English builds meaning by stacking nouns in front of one another, so that **graph colouring problem** says in three words what other languages might unfold into a whole clause. Master this structure and you will read mathematics faster and write it with the concision your readers expect.

a. Study the following word combinations and try to translate them into French:

- *error propagation analysis*
- *machine-learned classification model*

- Which word is the main one in each example?
- Which element(s) define (modify) this word?

b. See how the meaning of the following two noun phrases can be changed!

- *gas turbine* = turbine which is driven by gas
- *turbine gas* = gas which drives the turbine

c. Look at these **compounds** and at how long they are. Think of the strategies required to make these descriptions simpler. Then, reformulate them orally so that the information is clearly understood.



Florida man says he has no immediate plans on what he will be doing with his *newly acquired M61 Vulcan: a 250-pound, 6-foot-long pneumatically driven, six-barrel, air-cooled, electric rotary cannon which fires six thousand 20 mm rounds per minute.*

Vintage cobalt-blue glass orange juice reamer juicer (as described on eBay, i.e., even trivial objects described in a non-technical context can be defined by using tricky, complex compound forms)



A. Main compound forms

Nouns can be qualified in several different ways:

1. Adjectives	a linear / a general / a partial / a complex ...	equation
2. Possessive forms	Pythagoras's / Euler's ...	theorem
3. “-ing” participles	a converging / a smoothly increasing ...	sequence
4. Past participles	a well-defined / a rigorously proven ...	conjecture

However, it is very important to understand that nouns can also be qualified by other **nouns**, i.e. the qualifying nouns function as if they were adjectives. These constructions are called **compound nouns**.

5. Noun modifiers	a number / graph / probability ...	theory
-------------------	------------------------------------	--------

- **Compound nouns** are used to refer to specific, identifiable objects or concepts. They are **extremely common in scientific and mathematical English** because they make it possible for complex notions to be expressed in a condensed, elegant way.

Notice the efficiency and concision of: ► *prime number factorisation*

Compared to: ► **factorisation** *whose object is to decompose a number into its prime factors*

- Compound nouns are frequently difficult to understand. Why? It is because the order is the **reverse** of the one found in many languages. This is particularly clear in acronyms.

	English		French
Greatest Common Divisor	GCD	→	PGCD
Ordinary Differential Equation	ODE	→	EDO

B. Compound adjectives

There are three different forms:

■ The “-ing” form

The present participle of the verb can be used as a modifier.

- ▶ a **converging** series (*the series converges to a finite limit*) • a **decreasing** function • a **diverging** sequence • a **distinguishing** property ...

This form refers to a typical, defining characteristic.

■ The past participle may also be used.

- ▶ a **well-defined** set (*the set was defined rigorously by someone*) • a **well-known** theorem • a **pre-computed** value • **widely used** methods ...

This form usually has a **passive** meaning (i.e. you can add “by”) and refers to something already done, something in the past.

■ A small number of compound adjectives are formed by adding “-ed” to a noun

(the nouns act as if they were verbs).

- ▶ a two-**sided** polygon • a twin-**engined** plane • a multi-**valued** function • a closed-**formed** expression ...

C. Naming Theorems and Results

Mathematical results are usually named after the people who discovered them, and English offers **three** ways of attaching the name: with the **definite article** (*the Riemann hypothesis*), with the **possessive** (*Euler's theorem*), or with **no article at all** (*number theory*). The choice is systematic, and it is easy to get wrong — especially for a French speaker, because French reaches for the definite article far more readily than English does. This section sets the three patterns side by side.

1. *the* + Name + noun

The definite article with the bare surname is the default for conjectures, hypotheses and many named functions and objects. The name behaves like an adjective and stays uninflected.

- ▶ *the proof assumes **the Riemann hypothesis***
- ▶ *a counterexample to **the Poincaré conjecture***
- ▶ *expressed using **the Dirac delta function***

2. Name's + noun (possessive)

The possessive is fully idiomatic for theorems, lemmas, inequalities, algorithms and methods named after a single person. For many results it is interchangeable with pattern 1.

- ▶ *by **Fermat's little theorem**, $a^p - a$ is divisible by p*
- ▶ *we apply **Euclid's algorithm** to the pair*
- ▶ *it follows from **Harnack's inequality***

Interchangeable pairs are common: *Minkowski's theorem* = *the Minkowski theorem*; *Harnack's inequality* = *the Harnack inequality*; *Dirac's delta function* = *the Dirac delta function*. A few strongly prefer one slot — *the Riemann hypothesis* and *the Poincaré conjecture* are heard far more often than the possessive.

3. Zero article (no article at all)

Names of **fields** and **theories** take no article, and neither do labelled cross-references such as numbered chapters and equations. This is exactly where French interference bites hardest, because French keeps the article in every one of these cases.

- ▶ *she works in **number theory***
- ▶ *a standard result of **measure theory***
- ▶ *as shown in **Chapter one** / **Equation (7)***

4. Summary

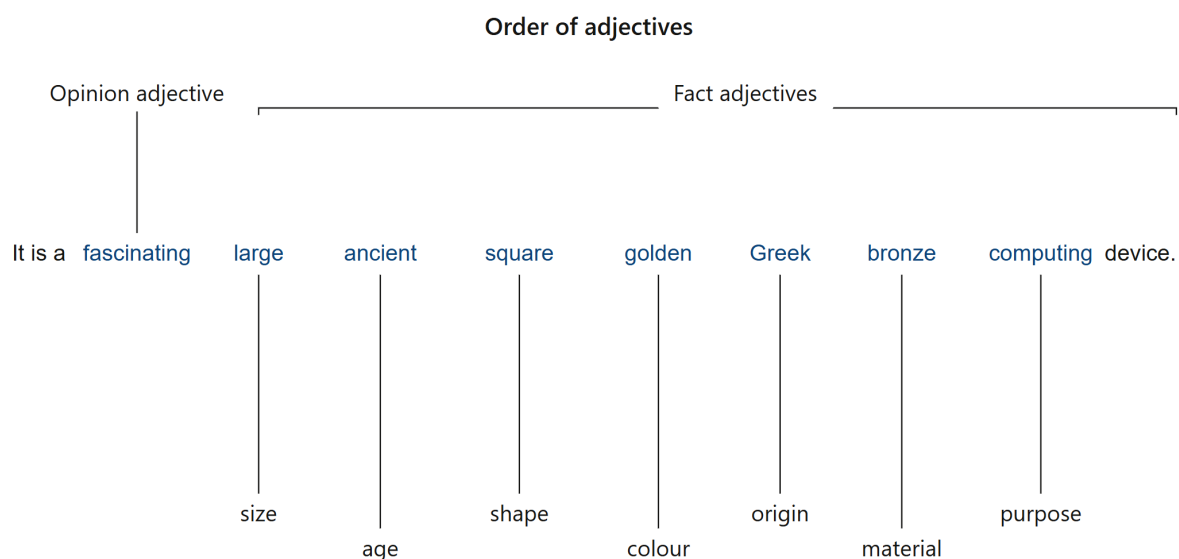
- Conjectures: *the*
- One-person results: *the* X = X's X
- Fields & labels: no article

D. Order of adjectives

When several adjectives are used together, they follow a fixed order*, the **OSASHCOMP** rule:

Opinion → **S**ize → **A**ge → **S**hape → **C**olour → **O**origin → **M**aterial → **P**urpose

Example: ► The Antikythera mechanism is a **fascinating large ancient square golden Greek bronze computing device**.



* Cambridge and several other ELT references put **shape** before **age**.

Exercise: Order of adjectives — reorder the scrambled adjectives before each noun.

Scrambled adjectives	Correct order
1. (hexagonal •metallic •vast •Russian •antique) lattice	
2. (German •cylindrical •new •steel •small) container	
3. (triangular •wooden •ancient •Italian •thin) ruler	
4. (elegant •circular •golden •tiny •Swiss) mechanism	
5. (rectangular •black •modern •plastic •flat) screen	

E. Compounds built on verbs and prepositions

Besides the noun- and adjective-based modifiers seen above, English forms compounds from **verbs** and **prepositions**. These are much rarer in scientific writing, but a handful are genuinely useful to mathematicians — especially in numerical analysis and computing.

verb + preposition	a round-off error • a trade-off • the cut-off value ...
preposition + verb	output • input • throughput ...
preposition + noun	overflow • underflow • the upper bound ...
noun + prep. phrase	a rule-of-thumb estimate • a state-of-the-art algorithm ...
adjective + verb (-ing)	dry-running a simulation • deep-learning methods ...

Notice that most of these are **fixed** expressions: you learn them as whole units rather than building them freely. Their spelling is often hyphenated, though usage varies (e.g. *round-off* / *roundoff*, *trade-off* / *tradeoff*).

WORTH KNOWING

Several of these compounds form a **verb** → **noun** pair: the same words give both a verb and a compound noun. The meaning is the same — only the grammar, spelling and **stress** change. The verb is two words, stressed on the particle; the noun is hyphenated, stressed on the first element:

to round **OFF** (verb, two words) → a **ROUND-off** (noun, hyphenated)

to trade **OFF** (verb, two words) → a **TRADE-off** (noun, hyphenated)

By contrast, patterns such as **noun + prepositional phrase** (*son-in-law*) or **noun + preposition** used as a person (*hanger-on*) belong to everyday English and almost never appear in mathematical texts — recognise them, but you will rarely need to produce them.

F. Compound formation conventions

In brief:

SPELLING

Compound spelling is a convention, **not a strict law**. Compounds can be:

- written as 1 word (**closed compound**) ► *eigenvalue*
- written as 2 words (**open compound**) ► *prime number*
- linked with a hyphen (**hyphenated compound**) ► *well-defined*



Here is a full list of [compounding rules](#) and [examples](#). Consult when in doubt.

Hyphenation:

There is considerable variation in the use of hyphens. The hyphen is used in British English more frequently than in American English. It is better to hyphenate:

■ Compound adjectives where the first element is:

- an **adjective**: ► a *short-term* solution • *full-time* research
- a **number**: ► a *3-dimensional* space • a *50-year-old* conjecture
- a **letter**: ► an *n-tuple* • an *x-axis*

■ Compound nouns where one of the elements is:

- a **past participle**: ► a *well-defined* set • a *rigorously proven* result (**exception**: rule [here](#))
- a **noun + “-ed”**: ► a *two-sided* polygon • a *multi-valued* function
- an **“-ing” form**: ► *problem-solving* strategies
- a **grammatical word** (non-, anti-, self-, etc.): ► *non-Euclidean* geometry • *self-adjoint* operator

Exercise: Can you guess? Keep it open, close it, or hyphenate it?

Each compound below is written as separate words. Decide whether it should stay as it is, be joined into one word, or be linked with a hyphen. More than one spelling can be accepted for some of them.

1. saddle point
2. cross hair
3. lowest common denominator
4. half life
5. truth table
6. eigen vector
7. break even
8. scatter plot
9. feed back
10. one to one
11. growth rate
12. non zero

THE “-S” TRAP: COUNTING WORDS STAY INVARIABLE

When a noun is used as a modifier (i.e. it **functions like an adjective**), it becomes **invariable** — it stays singular in form, whatever its meaning.

1. Without numerals — the modifying noun is always singular

Even when the modifier refers to many things, it does not take an “-s”:

- ▶ a **root** finder (a tool that finds the roots of an equation)
- ▶ an **error** term (a term accounting for several errors)
- ▶ a **vector** space (a space made of vectors)

2. With numerals — the invariability still holds

The same rule applies **even after a numeral**. In practice this works in two ways:

2a) The plural “-s” is dropped

- ▶ a four-**digit** number (not four-digits)
- ▶ a three-**step** proof (not three-steps)
- ▶ a five-**variable** system (not five-variables)

2b) Irregular plurals revert to the singular

- ▶ a two-**foot** ruler (not two-feet)
- ▶ a three-**man** research team (not three-men)
- ▶ a six-**woman** committee (not six-women)

Exercise: Compound nouns — mind the plural!

Rewrite each description as a compound noun or compound adjective. Watch the plural endings carefully — the modifying noun stays invariable.

Description	Compound
1. A model that depends on many parameters .	
2. A solid that has twenty faces .	
3. A survey covering two thousand people .	
4. A function that handles errors .	
5. A matrix with three rows .	
6. A scatter plot with two axes .	
7. A graph with six vertices .	
8. A sequence defined by twelve terms .	
9. An experiment which requires two dice .	
10. A puzzle relay solved in four rounds .	

G. Summary of all compound form combinations

Formation		Compound	Example
noun + noun	Open compound	bus stop	Is this the bus stop for the number 12 bus?
	Hyphenated compound	fire-fly	In the tropics you can see fire-flies at night.
	Closed compound	football	Shall we play football today?
adjective + noun		full moon	I always feel crazy at full moon .
		blackboard	Clean the blackboard please.
		software	I can't install this software on my PC.
verb (-ing) + noun		breakfast	We always eat breakfast at 8 a.m.
		washing machine	Put the clothes in the red washing machine .
		swimming pool	What a beautiful swimming pool !
noun + verb (-ing)		sunrise	I like to get up at sunrise .
		haircut	You need a haircut .
		train-spotting	His hobby is train-spotting .
verb + preposition		check-out	Please remember that check-out is at 12 noon.
preposition + verb		output	Last year British manufacturing output fell by 14%.
noun + prepositional phrase		mother-in-law	My mother-in-law lives with us.
preposition + noun		underworld	Do you think the police accept money from the underworld ?
noun + preposition		hanger-on	Wherever there is royalty, there are always hangers-on .
noun + adjective		trunkful	We need 10 trunkfuls of bricks.
adjective + verb (-ing)		dry-cleaning	He owns a dry-cleaning business.

Exercise: Identify how each compound is formed. Tick the matching column.

	noun + noun	adjective + noun	verb (-ing) + noun	noun + verb (-ing)	verb + preposition	noun + prepositional phrase	preposition + noun	noun + adjective	noun + preposition	adjective + verb	preposition + verb
1. Eating habit											
2. Train station											
3. Bluebird											
4. Dry-cleaning											
5. Underbelly											
6. Whiteboard											
7. Overheads											
8. Split end											
9. Son-in-law											
10. Railway											
11. Hanger-on											
12. Spoilsport											
13. Hardware											
14. Check-in											
15. Sunset											
16. Washing powder											
17. Underpants											
18. Fire-eater											
19. Aftershave											
20. Sunshine											
21. Spoonful											
22. Brother-in-law											
23. Sunshade											

H. Paraphrasing complex compound forms

Compound nouns can often be **paraphrased** by reversing the order and inserting a preposition. This reveals the hidden relationship between the two nouns:

	Noun-noun combination	Paraphrase
of-phrase	vector length	length of the vector
	number theory	theory of numbers
prepositions	step response	response to a step input
	rounding function	function for rounding sth.
	error tolerance	tolerance for error
	phase transition	Transition between two phases
	boundary value problem	problem about values at the boundary
	noise filter	filter against noise
	data points	points in a dataset
	search space	space in which sth. is searched for

You can of course also paraphrase by creating full definitions (see “Defining & Describing”) e.g. *They have introduced new “**aviation fuel tests**”.*

That is to say, **tests** which are carried out on **fuel** that is used in **planes**.

I. Discipline-specific terminology

Beyond the descriptive compounds that can be paraphrased for clarity, many **established compound forms** exist in mathematical language that cannot — and must not — be reworded, even for the sake of comprehension (e.g. “a fixed point”, “the central limit theorem”, “a partial differential equation”).

These terms function as a single, unalterable concept within a specific context, which is why you must **never paraphrase established discipline-specific terminology**. If you notice that certain words are always used in a fixed way in mathematical publications and oral presentations, there is a high chance that these terms are widely used and accepted by the mathematical community. Leave them as such.

Using precise terminology strengthens your credibility and conveys professionalism. Moreover, defining terminology gives the reader deeper insight into critical concepts, helps them distinguish between related and unrelated concepts, and grounds the study in established knowledge.

III – PRACTISE

1. Expand the 8 compound forms below by paraphrasing (giving full definitions).^{xiv}

- a. The optimization algorithm converged to a saddle point solution.
- b. The structure was supported by a 50-tonne steel girder.
- c. The satellite completes a 90-minute orbital period.
- d. Space shuttles use heat-resistant tiles during atmospheric re-entry.
- e. They use data-compression algorithms to reduce storage requirements.
- f. The integral is evaluated over a closed interval domain.
- g. The telescope mirror consists of a borosilicate glass disc.
- h. They are developing new noise-filtering techniques for signal analysis.

2. Put the 8 compound forms in the correct column in the box.

<i>purpose (what is it for?)</i>	<i>is made of / contains</i>	<i>location / time</i>

3. Create compound forms with hyphens.

- a. A week that lasts five days:
- b. An insect that has four wings:
- c. A rate that increases fast:
- d. A text that is written well:

4. Reformulate the following phrases to create compound forms.

- a. A tube made out of Plexiglas which is four inches in diameter.
- b. A trend that is nationwide and is growing.
- c. A programme for training that lasts six months.
- d. A manufacturer of equipment which produces rays that are in the category X.
- e. A programme for the evaluation of cancer on a scale that is large.
- f. A system for classification which is used very widely.
- g. Staff whose job it is to care for the health of prisoners.

5. Rephrase these four sentences to create more compound forms.

*You need to create CFs with words that are already (basic) compound forms themselves!
However, this does not change the construction process.*

- a. A programme designed to provide emotional assistance to housewives
- b. Emissions of greenhouse gas which are provoked by seafood production
- c. A dispenser of washing-up liquid that is nearly broken
- d. A company that produces machines to wash clothes

IV – PRACTISE AUDIO: LIFE SCIENCES LABORATORY EQUIPMENT (LSLE)

I. Listen to the following recording and answer the questions... if you can!

1. Give a detailed description of the AFS-2.
2. What was it designed for?
3. How heavy is it?
4. How is it powered? Provide a thorough answer.
5. Sensors (fill in the missing information):
 - a. Blood volume Pulse (_____)
 - b. Skin Temperature (_____)
 - c. _____ (0.5 to 50 μmhos^* $\pm$ 2 %)
 - d. Respiration (_____)
 - e. Electrocardiography (_____)
 - f. Acceleration (_____)

* 1 mho ($\mathcal{U}$) = 1 siemens (S) = reciprocal of one ohm (Ω^{-1}). See [Siemens](#).

II. Why was it hard if not impossible to understand the information?

III. What should be done to make the information easy to understand? Discuss in small groups then think about how to present the information.

IV. Create a short speech based on the answer key (on Moodle) and present it to the class.

V – DESCRIBING FIELD-SPECIFIC TERMINOLOGY (PROJECT)

1. Identify 4 compound forms found in your book chapters and define them.

a. Compound form:

Definition:

b. Compound form:

Definition:

c. Compound form:

Definition:

d. Compound form:

Definition:

VI – IDENTIFYING KEY BACKGROUND INFORMATION (PROJECT)

Now that most concepts in your documents should be understood, it is time for you to decide what you intend to include in your introduction in terms of **background information**.

Background information consists of the (basic) **knowledge required to understand your topic**. In this course, you will mainly find this information in the two book chapters you selected, but you are also allowed to complement with **additional resources** of your choice.

Keep in mind that this information is **limited to 50 % of your presentation**. In real life, most people do not attend conferences to receive background information, but novel research grounded in theory, which is why you need to leave the other 50% to your scientific paper, popular science article (and potentially your audiovisual resource).

First, list what it is necessary to include in your background and provide reasons for your choice.

Background information	Reason for including it in your presentation

Then, walk around the room and ask if your fellow students agree with your choices.

1. False friends

False friends are deceptive because they look familiar but mean something different in English than in your native language. In mathematics and physics, this can lead to serious misunderstandings.

Examples (ENG-FR):

- **Actual vs. Actuel:** In English, *actual value* means the real value, not the current one.
- **Eventual vs. Éventuel:** *Eventual result* means the final outcome, not something "possible."
- **Formula vs. Formule:** In English, *formula* refers to a mathematical expression, while in French *formule* can also mean a phrase or wording.

Why it matters:

Misinterpreting these terms can distort the meaning of proofs, definitions, or experimental results.

What you should do:

In doubt, always look up in the dictionary. There are also lists that can be studied (although it is challenging to find an exhaustive one, the [Wiktionary list](#) is already a good start).

2. Types of Acronyms & Abbreviations in mathematics and physics

A. Mathematical Methods

- **FFT** → *Fast Fourier Transform*
- **PDE** → *Partial Differential Equation*
- **ODE** → *Ordinary Differential Equation*
- **LHS / RHS** → *Left-Hand Side / Right-Hand Side*
- **SVD** → *Singular Value Decomposition*
- **MLE** → *Maximum Likelihood Estimation*
- **CDF / PDF** → *Cumulative Distribution Function / Probability Density Function*

B. Physical Quantities & Constants

- **c** → *Speed of light*
- **h** → *Planck's constant*
- **G** → *Gravitational constant*
- **μ** → *Magnetic permeability or mean (context matters!)*
- **σ** → *Standard deviation or cross-section*
- **ΔE** → *Change in energy*

C. Theories & Models

- **QM** → *Quantum Mechanics*
- **GR** → *General Relativity*
- **SM** → *Standard Model*
- **QFT** → *Quantum Field Theory*
- **LHC** → *Large Hadron Collider*

D. Organisations & Journals

- **BIPM** → *International Bureau of Weights and Measures* ('Bureau International des Poids et Mesures')
- **AMS** → *American Mathematical Society*
- **APS** → *American Physical Society*

Why They Cause Problems

- Context sensitivity: PDF in physics often means "Probability Density Function", but in computing it is a file format.
- Multiple meanings: μ can mean "micro" (prefix), "mean," or "magnetic permeability".
- Implicit knowledge: Authors rarely define common acronyms in advanced papers.

E. Latin abbreviations

Also, scientific papers often use Latin abbreviations without explanation. Understanding them is crucial for reading proofs and theoretical arguments.

Examples:

- **i.e. (id est)**: "that is" → clarifies a statement.
Example: "The set is finite (i.e., it contains a limited number of elements)."
- **e.g. (exempli gratia)**: "for example."
Example: "Consider a prime number (e.g., 2 or 3)."
- **cf. (confer)**: "compare."
Example: "cf. Theorem 2 for a similar result."
- **et al. (et alii)**: "and others."
Example: "As shown by Smith et al."

3. Pronouncing terminology

Pronunciation matters for oral presentations and collaboration. Many terms come from Greek or Latin, and names of mathematicians and physicists often defy spelling-based expectations.

Examples:

- **Euler** → pronounced "Oiler," not "Yooler."
- **Fourier** → "Foo-ree-ay" (French origin).
- **Lagrangian** → stress on second syllable: "La-GRAN-jian."

Why it matters:

Mispronouncing key terms can cause confusion in seminars or when discussing proofs.

Exercise:

Say these words aloud and check their correct pronunciation:

Mathematicians & Physicists	Terminology
• Euler → <i>Oiler</i> • Fourier → <i>Foo-ree-ay</i> • Gauss → <i>Gowss</i> (rhymes with "house") • Poisson → <i>Pwah-son</i> (French origin) • Lagrange / Lagrangian → <i>La-GRANJ / La-GRAN-jian</i> • Leibniz → <i>Libe-nits</i> • Riemann → <i>Ree-mahn</i> • Hilbert → <i>Hil-bert</i> (stress on first syllable) • Noether → <i>Ner-ter</i> (German origin) • Dirichlet → <i>Dee-ree-klay</i> (often mispronounced as "Dirich-let") • Hermite → <i>Er-meet</i> • de Broglie → pronounced as if it were spelled "de Breuil" • Poincaré → <i>Pwan-ka-ray</i> • Laplace → <i>La-plahs</i> • Chebyshev → <i>Cheb-ee-shev</i> • Kolmogorov → <i>Kol-mo-go-rov</i> • Feynman → <i>Fine-man</i> • Heisenberg → <i>Hi-zen-berg</i> • Schrödinger → <i>Shrer-ding-er</i> (with German "ö")	• Asymptote → <i>ASS-im-toht</i> (not "a-sym-pote") • Epitome → <i>ih-PIH-tuh-mee</i> • Hyperbola → <i>hy-PER-bo-la</i> (not "hyper-bowl-a") • Homogeneous → <i>ho-mo-JEE-nee-us</i> • Eigenvalue / Eigenvector → <i>Eye-gen</i> (from German "eigen") • Quotient → <i>KWO-shunt</i> • Integral → <i>IN-ti-gruhl</i> (stress on first syllable) • Tensor → <i>TEN-sor</i> • Manifold → <i>MAN-i-fold</i> • Topology → <i>to-PO-lo-gee</i> • Schwarzschild radius → <i>Shvarts-schild</i> • Gyroscope → <i>JAI-ro-scope</i> (not "guy-ro") • Momentum → <i>mo-MEN-tum</i> (stress on second syllable)

Tip: Use a pronunciation tool (Howjsay, Oxford, etc.) to check.

VIII – KEY DISCIPLINE-SPECIFIC TERMINOLOGY

You can find a list of key discipline-specific terminology for mathematics on Moodle. Study it carefully.

IX – HOMEWORK FOR NEXT CLASS

1. GRAMMAR: Revise Past & Perfect Tenses

- Grammar book, Section 4 (Past tenses), pp. 39–46.
- Grammar book, Section 5 (Perfect tenses), pp. 47–53.
- Coursebook
- Anglalink [Video 1](#)
- Anglalink [Video 2](#)
- Anglalink [Video 3](#)

2. Pronunciation of -ed

- Coursebook
- Anglalink [video](#)
- Woodward [video](#)

3. PRACTISE FOR YOUR PROJECT: Presenting a concept/theory.

- Explain your scientific paper briefly.
- Identify a central concept in your **scientific paper**.
- To do so, revise and use vocabulary from “Defining, Describing & Qualifying” and from “Compound Forms & Discipline-specific Terminology”.

Try to answer to following questions:

- What sort of concept/theory is it?
 - What are the basic elements of this concept/theory?
 - Where is it used/ Where was it established?
 - If relevant: does this concept/theory involve anything which can be described in terms of colour, shape, etc.?
 - What is it used for?
 - How was it designed?
 - How is it used?
 - Was it designed specifically for certain tasks? Is it common in Maths?
- Create a description of the concept based on the questions above. Make your descriptions formal by using impersonal forms and a few compound forms. Paraphrase adequately.
 - Explain your concept to 3-4 students orally. Paraphrase and **stick to simple sentences and vocabulary**.
 - Listeners: take notes of what you understand

Central concept:

Description:

<u>Student 1:</u>	<u>Student 2:</u>
<u>Student 3:</u>	<u>Student 4:</u>

UNIT 3: PAST & PERFECT TENSES^{xv}

I – TENSES AT A GLANCE

Tense	Form	Mathematical example	Use
Past simple	<i>proved</i>	<i>Wiles proved the theorem in 1994.</i>	a finished action at a stated past time
Past continuous	<i>was proving</i>	<i>He was working when a referee found a gap.</i>	an action in progress / interrupted in the past
Present perfect	<i>has proved</i>	<i>No one has proved the Riemann Hypothesis.</i>	past with present relevance, with a completed result; no stated time
Present perfect continuous	<i>has been proving</i>	<i>They have been working on it for decades.</i>	duration of an ongoing activity
Past perfect	<i>had proved</i>	<i>By 1995 he had repaired the proof.</i>	the earlier of two past actions
Past perfect continuous	<i>had been proving</i>	<i>He had been working in secret for seven years.</i>	duration before a later past moment

Signal words often point to the right tense:

PAST SIMPLE

yesterday • last week • in 1994 • ago • then • when

PRESENT PERFECT

already • yet • just • ever / never • since • for • so far

In context:

- ▶ Euler **published** his solution in 1748. (finished + stated date → past simple)
- ▶ Mathematicians **have studied** prime numbers since antiquity. (continues up to now → present perfect)

N.B. – A note for French speakers

- English splits one French passé composé into two tenses. A finished action with a stated past time takes the past simple (Euler published it in 1748), not the present perfect.
- depuis = **for** + duration (for two hours) or **since** + starting point (since 2015) and usually needs the present perfect: I have been working for two hours.
- Pendant = **for** + defined, measurable duration (for two hours) + Past = **during** + specific, named time block (during the weekend).
- il est né → he **was** born (a past event, never “he is born”).

II – READING: THE 350-YEAR PROBLEM

Read the text. The target verb forms are in bold to help you with the analysis tasks that follow.

In 1637, the French jurist and amateur mathematician Pierre de Fermat **was reading** a copy of Diophantus's *Arithmetica*. In the margin he **wrote** that he **had discovered** a remarkable proof that the equation $x^n + y^n = z^n$ has no positive integer solutions when n is greater than 2. Unfortunately, he **added**, the margin **was** too small to contain it. Fermat **died** in 1665, and no one ever **found** the proof he **had claimed** to possess.

For the next three and a half centuries, the statement — now known as Fermat's Last Theorem — **resisted** every attack. Euler **proved** the case $n = 3$, and by the nineteenth century others **had settled** several more cases, but a general proof **remained** out of reach. While number theorists **were struggling** with the problem, they **were** also **developing** powerful new tools, and much of modern number theory **grew** out of these failed attempts.

By the early 1990s the theorem **had become** the most famous open problem in mathematics. The English mathematician Andrew Wiles **had dreamed** of solving it since childhood. Working almost entirely in secret, he **had been developing** an approach based on elliptic curves and modular forms for seven years. In June 1993 he **announced** that he **had** finally **proved** it.

However, while the referees **were checking** the manuscript, one of them **found** a gap in the argument. For more than a year Wiles **struggled** to repair it. He **had** almost **given up** when, in September 1994, he and his former student Richard Taylor **found** a way to close it. The proof **was published** in 1995, and the problem that **had defeated** mathematicians for over 350 years **was** solved at last.

Since then, the techniques Wiles introduced **have transformed** number theory, and mathematicians **have been building** on them ever since. Fermat's own proof — if it ever existed — **has never been found**.

1. Comprehension. Answer in full sentences.

1. Why couldn't Fermat write his proof in the book?
2. What had other mathematicians achieved before Wiles, and what remained unsolved?
3. What happened after Wiles's first announcement in 1993?
4. How long had Wiles been interested in the theorem before he solved it?
5. What lasting effect has the proof had on mathematics?

2. Tense spotting. Find one example of each tense in the text and copy out the verb phrase.

1. past simple — a finished, dated action:	
2. past continuous — an action in progress in the past:	
3. past perfect — an action completed before another past action:	
4. past perfect continuous — a duration leading up to a past moment:	
5. present perfect — a past action with present relevance:	
6. present perfect continuous — an activity continuing up to now:	

3. Justify the choice. Explain why each form was used, not an alternative.

1. Why "Fermat *died* in 1665" and not "has died"?
2. Why "others *had settled* several cases" and not "settled"?
3. Why "mathematicians *have been building* ... ever since" and not "have built"?

III – GRAMMAR EXERCISES

1. Past simple or present perfect? Put the verb in brackets into the correct tense.

1. Euler _____ (publish) more than 800 papers during his lifetime.
2. Mathematicians _____ (not / find) a proof of the Riemann Hypothesis yet.
3. Gauss _____ (discover) the method of least squares when he was still a teenager.
4. Since 1900, several of Hilbert's problems _____ (be) solved.
5. In 1976, Appel and Haken _____ (prove) the Four Colour Theorem with a computer.
6. I _____ (already / check) the first three cases; the rest are still to do.
7. Cantor _____ (introduce) different sizes of infinity in the 1870s.
8. This is the most elegant argument I _____ (ever / read) .
9. We _____ (finish) the problem sheet last Thursday.
10. The conjecture _____ (remain) open for over a century, and it is still open today.

2. for or since? Complete each gap.

1. Mathematicians have studied prime numbers _____ antiquity.
2. The Riemann Hypothesis has been open _____ more than 160 years.
3. I have been working on this integral _____ two hours.
4. She has taught topology at Mons _____ 2015.
5. We have known about irrational numbers _____ the time of the ancient Greeks.
6. They have been collaborating _____ several years.
7. No one has found a counterexample _____ the conjecture was first stated.
8. The result has been known _____ Euler's day.

3. Past perfect — ordering two past events. Put one verb in the past simple and the other in the past perfect.

1. By the time Wiles _____ (announce) his result, he _____ (work) on it for seven years.
2. When I _____ (arrive) at the lecture, the professor _____ (already / write) the theorem on the board.
3. Perelman _____ (post) his proof online after he _____ (develop) Hamilton's programme further.
4. No one _____ (understand) Galois's ideas until decades after he _____ (die) .
5. By the time the result _____ (be) published, three groups _____ (already / prove) it independently.
6. She _____ (realise) that she _____ (make) a sign error on the previous page.

4. Past simple or past continuous?

1. While Newton _____ (develop) calculus, Leibniz _____ (work) on the same ideas in Germany.
2. Ramanujan _____ (send) his first letter to Hardy while he _____ (work) as a clerk in Madras.
3. The students _____ (discuss) the proof when the professor _____ (point out) a mistake.
4. I _____ (solve) the equation when I suddenly _____ (realise) my error.
5. What _____ (you / do) when the seminar _____ (end) ?

5. Present perfect continuous. Put the verb into the correct form.

1. Researchers _____ (try) to prove the twin prime conjecture for decades.
2. I _____ (work) on this problem set all afternoon — and I am still not finished.
3. She _____ (teach) here since she completed her PhD.
4. How long _____ (you / study) functional analysis?
5. They _____ (collaborate) on the paper, and they have nearly completed it.

6. Error correction. Each sentence contains (at least) one tense mistake. Write the correct version.

1. *I have finished the proof yesterday.*
2. *Euler has lived in the eighteenth century.*
3. *I am working on this problem since two hours.*
4. *We study group theory since last semester.*
5. *When I have arrived, the lecture already started.*
6. *Gauss is born in 1777.*
7. *I know my supervisor since we were undergraduates.*
8. *Mathematicians try to solve this problem for 150 years.*
9. *The professor explains the theorem since twenty minutes.*
10. *How long do you work on your thesis? — About three years now.*

7. Present perfect simple or present perfect continuous? Choose the better form.

1. So far this year, our seminar group _____ (publish) two papers.
2. She _____ (work) on the same proof all week and still hasn't finished.
3. I _____ (read) four chapters of the textbook so far.
4. How long _____ (you / use) this software?
5. My eyes are tired — I _____ (stare) at this diagram for an hour.
6. We _____ (just / find) a neat shortcut for the calculation.
7. He _____ (try) to generalise the result for months, but without success.
8. The proof is finally complete: I _____ (check) every step twice.

8. Recap. Put each verb in brackets into the correct tense. Every tense from this unit appears at least once.

Grigori Perelman and the Poincaré conjecture

Henri Poincaré (state) _____ ¹ his famous conjecture in 1904. By the 1980s, topologists (try) _____ ² to prove it for decades, without success. Then the American mathematician Richard Hamilton (introduce) _____ ³ a powerful technique known as Ricci flow.

While most mathematicians (work) _____ ⁴ on other approaches, the Russian mathematician Grigori Perelman (study) _____ ⁵ Hamilton's ideas in depth. By the time he posted his first paper online in 2002, he (develop) _____ ⁶ a complete strategy. He (announce) _____ ⁷ his proof not in a journal, but on the internet.

When specialists finally checked the argument, they confirmed that he (solve) _____ ⁸ one of the deepest problems in mathematics. Since 2006, the mathematical community (accept) _____ ⁹ the proof as correct. Remarkably, Perelman (turn down) _____ ¹⁰ every prize he (win) _____ ¹¹, including the Fields Medal. Mathematicians (build) _____ ¹² on his methods ever since, and topology (never / be) _____ ¹³ the same.

III – SPEAKING

1. Pair interview: “What have you been working on?”

Work in pairs. Student A is a science journalist; Student B is a mathematician who has just proved a theorem. Then swap roles. Mind the tense each prompt forces.

1. “When did you first become interested in this problem?” → *past simple*
2. “How long have you been working on it?” → *present perfect continuous + for / since*
3. “What methods have you tried so far?” → *present perfect*
4. “What were you doing when you found the key idea?” → *past continuous*
5. “Had anyone attempted this approach before you?” → *past perfect*

2. “Have you ever...?” — mathematical experience.

Mingle and ask classmates. If someone answers “yes”, follow up in the past simple: “When did you...? What happened?”

1. Have you ever solved a problem you thought was impossible?
2. Have you ever found a mistake in a textbook or a lecturer's proof?
3. Have you ever spent more than three hours on a single exercise?
4. Have you ever read a famous mathematical paper in English?
5. Have you ever changed your mind about which area of maths you like best?

3. Biography role-play (mini presentation).

Choose a mathematician (Gauss, Noether, Ramanujan, Cantor, Turing...). In 90 seconds, present “their” life in the first person, using: two past-simple facts, one past-perfect link, and one present-perfect statement about their lasting influence (“My work has influenced...”).

4. Discussion

In groups of 3-4, discuss the following questions:

1. Which mathematical results have been proved in your lifetime? When were they proved?
2. Some great proofs had remained open for centuries before they were solved. Why do some problems resist for so long?
3. Does a computer-assisted proof (like the Four Colour Theorem) have the same status as a proof a human has checked by hand? Why / why not?

IV – HOMEWORK FOR NEXT CLASS

1. Grammar book, Appendix 7 (Numbers), pp. 325–326.
2. U4: Measuring & Quantifying: Preparatory Tasks (Moodle)
3. U4: Measuring & Quantifying: Test Yourself! (Moodle)
4. U4: Frequency: Preparatory Tasks (Moodle)
5. U4: Frequency: Test Yourself! (Moodle)
6. [To relax after hard homework](#)

UNIT 4: QUANTIFICATION, MEASUREMENT & FREQUENCY



This unit introduces students to the language of **quantitative description in mathematics and the sciences**, integrating:

- measurement (values, dimensions, units)
- frequency (rates, occurrence, repetition)
- data interpretation (graphs, trends, distributions)

I – VOCABULARY (1): MEASURING & QUANTIFYING^{xvi}

N.B. – Many of the words in this section have multiple grammatical forms, e.g.

NOUN: ► a high **value**

VERB: ► the function is **valued** at each point

1. Adjectives

deep ≠ shallow • far ≠ near • fast ≠ slow • heavy ≠ light • high ≠ low • long ≠ short •
odd ≠ even • thick ≠ thin • wide / broad ≠ narrow

► All prime numbers greater than 2 are **odd** numbers.

accurate ≠ inaccurate • average / mean • standard ≠ sub-standard •
exact ≠ approximate

► The **mean** of a normal distribution coincides with its median and mode.

2. Nouns

amount • extent • measurement • range • size • span • speed

accuracy • average • level • mean • rate • scale • stage • step

► The **rate** of convergence of this series is exponential.

check • study • survey • estimate

area • circumference • cross-section • diameter • radius • perimeter

► The **circumference** of a circle equals $2\pi r$, where r is the radius.

► The **cross-section** of the wire is 0.22 mm^2 (nought point two two square millimetres)

■ Rules for noun formation — suffixes

ADJ / VERB + -th / -t (+ vowel change)

depth • height • long / length • weight • width

ADJ + -ness

flatness • nearness • thickness • smoothness • heavy / heaviness

VERB + -ment

to refine / refinement • measurement • displacement • movement

3. Verbs

■ Rules for forming verbs

NOUN / ADJ + Ø (no change)

to narrow / to thin • to range / to span / to extend / to reach •
to rate / to check / to monitor • to record / to map / to plot

► The partial sums were **plotted** against n .

NOUN / ADJ + -en

to deepen • to lengthen • to shorten • to thicken • to widen

► The confidence interval **widens** as the sample size decreases.

NOUN / ADJ + adv particle

to check **up** • to round **up** • to level **off** • to work **out** •
to scale **down** ≠ to scale **up** • to slow **down** ≠ to speed **up**

► As n grows large, the sequence **levels off** towards its limit.

4. Structures

Dimensions can be expressed by 4 different structures.

The Leaning Tower of Pisa is 56 m	in height in width in depth in diameter
-----------------------------------	---

It is 56 m	high wide long thick
------------	--------------------------------------

Its	radius is 5 m length circumference cross-section is $\pi r^2 \approx 78.5 \text{ m}^2$
-----	---

It has	a diameter of 10 m a length of ... a volume of ... a weight/mass of ...
--------	--



Do you know other facts about the Leaning Tower of Pisa?

5. Other measurements

■ Area

To obtain the area, you **multiply** the length **by** the width.

It measures 10 cm **by** 10 cm. The area is 100 cm^2 (a hundred **square** cm).

πr^2 (**pi r squared**) • $\sqrt{x}$ (the **square root** of x)

■ Volume

The volume is $1,000 \text{ cm}^3$ (a thousand **cubic centimetres**).

x^3 (x **cubed**) • $\sqrt[3]{y}$ (the **cube root** of y)

■ Power

x^9 (x **to the power** nine / x **to the** ninth)

x^{-9} (x **to the power minus** nine / x **to the minus** ninth)

e^x (e **to the** x / the **exponential** of x) • $\log_2 x$ (log **base** two of x)

6. Approximate measurements

These can be expressed by means of **adverbial modifiers**.

It is	approximately 5 cm long about / roughly / more or less almost / nearly a little over / slightly under of the order of 10^3
-------	---



7. Questions

Note the question forms.

It equals 10	How much is it? / What is its value?
It is 5 cm long	How long is it? / What is its length?
It weighs 10 kg	How heavy is it? / How much does it weigh ? What does it weigh ?
It is 5 km away	How far (away) is it? / How many kilometres away is it? What is the distance ?
It grows quickly	How fast does it grow? / At what rate does it grow?



Flashcards: [Revision flashcards](#)

8. Customary & Metric Units

The metric (SI) system is standard in science; customary units turn up mainly in problems set in the US or UK, and in patient-reported height and weight.

ONE RULE Convert the customary figure to **SI** first, then do all your working in SI.

LENGTH

SI unit: metre (m)

1 inch (in) = **2.54 cm** (25.4 mm)
1 foot (ft) = 12 in = **0.30 m** (30.48 cm)
1 yard (yd) = 3 ft = **0.91 m**
1 mile (mi) = 1 760 yd = **1.61 km**

REVERSE 1 cm $\approx$ 0.39 in · 1 m $\approx$ 3.28 ft · 1 km $\approx$ 0.62 mi

VOLUME · CAPACITY

SI unit: m³ · the litre (L) in practice

1 fluid ounce (fl oz) = **29.6 mL**
1 cup = 8 fl oz = **237 mL**
1 pint (pt) = 2 cups = **0.47 L** (473 mL)
1 quart (qt) = 2 pt = **0.95 L**
1 gallon (gal) = 4 qt = **3.79 L**

REVERSE 1 L $\approx$ 1.06 US qt $\approx$ 2.11 US pt

US VS UK The UK (imperial) pint is ~20% larger: 1 UK pint $\approx$ 568 mL, 1 UK gallon $\approx$ 4.55 L.

MASS · WEIGHT

SI unit: kilogram (kg)

1 ounce (oz) = **28.35 g**
1 pound (lb) = 16 oz = **453.6 g** (0.454 kg)
1 stone (st) = 14 lb · UK = **6.35 kg**
1 ton (T) = 2 000 lb = **907 kg** (0.91 t)

REVERSE 1 g $\approx$ 0.035 oz · 1 kg $\approx$ 2.2 lb

In the lab you measure **mass** (g, kg);
“weight” in pounds is the everyday word
for the same quantity here.

TEMPERATURE

SI unit: kelvin (K) · °C in practice ·
 $K = ^\circ C + 273.15$

Water freezes: 32 °F = 0 °C = 273 K

Water boils: 212 °F = 100 °C = 373 K

Body temperature: 98.6 °F = 37 °C = 310 K

°F → °C $^{\circ}C = (^{\circ}F - 32) \div 1.8$

°C → °F $^{\circ}F = (^{\circ}C \times 1.8) + 32$

The 1.8 is 9/5. A 1 K change equals a 1 °C change — only the zero point differs.

1. Adverbs

The adverbs of frequency form a continuum from always to never:

always ◀	constantly steadily non-stop	often frequently usually	sometimes occasionally periodically from time to time now and then	rarely seldom hardly ever	▶ never
-----------------	------------------------------------	--------------------------------	--	---------------------------------	----------------

► A determinant **seldom** vanishes for a randomly chosen matrix.

► **From time to time**, an iterative method converges only after very many steps.

GRAMMAR: POSITION OF FREQUENCY ADVERBS

➤ IN SIMPLE TENSES (e.g. present simple, past simple ...)

- 'BE': after 'be'
e.g. I am hardly ever late.
Are you often late?
- OTHER VERBS: between the subject and the verb
e.g. I never arrive late.

➤ IN COMPOUND TENSES (e.g. present perfect, will-future...)

The frequency adverb goes **after the first auxiliary and before the main verb**.

e.g. He does not often go skiing.

He doesn't often go skiing.

Do you always remember to do your homework?

I have never been to Australia.

Have you ever been to Australia?



Exception: 'Now and then': at the **beginning or the end** of the sentence

e.g. Now and then, she goes to the cinema.

She goes to the cinema now and then.

2. Adjectives

- regular • steady • even • constant • non-stop
- irregular • random • unpredictable
- periodic • intermittent • recurrent • repeated • cyclic
- oscillating • alternating • fluctuating

- ▶ The decimal expansion of $1/7$ is **recurrent**, repeating every six digits.
- ▶ The sample points have a relatively **even** distribution across the interval.

3. Nouns

- an oscillation • wave • wavelength • pulse • beat
- a group • set • series • cluster • range • array • pattern

- ▶ A **series** of nested intervals shrinks to a single point.
- ▶ The eigenvalues form a tight **cluster** near the origin.

4. Verbs

- to recur • repeat • reduplicate • echo
- to oscillate • fluctuate • vibrate • alternate • beat • swing

- ▶ The sign of the terms periodically **alternates** between positive and negative.
- ▶ If the same remainder should **recur**, the decimal expansion is periodic.

5. Word formation

NOUN (time unit) + -ly	re- + VERB
hourly	renew
daily	rebuild
weekly	replace
yearly	rearrange
	reorganise
	reproduce

- ▶ A **weekly** problem class on number theory.
- ▶ We **rearranged** the terms of the absolutely convergent series.

6. Fixed expressions

Fixed expressions are also used to indicate recurrence and ratio.

■ Recurrence

every	month two weeks / three weeks (every) other / second / third day
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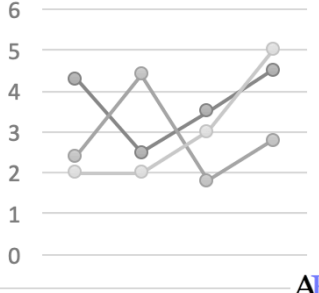
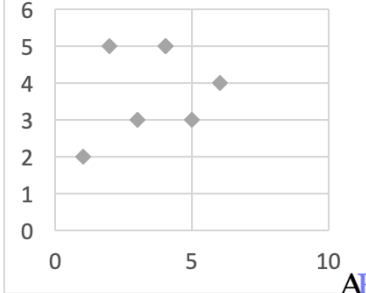
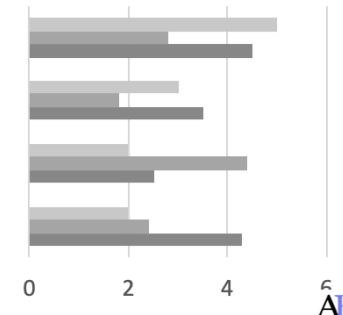
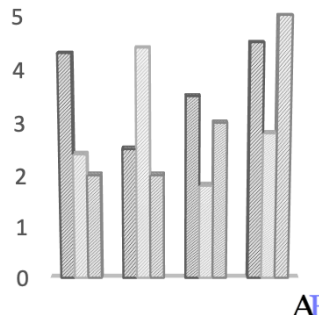
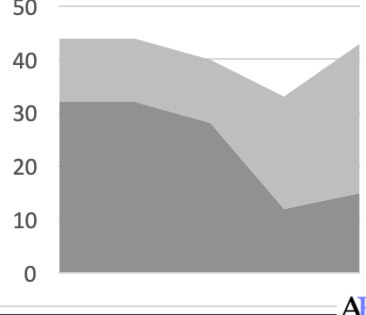
once twice three times	a / per	minute day week
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- ▶ The solver completes 100 iterations **a / per** second.
- ▶ The sequence changes sign **every other** term.

■ Ratio

- ▶ **Two out of three** integers are not divisible by three.
- ▶ **Three in ten** of the sampled matrices are singular.
- ▶ A common approximation gives the ratio **22 : 7** (twenty-two **to** seven) for π .
- ▶ The recorded masses ranged from 0.6-63 mg (nought point six **to** sixty-three milligrams).

1. Types of graphs

		
1. Pie Chart	2. Line Graph	3. Scatter Plot / Graph
		
4. Bar Chart	5. Column Chart	6. Area Chart

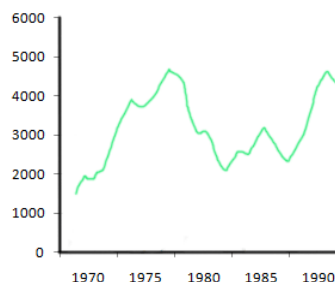
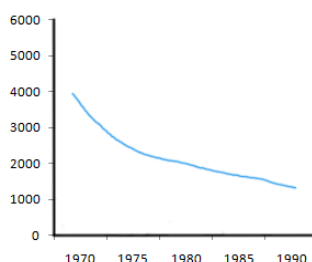
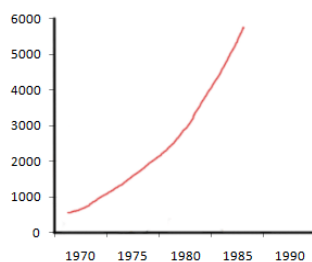
2. Introducing the graph

The graph/table/pie chart/bar chart/diagram ...

- gives information about/on ...
- provides information about/on ...
- shows ...
- illustrates ...
- compares ...
- explains why ...
- describes ...
- draws the conclusion of (a survey) ...

→ **e.g.** The pie charts provide information on the proportion of males and females working in the agricultural sector.

3. Types of changes



NOUNS

a rise (in)*
an increase (in)
a growth (in)
a peak (in)
a surge (in)

→ **e.g.** a rise in prices

a fall (in)*
a decrease (in)
a decline (in)
a dip (in)

→ **e.g.** a fall in prices

a fluctuation (in)
a variation (in)

→ **e.g.** a fluctuation in prices

VERBS

➤ Rises:

to rise (by)*
to increase
to grow

→ **e.g.** Prices rose by 3%.

➤ Falls:

to fall (by)
to decrease
to decline
to dip (to)*

→ **e.g.** Prices fell by 3%.
Prices dipped to £3.

➤ Fluctuations:

to fluctuate (between...and)*
to vary (from...to)*

→ **e.g.** Prices vary from \$2 to \$3

➤ Large rises:

to rocket
to soar
to leap (->leapt)
to peak (at)*
to surge

➤ Large falls:

to dive
to plunge
to plummet

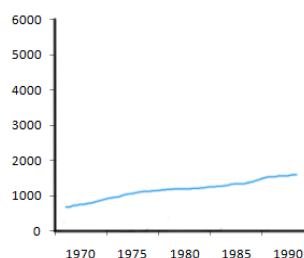
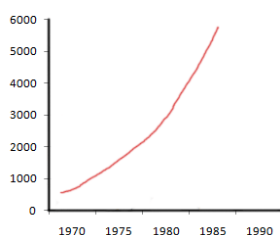
➤ Other:

to stand (at)*
to start (at)
to finish (at)
to stabilise (at)

→ **e.g.** The price of a croissant stood at €2 in 2024.

* Several prepositions can be used in different trend description contexts. Learn more about the use of prepositions in trend description on this [page](#).

4. Description of changes



ADVERBS

sharply
suddenly
rapidly
abruptly
dramatically
significantly
considerably
markedly
wildly

→ **e.g.** Prices rose sharply

slightly
gently
gradually
steadily
modestly
marginally

→ **e.g.** Prices increased modestly

ADJECTIVES

sharp
sudden
rapid
abrupt
dramatic
steep
significant
considerable
marked
substantial
spectacular

→ **e.g.** there was considerable growth

slight
gentle
gradual
steady
consistent
modest
marginal

→ **e.g.** there was a gradual decline

5. Useful Phrases

a small fraction, a small number, a small minority

a large portion, a significant majority

nearly a fifth, almost 10%, in the region of 40%, more than half, over a quarter, around two thirds, more or less three quarters, exactly one in ten, approximately a third

IV – PRACTISE

1. Answer the following questions.

- a. What can be odd or even?
- b. How many kilometres is a mile? 1 mile = km
- c. To span = (synonym)
e.g. A series of bridges spans the river.
- d. To speed up ≠
e.g. They have speeded up production of the new car.

2. Write sentences to describe the following landmarks. [1 ft = 0.3m ; 1 lb = 453.6g]

➤ Big Ben



Height: 97.5m

Length: 12m

Width: 12m

Weight (bell): 13.7 tonnes

Weight (pendulum): 310 kg

Length (pendulum): 4.4m

Depth (below ground): 4m

.....

.....

.....

.....

.....

.....

.....

.....

.....

➤ **Queensferry Crossing**

River: Firth of Forth

Length: 1.7 miles (2.7 km)

Height: 696 ft (212 m)

Weight (bridge deck): 35,000 tonnes



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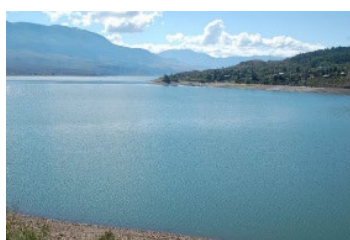
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3. Look at the pictures and write questions with:

- **how + adjective** e.g. How big...?
- **what** e.g. What is the height...?



Picture 1



Picture 2



Picture 3



Picture 4

Picture 1:?

Picture 2:?

Picture 3:?

Picture 4:?

4. Habits: then and now.

Think about the following questions for our next class (and write a few key words below).

1. What good habits did you have before university?

2. How about now?

3. What has changed?

5. Place the following adverbs on the frequency line.

always – never – frequently - now and then - hardly ever - constantly



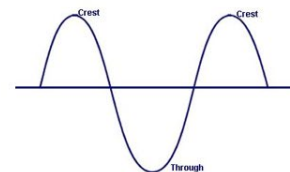
6. Find synonyms for each adverb.

- constantly =
- frequently =
- hardly ever =
- now and then =

7. Where does the frequency adverb go in the sentence? Use a cross (X) to indicate the right position. See section 'Grammar: Position of frequency adverbs' above.

- a. Are you late? (frequently)
- b. I arrive late. (never)
- c. Even simple chains are quite difficult. (often)
- d. The evidence has been contradictory. (often)
- e. Temperatures drop below freezing at night in winter. (frequently)
- f. Do science experiments provide the data needed to definitively answer a question? (always)
- g. Statistics must be used with care. (always)
- h. It doesn't work. (often)
- i. I have been to Australia. (never)
- j. Have you been to Australia? (ever)
- k. We need a break. (now and then)

8. What do these pictures refer to?



9. Walk around in the classroom and ask your peers the following questions. Make full sentences.

- a. How often do you exercise?
- b. How often do you go abroad?
- c. How often do you work for your English class?
- d. How often do you watch programmes in English on TV or on your computer?
- e. How often do you read scientific news on the internet?
- f. How often do you use your smartphone?
- g. What's your best or worst habit?
- h. Tell us about a habit you'd like to start having and why.
- i. Describe bad habits you had before university and that you managed to change.
- j. Describe good habits you had before university and you have not kept.
- k. Your turn! Write 3 more questions and ask around.

-

-

-

GENERAL NOTIONS

■ 0

The following are different ways of saying **0** in English for various contexts:

Notation	Context	Example	Read aloud
OH	After a decimal point	5.03	five point oh three
	In telephone numbers	67 01 75	six seven oh one seven five
	In bus numbers	No. 703	get the seven oh three
	In hotel room numbers	Room 206	room two oh six
	In years	1904	nineteen oh four
NOUGHT	Before a decimal point	0.03	nought point oh three
ZERO	For the integer	0	the number zero
	For temperature	-8°C	eight below zero
NIL	In football scores	5-0	Belgium beat Andorra five nil
LOVE	In tennis	15-0	fifteen love to Alcaraz

■ Integers

0	zero	10	ten	20	twenty
1	one	11	eleven	30	thirty
2	two	12	twelve	40	forty
3	three	13	thirteen	50	fifty
4	four	14	fourteen	60	sixty
5	five	15	fifteen	70	seventy
6	six	16	sixteen	80	eighty
7	seven	17	seventeen	90	ninety
8	eight	18	eighteen	100	one hundred
9	nine	19	nineteen	1000	one thousand

Notation	Read aloud
- 245	minus two hundred <u>and</u> and forty-five
336	three hundred <u>and</u> and thirty-six
1 074	one thousand and seventy-four
2 469	two thousand <u>four</u> hundred <u>and</u> and sixty-nine
22 731	twenty-two thousand seven hundred and thirty-one
250 000	a quarter of a million OR two hundred <u>and</u> fifty thousand <u>and</u>
331 749	three hundred <u>and</u> thirty-one thousand <u>seven</u> hundred <u>and</u> and forty-nine
1 000 000	one million
3 146 658	three million <u>one</u> hundred and forty-six thousand <u>six</u> hundred <u>and</u> and fifty-eight
56 000 000	fifty-six million <u>and</u>
7 000 000 000	seven billion <u>and</u> [US usage, now universal]
1 000 000 000 000	one trillion [US usage, now universal]
3 000 000 000 000	three trillion <u>and</u> [US usage, now universal]
1 000 000 000 000 000	one quadrillion

N.B. – English *billion* = French *milliard* (10^9), English *trillion* = French *billion* (10^{12}).

■ **Fractions [= Rational Numbers]**

Notation	Read aloud
$1/2$	one half
$1/3$	one third
$1/4$	one quarter [= one fourth]
$1/5$	one fifth
$-1/17$	minus one seventeenth
$3/8$	three eighths
$26/9$	twenty-six ninths
$1/1024$	one over one thousand and twenty-four
$-5/34$	minus five thirty-fourths
$2\frac{3}{7}$	two and three sevenths

■ Ordinal numbers

1 st	first	4 th	fourth	11 th	eleventh
2 nd	second	5 th	fifth	12 th	twelfth
3 rd	third	6 th	sixth	13 th	thirteenth
		7 th	seventh	14 th	fourteenth
		8 th	eighth	15 th	fifteenth
		9 th	ninth	...	
		10 th	tenth		
20 th	twentieth	100 th	hundredth		
21 st	twenty-first	1000 th	thousandth		
22 nd	twenty-second	1 000 000 th	millionth		
23 rd	twenty-third				
24 th	twenty-fourth				
25 th	twenty-fifth				
...					

■ Real Numbers

Notation	Read aloud
-0.067	minus nought point oh/zero six seven
1.1	one point one
1.23	one point two three
2.666...	two point six recurring
21.57	twenty-one point five seven
81.59	eighty-one point five nine
$-2.3 \cdot 10^6$	minus two point three times ten to the sixth/to the power six
$4 \cdot 10^{-3}$	four times ten to the minus three
2.5 million	two point five million
π [= 3.14159...]	pi (pronounced "pie")
e [= 2.71828...]	e (base of the natural logarithm)

■ Complex Numbers

Notation	Read aloud
i	i
$3 + 4i$	three plus four i
$1 - 2i$	one minus two i
$\text{conj}(1 - 2i) = 1 + 2i$	the complex conjugate of one minus two i equals one plus two i

The *real part* and the *imaginary part* of $3 + 4i$ are equal, respectively, to 3 and 4.

■ Basic arithmetic operations

Notation	Read aloud
$3 + 5 = 8$	three plus five equals [= is equal to] eight
$3 - 5 = -2$	three minus five equals minus two
$3 \cdot 5 = 15$	three times five equals fifteen
$3/5 = 0.6$	three divided by five equals zero point six
$(2 - 3) \cdot 6 + 1 = -5$	two minus three in brackets times six plus one equals minus five
$(1-3)/(2+4) = -1/3$	one minus three over two plus four equals minus one third
$4! \quad [= 1 \cdot 2 \cdot 3 \cdot 4]$	four factorial

■ Exponentiation, Roots

Notation	Read aloud
5^2	five squared
5^3	five cubed
5^4	five to the (power of) four
5^{-1}	five to the minus one
$\sqrt{3}$	the square root of three
$\sqrt[3]{64}$	the cube root of sixty-four
$\sqrt[5]{32}$	the fifth root of thirty-two
$(1+2)^{2+2}$	one plus two, all to the power of two plus two
$e^{\pi} = -1$	e to the (power of) π equals minus one

■ Divisibility

The **multiples** of a positive integer a are the numbers $a, 2a, 3a, 4a, \dots$. If b is a multiple of a , we also say that a **divides** b , or that a is a **divisor** of b (notation: $a \mid b$). This is equivalent to b/a being an integer.

■ Division with remainder

For arbitrary positive integers a, b we can in general divide b by a only with a **remainder**. If qa is the largest multiple of a less than or equal to b , then $b = qa + r$, with $r = 0, 1, \dots, a - 1$. The integer q is the **quotient** and r the **remainder** of the division of b by a .

■ Euclid's algorithm

This algorithm computes the **greatest common divisor** $\gcd(a, b)$ of two positive integers. It replaces the pair a, b (with $a \leq b$) by r, a , where r is the remainder of dividing b by a ; this preserves the $\gcd$ and is repeated until $r = 0$.

► $\gcd(12, 44)$: $44 = 3 \cdot 12 + 8$, $12 = 1 \cdot 8 + 4$, $8 = 2 \cdot 4 + 0 \rightarrow \gcd = 4$

If $\gcd(a, b) = 1$, we say that a and b are **relatively prime**.

Algebra

■ Algebraic Expressions

Notation	Read aloud
$A = a^2$	capital a equals small a squared
$a = x + y$	a equals x plus y
$c = x \cdot y \cdot z = xyz$	c equals x times y times z = c equals x y z
$(x + y)z + xy$	x plus y in brackets times z plus x y
$x^2 + y^3 + z^5$	x squared plus y cubed plus z to the (power of) five
$x^n + y^n = z^n$	x to the n plus y to the n equals z to the n
$(x - y)^{3m}$	x minus y, all to the (power of) three m
$2^x \cdot 3^y$	two to the x times three to the y
$ax^2 + bx + c$	a x squared plus b x plus c
$\sqrt{x} + \sqrt[3]{y}$	the square root of x plus the cube root of y
$\sqrt[n]{x + y}$	the n-th root of x plus y
$(a+b)/(c-d)$	a plus b over c minus d
$C(n, m)$	n over m (the binomial coefficient)

■ Indices

Notation	Read aloud
x_0	x zero ; x nought
$x_1 + y_i$	x one plus y i
R_{ij}	(capital) R subscript i j ; (capital) R lower i j
M_{ij}^k	(capital) M upper k lower i j ; M superscript k subscript i j
$\sum a_i x^i \quad (i = 0 \dots n)$	sum of a i x to the i for i from zero to n
$\prod b_m \quad (m = 1 \dots \infty)$	product of b m for m from one to infinity
$\sum a_{ij} b_{jk} \quad (j = 1 \dots n)$	sum of a i j times b j k for j from one to n

■ Inequalities

Notation	Read aloud
$x > y$	x is greater than y
$x \geq y$	x is greater (than) or equal to y
$x < y$	x is smaller than y
$x \leq y$	x is smaller (than) or equal to y
$x > 0$	x is positive
$x \geq 0$	x is positive or zero ; x is non-negative
$x < 0$	x is negative
$x \leq 0$	x is negative or zero

■ Polynomial equations

A polynomial equation of degree $n \geq 1$ with complex coefficients, $f(x) = a_0 x^n + \dots + a_n = 0$ ($a_0 \neq 0$), has n complex **solutions** (= **roots**), counted with multiplicities. A **quadratic equation** $ax^2 + bx + c = 0$ can be solved by *completing the square*.

► $x_{1,2} = (-b \pm \sqrt{\Delta}) / 2a$, where $\Delta = b^2 - 4ac$ is the discriminant

If a, b, c are real, the sign of Δ is decisive: $\Delta = 0$ gives a **double root**; $\Delta > 0$ gives two distinct real roots; $\Delta < 0$ gives a pair of complex conjugate (non-real) roots.

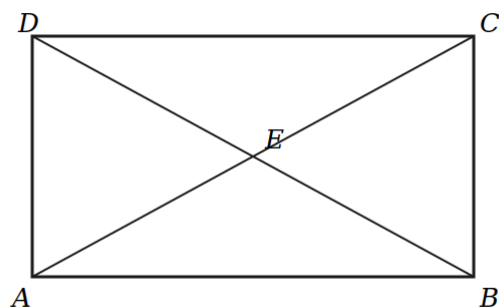
■ Congruences

Two integers a, b are **congruent modulo** a positive integer m if they leave the same remainder on division by m (equivalently, if m divides $a - b$).

Notation	Read aloud
$a \equiv b \pmod{m}$	a is congruent to b modulo m

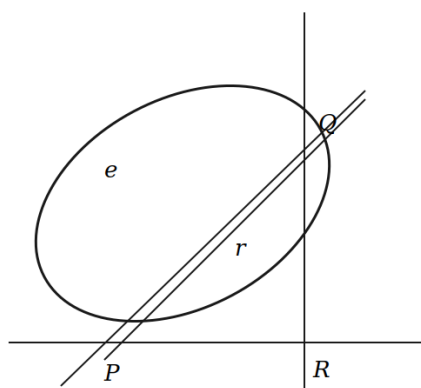
Geometry

■ Angles in a rectangle



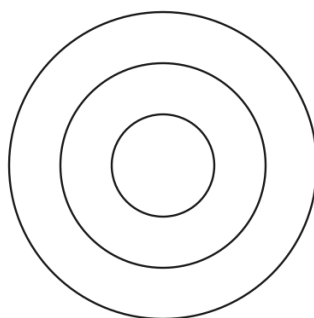
Let E be the intersection of the diagonals of the rectangle ABCD. The lines (AB) and (CD) are **parallel** to each other (and similarly (BC) and (DA)). The figure shows several *acute angles* ($\angle EAD$, $\angle EAB$, ...), *right angles* ($\angle ABC$, $\angle BCD$, ...) and *obtuse angles* ($\angle AEB$, $\angle CED$).

■ Polar of a point

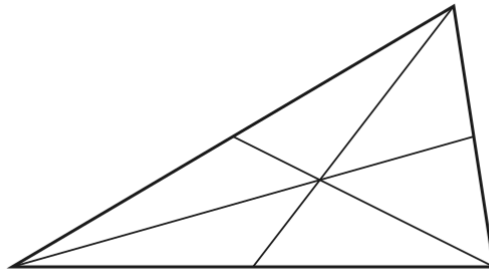


Let P and Q be two points lying on an ellipse e . Denote by R the intersection of the tangent lines to e at P and Q. The line r **passing through** P and Q is called the *polar* of the point R with respect to the ellipse e .

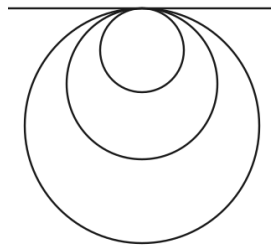
■ Concentric circles, medians, tangent circles



Three **concentric** circles with respective radii 1, 2 and 3.



A line through each **vertex** of a triangle and the **midpoint** of the opposite side gives three lines meeting at the *barycentre* (= centre of gravity).



Three circles with a common **tangent** at their unique intersection point.

■ **Euler's Formula**

For a convex **polyhedron** P , Euler's formula asserts $V - E + F = 2$, where V , E , F are the numbers of *vertices*, *edges* and *faces* of P .

The five regular polyhedra are the tetrahedron, cube, octahedron, dodecahedron and icosahedron. For instance, an **icosahedron** has 20 faces, 30 edges and 12 vertices; its face midpoints form a dual dodecahedron.

Mathematical Arguments

■ Set theory

Notation	Read aloud
$x \in A$	x is an element of A ; x lies in A ; x belongs to A
$x \notin A$	x is not an element of A ; x does not belong to A
$x, y \in A$	(both) x and y are elements of A ; ... belong to A
$x, y \notin A$	(neither) x nor y is an element of A
$\emptyset$	the empty set (= set with no elements)
$A = \emptyset$ / $A \neq \emptyset$	A is an empty set / A is non-empty
$A \cup B$	the union of A and B ; A union B
$A \cap B$	the intersection of A and B ; A intersection B
$A \times B$	the product of A and B ; A times B
$A \cap B = \emptyset$	A is disjoint from B
$\{x \mid \dots\}$	the set of all x such that ...
$\mathbb{C} / \mathbb{Q} / \mathbb{R}$	the set of all complex / rational / real numbers

$A \times B$ contains the **ordered pairs** (a, b) with $a \in A$ and $b \in B$; A^n contains all ordered **n-tuples** of elements of A.

■ Logic

Notation	Read aloud
$S \vee T$ / $S \wedge T$	S or T / S and T
$S \Rightarrow T$	S implies T ; if S then T
$S \Leftrightarrow T$	S is equivalent to T ; S iff T
$\neg S$	not S
$\forall x \in A$	for each [= for every] x in A
$\exists x \in A$	there exists [= there is] an x in A (such that)
$\exists! x \in A$	there exists a unique x in A (such that)
$\nexists x \in A$	there is no x in A (such that)

Functions

■ *Formulas / Formulae*

Notation	Read aloud
$f(x)$	f of x
$g(x, y)$	g of x (comma) y
$h(2x, 3y)$	h of two x (comma) three y
$\sin(x) / \cos(x) / \tan(x)$	sine x / cosine x / tan x
$\arcsin(x) / \arccos(x) / \arctan(x)$	arc sine x / arc cosine x / arc tan x
$\sinh(x) / \cosh(x) / \tanh(x)$	hyperbolic sine / cosine / tan x
$\sin(x^2)$	sine of x squared
$\sin(x)^2$	sine squared of x ; sine x, all squared
$(x+1)/\tan(y^4)$	x plus one, all over tan of y to the four
$3^x - \cos(2x)$	three to the (power of) x minus cosine of two x
$\exp(x^3 + y^3)$	exponential of x cubed plus y cubed

■ *Intervals*

Notation	Read aloud
(a, b)	open interval a b
$[a, b]$	closed interval a b
$(a, b]$	half-open interval a b (open on the left, closed on the right)
$[a, b)$	half-open interval a b (open on the right, closed on the left)

■ Derivatives

Notation	Read aloud
f'	f dash ; f prime ; the first derivative of f
f''	f double dash ; f double prime ; the second derivative of f
$f^{(3)} / f^{(n)}$	the third / the n-th derivative of f
dy/dx	d y by d x ; the derivative of y by x
d^2y/dx^2	the second derivative of y by x ; d squared y by d x squared
$\partial f / \partial x$	the partial derivative of f by x (with respect to x)
∇f	nabla f ; the gradient of f
Δf	delta f

The **gradient** ∇f is the vector of partial derivatives; the **Laplace operator** Δ sums the second partial derivatives; the **Jacobian matrix** of $g(x,y)$, $h(x,y)$ collects their partial derivatives in a 2×2 array.

■ Integrals

Notation	Read aloud
$\int f(x) dx$	integral of f of x d x
$\int_a^b t^2 dt$	integral from a to b of t squared d t
$\iint_S h(x, y) dx dy$	double integral over S of h of x y d x d y

■ Differential equations

An **ordinary** (resp. **partial**) differential equation — ODE (resp. PDE) — involves an unknown function of one (resp. more than one) variable and its derivatives. Its **order** is the maximal order of derivative appearing; it is **linear** if f and its derivatives appear linearly.

Notation	Read aloud
$f' + xf = 0$	first order linear ODE
$f'' + \sin(f) = 0$	second order non-linear ODE
$\Delta f = 0$	the Laplace equation
$\Delta f = \lambda f$	the Helmholtz equation
$\Delta g = \partial g / \partial t$	the heat equation
$\Delta g = \partial^2 g / \partial t^2$	the wave equation

■ Greek letters

Small Greek letters used in mathematics

α	alpha	β	beta	γ	gamma	δ	delta
ϵ	epsilon	ζ	zeta	η	eta	θ, ϑ	theta
ι	iota	κ	kappa	λ	lambda	μ	mu
ν	nu	ξ	xi	$\omicron$	omicron	π, w	pi
ρ	rho	σ	sigma	τ	tau	υ	upsilon
ϕ, ϕ	phi	χ	chi	ψ	psi	ω	omega

Capital Greek letters used in mathematics

$\mathbf{B}$	Beta	$\mathbf{\Gamma}$	Gamma	$\mathbf{\Delta}$	Delta	$\mathbf{\Theta}$	Theta
$\mathbf{\Lambda}$	Lambda	$\mathbf{\Xi}$	Xi	$\mathbf{\Pi}$	Pi	$\mathbf{\Sigma}$	Sigma
$\mathbf{Y}$	Upsilon	$\mathbf{\Phi}$	Phi	$\mathbf{\Psi}$	Psi	$\mathbf{\Omega}$	Omega

Sequences, Series

■ Convergence criteria

An infinite **series** $\sum a_n$ of complex numbers **converges** (to s) if the sequence of **partial sums** $s_n = a_1 + \dots + a_n$ has a finite limit; otherwise, it **diverges**.

Fact 1. If $\sum |a_n|$ converges, so does $\sum a_n$ (the series is *absolutely convergent*).

Fact 2. If $0 \leq a_n \leq b_n$ for all large n and $\sum b_n$ converges, so does $\sum a_n$.

The ratio test (d'Alembert). If $|a_{n+1}| \leq r |a_n|$ for some $0 < r < 1$ and all large n , then $\sum a_n$ converges absolutely.

The root test (Cauchy). If $\sqrt[n]{|a_n|} \leq r$ for some $0 < r < 1$ and all large n , then $\sum a_n$ converges absolutely.

■ What is $1 + 2 + 3 + \dots$ equal to?

The partial sums $1, 3, 6, 10, \dots$ **tend towards** plus infinity, so $1 + 2 + 3 + \dots = +\infty$. Yet several methods of *regularising* this divergent series all assign it the surprising value $-1/12$.

► Writing $c = 1 - 1 + 1 - \dots$, $b = 1 - 2 + 3 - \dots$, $a = 1 + 2 + 3 + \dots$ gives $c = 1/2$, $b = 1/4$, and finally $a = -1/12$.

Notation

$$\lim_{x \rightarrow 1} f(x) = 2$$

Read aloud

the limit of f of x as x tends to one is equal to two

VI – PRACTISE: NUMBERS AND MATHEMATICAL SIGNS^{xx}

1. Read the following calculations out loud.

$$9^2 = 81$$

$$5^3 = 125$$

$$10 \text{ km}^2 = 1 \text{ km} \times 10 \text{ km}$$

$$\sqrt[3]{216} = 6$$

$$3^4 = 81$$

$$4^{-5} = \frac{1}{1024}$$

2. Read the following out loud, paying particular attention to stress patterns:

1 km = 0.621 mile	³ / ₈ s of a pound	2.718281828	\$50.25
0.00000001	1 cm = 0.39 inches	45 mins = ?	1 kg = 2.2 pounds
€25,000.21	20 mins = ¹ / ₃ hr	3.14159	1 m = 1.09 yards
9.11 secs	1 g = 0.035 ounces	19.07%	53.6%

3. Listening comprehension: Take notes of the numbers and mathematical signs that you hear.¹

- | | | | | |
|----|----|-----|----|------|
| I. | a. | II. | a. | III. |
| | b. | | b. | |
| | c. | | c. | |
| | d. | | | |
| | e. | | | |

¹ Source: Unit 2: ex. 9

4. Listening comprehension: Fill in the correct numbers (in digits) for each statement relating to a scientific experiment or finding.²

1. As this was a dosage of _____ milligrams of fluoride per kilogram of body weight, the equivalent amount needed to achieve a similar peak in a _____ -kilogram child would be _____ milligrams of fluoride.
2. The sensitivity of the assay was _____ picomoles.
3. The output impedance is about _____ ohms at the _____ volt end and _____ ohm at the _____ volt end of the range.
4. _____ -amp _____ -core mains flex is used for the mains input which connects straight to the PCB.
5. Inserting a few atoms of potassium makes the compound a superconductor which, below a critical temperature of about _____ kelvins, conducts electricity with no resistance.
6. This shows that where _____ moles of hydrogen gas combine with _____ mole of oxygen gas to form _____ moles of liquid water, at a pressure of _____ atmosphere, and a temperature of _____ kelvins, the enthalpy change is _____ kilojoules.
7. Isolated young mice squeak repeatedly at frequencies of _____ kilohertz to _____ kilohertz, until their mother comes and returns them to the nest.
8. In a similar form of these experiments, conventional, _____ -nanosecond laser pulses were used.

² Source: Unit 2: ex. 10

A. LISTENING COMPREHENSION

➤ ***1st listening***

Take notes of the gist.

➤ **2nd & 3rd listenings**

Formulate the questions correctly and then answer.

1. . What do we already know about *Dromornis stirtoni*?

 . What kind of animal was it?

 . Where did it live?

 . When?

 . Weight?

 . Wingspan?

 . Height?

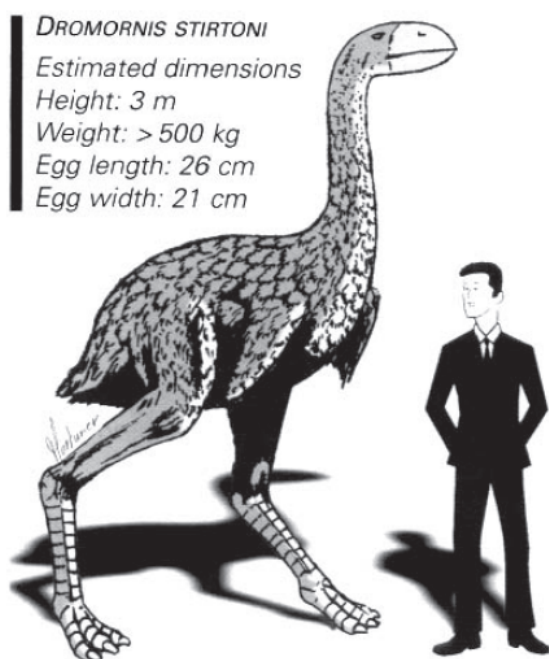
 . Diet?
2. Why couldn't *Dromornis stirtoni* fly?
3. Why do we know so much about the bird?
4. How do we know it's not the same species as an ostrich?
5. What makes it possible for an ostrich to run so fast?

B. VOCABULARY

Find words in the text that have the same meaning.

- a. approximately, more or less:
- b. a number of, quantity:
- c. to estimate, to calculate, to assess:
- d. to get, to receive:
- e. extremely large, enormous:

Dromornis stirtoni, an extinct flightless bird, lived in Australia roughly 8 million years ago. It was probably the heaviest bird in the history of evolution, with a weight of slightly more than 500 kg although its wingspan was very small. A considerable amount of information has been obtained from recent fossil finds in Queensland, enabling scientists to work out basic measurements. From a morphological point of view, *Dromornis stirtoni* appears to be similar to an emu or an ostrich. However, scientists now believe that it is related to the duck species, as the massive dimensions of the head show. The bird attained a height of over 3 metres. The large head and formidable beak suggest that the bird was carnivorous. The cross-section of fossils of the leg bones reveals that the bird had short, thick legs indicating that it could not have run as fast as the ostrich. The width of the body was about the same as the length of the neck and legs.



➤ **Brainstorming:**

What do you expect to hear about? What words do you expect to hear?

➤ **1st viewing – WITHOUT picture**

1. What numbers do you hear and what do they correspond to?
2. What frequency expressions do you hear?

Numbers	Frequency expressions

➤ **2nd & 3rd viewings – WITHOUT picture**

What are the positive and negative effects of being in space on the human body?

<i>Positive effects</i>	<i>Negative effects</i>

IX – HOMEWORK FOR NEXT CLASS

1. PRACTISE FOR YOUR PROJECT: presenting specific data

You should focus on specific data. This involves mainly proofs, demonstrations, statistics, proportions, percentages, ratios and rates. However, any visual mathematical content you find relevant to your project is acceptable.

Instructions

1. prepare to describe at least one figure from your project orally using the vocabulary from the subunits 'Frequency' and 'Qualifying'. If relevant, refer to 'III – VOCABULARY (3): DESCRIBING GRAPHS' above and [Present Your Data Like a Pro](#) to understand better how to present data on a slide.



Tips:

- Make sure to present this figure meaningfully, that is to say, put it in perspective (remember 'Purpose & Process' - facts are not interesting unless we understand why they matter in a specific context).
- Do not use sentences on your slides and do not read a text.
- Use illustrations, keywords and bullet points only.

2. Use your shared project PowerPoint.

2. PRACTISE FOR YOUR PROJECT: are my documents acceptable?

See [MATH - Project topics + article selection - 26-27](#)

In groups of 3. Please do a first check before the next class.

3. GRAMMAR: Revise Future Tenses

- Grammar book, Section 3 (Future), pp. 25–32.
- Coursebook
- AngloLink [Video 1](#)
- AngloLink [Video 2](#)
- [Future Time clauses](#)

X – PRACTISE FOR YOUR PROJECT: PRESENTING SPECIFIC DATA (IN-CLASS ACTIVITY)

1. Presenter: Present your figure to four different students. Do not read a text. Use a few notes only (keywords on a small piece of paper) and a POWERPOINT slide.
2. Listeners: Take notes of the key graph information in the table below.
3. Be ready to report back on other students' figures.

<u>Figure 1:</u>	<u>Figure 2:</u>
<u>Figure 3:</u>	<u>Figure 4:</u>

4. Discussion: Feedback about the presentations.

UNIT 5: FUTURE TENSES^{xxiii}

I – TENSES AT A GLANCE

Form	Structure	Mathematical example	Use
will future	will + infinitive	<i>If we double n, the error will halve.</i>	prediction from reasoning; instant decision; offer / promise
be going to	am/is/are going to + inf.	<i>Look at the residuals — the model is going to fail.</i>	intention / plan; prediction from present evidence
Present continuous	am/is/are + -ing	<i>I'm presenting my poster on Friday.</i>	a fixed (personal) arrangement
Present simple	base form / -s	<i>The lecture starts at nine.</i>	a timetable or scheduled event
Future continuous	will be + -ing	<i>This time next year I'll be writing up.</i>	an action in progress at a future time
Future perfect	will have + past participle	<i>By June I'll have collected the data.</i>	an action completed before a future time

Two questions decide the form: is it a prediction or a plan? And what is the evidence?

WILL — prediction & decision

I think • I expect • probably • certainly
• perhaps • OK, I'll...

BE GOING TO — intention & evidence

I'm going to • I plan to • Look! •
the data show • it's about to

In context:

- ▶ *If we double n , the error **will halve**.* (prediction by calculation → will)
- ▶ *Look at the residuals — the model **is going to fail**.* (present evidence → be going to)
- ▶ *I **'m presenting** my results at 3 p.m. on Friday.* (fixed arrangement → present continuous)
- ▶ *The congress **opens** on 23rd July.* (a fixed timetable → present simple)

N.B. – A note for French speakers

- After when, as soon as, before, until and if, English uses the present, not will: "When the program finishes (not 'will finish'), it will display the result." (French *quand + futur*)
- je vais + infinitif → be going to (intention). Do not translate every French future with will.
- For a fixed arrangement, English prefers the present continuous: "I'm meeting my supervisor tomorrow" (French uses the present, *je vois mon directeur demain*).

II – READING: THE NEXT GREAT THEOREM?

Read the text. Underline the future forms to help you with the analysis tasks that follow.

No one knows which famous problem mathematicians will solve next, but almost everyone has a prediction. Many believe that artificial intelligence is going to change the way proofs are written, because the evidence is already visible: powerful proof assistants such as Lean are checking more and more arguments line by line.

Next July, the International Congress of Mathematicians is taking place in Philadelphia — the first time in the United States for forty years. The congress opens on 23rd July, and during the week the organisers will award the Fields Medals to the brightest young researchers. Mathematicians will gather from all over the world, and many of them will be presenting results that no human has fully checked by hand.

By the middle of the century, some experts argue, computers will have formalised most of undergraduate mathematics, and students will be learning to write proofs in a language a machine can read. Others disagree. They predict that deep new ideas will always come from people, and that the hardest questions — the Riemann Hypothesis, or whether P equals NP — will resist machines for decades.

What is certain is that the field will keep changing. If quantum computers become practical, much of today's cryptography will collapse, and as soon as that happens, mathematicians will have to rebuild it. Whatever comes next, the next generation is going to inherit a subject that looks very different from the one in this book.

1. Comprehension (in pairs). Read again quickly and answer orally.

1. Why do some people think artificial intelligence is going to change mathematics?
2. What is special about this year's International Congress of Mathematicians?
3. What do optimists expect computers to have done by the middle of the century?
4. What do the sceptics predict instead?
5. According to the text, what will happen to cryptography if quantum computers become practical?

2. Form spotting. Find one example in the text of each form and copy out the verb phrase.

1. **will future** — a prediction from reasoning: _____
2. **be going to** — intention or evidence-based prediction: _____
3. **present continuous** — a fixed future arrangement: _____
4. **present simple** — a timetable / scheduled event: _____
5. **present after when / if / as soon as** — a time clause referring to the future: _____

3. Justify the choice. Explain why each form was used, not an alternative.

1. Why "the congress opens on 23rd July" and not "will open"?
2. Why "artificial intelligence is going to change the way proofs are written" and not "will change"?
3. Why "as soon as that happens" and not "will happen"?

III – GRAMMAR EXERCISES

1. will or be going to? Put the verb into the better form.

1. The data are all over the place — this regression _____ (fail) .
2. I'm sure the proof _____ (work) ; the logic is sound.
3. "Someone needs to check the references." — "OK, I _____ (do) it."
4. She has made up her mind: she _____ (specialise) in topology.
5. Look at the clock — we _____ (run) out of time!
6. By my calculation, the algorithm _____ (terminate) in n steps.
7. They _____ (rewrite) the whole section; they announced it yesterday.
8. "The projector isn't working." — "No problem, I _____ (use) the board."

2. Present continuous or present simple? Both refer to the future here.

1. The lecture _____ (start) at 9 a.m. sharp.
2. I _____ (meet) my supervisor at three this afternoon.
3. The train to the conference _____ (leave) at 7.45.
4. We _____ (present) our poster on Friday morning.
5. The exam _____ (begin) at noon and _____ (finish) at three.
6. What _____ (you / do) after the seminar?
7. The congress _____ (open) on 23 July.
8. She _____ (give) a talk at the workshop next week.

3. Present or will? After when / if / as soon as / until / before, use the present.

1. When the program _____ (finish) , it _____ (display) the result.
2. If we _____ (increase) n , the error _____ (decrease) .
3. I _____ (send) you the file as soon as I _____ (check) it.
4. The sequence _____ (converge) if the ratio _____ (be) less than one.
5. We _____ (not / start) until everyone _____ (arrive) .
6. As soon as the proof _____ (be) complete, we _____ (submit) it.
7. If quantum computers _____ (become) practical, much cryptography _____ (collapse) .

4. Functions of will. Label each sentence: prediction, instant decision, promise, offer, refusal or threat.

1. "I'll help you with that integral." _____
2. "I'll definitely finish it by Friday — you have my word." _____
3. "The algorithm will run in linear time." _____
4. "Fine — I'll redo the whole calculation." _____
5. "My laptop won't connect to the projector." _____
6. "Submit it on time, or the system will lock you out." _____

5. Error correction. Each sentence contains one mistake with a future form. Write the correct version.

1. When I will finish my degree, I will look for a PhD position.
2. If the series will converge, the sum will be finite.
3. Look at those clouds — it will rain.
4. As soon as I will receive the data, I will start the analysis.
5. Tomorrow, I work in the library all day.
6. Don't worry, I help you with the calculation.
7. We will give our presentation on Friday; it's all arranged.
8. The lecture is starting at nine every Monday.

6. Recap. Put each verb in brackets into the correct future form.

Mathematics in fifty years

No one knows exactly what mathematics (look) _____ ¹ like in 2075, but almost everyone has a prediction. Most experts believe that artificial intelligence (transform) _____ ² the subject, and the first signs are already here: if you (open) _____ ³ a research journal today, you (find) _____ ⁴ proofs that a computer has helped to check.

Big meetings (play) _____ ⁵ their part too. This July, the International Congress of Mathematicians (take) _____ ⁶ place in Philadelphia — the first time in the United States for forty years. The congress (open) _____ ⁷ on 23 July, and during the week the organisers (award) _____ ⁸ the Fields Medals to the brightest young researchers.

What (happen) _____ ⁹ after that? Many believe that, if computers (become) _____ ¹⁰ powerful enough, they (take over) _____ ¹¹ much of the routine work, while people (concentrate) _____ ¹² on the deepest ideas.

Yet many problems (resist) _____ ¹³ even the cleverest machines. As soon as someone (solve) _____ ¹⁴ the Riemann Hypothesis, a dozen new questions (appear) _____ ¹⁵. Whatever happens, the mathematics of the future (be) _____ ¹⁶ as surprising as its past.

IV – SPEAKING

1. Extrapolate the graph.

Your teacher is going to show partial plots (a convergence curve, an exponential, a diverging series). Predict what happens next. If you reason from the maths, use *will* ("if we double n , the error *will* halve"); if you read it straight off the evidence, use *be going to* ("the curve *is going to* cross the axis"). Then the real continuation is revealed.

2. Week programme role-play.

Using your timetable for this week, ask and answer in pairs. Use the *present simple* for the schedule ("When/where does each class start?") and the *present continuous* for your own arrangements ("What are you doing after dinner?" — "I'm playing football with Mike at 7 p.m.")

3. Forecast the future of mathematics.

Discuss in groups: Will AI prove a major theorem before humans? Will $P = NP$ ever be resolved? Will we still teach proofs by hand in twenty years? Predict with *will* / *won't*, then hedge with academic language: *is likely to*, *will probably*, *might well*, *is unlikely to*.

4. Spontaneous-decisions lab.

Your teacher throws sudden situations at the room. Respond with an instant *will* decision, then restate it as a prior plan with *be going to*:

- "The projector just died." → "I'll switch to the whiteboard." → "I'm going to email everyone the slides afterwards."
- "There's a sign error in line 3." → "I'll re-derive it." → "I'm going to recheck the whole proof tonight."

V – HOMEWORK FOR NEXT CLASS

- U6: Linking your ideas: Preparatory Tasks (Moodle)

UNIT 6: LINKING YOUR IDEAS



I – STARTER

1. Why is it important to link your ideas?
2. How can you show a link between your ideas?

Connectors and discourse markers

1. Giving additional information

and • not only / but • as well as
in addition • furthermore • moreover • besides • apart from

► The proof is short; **furthermore**, it requires no extra hypotheses.

2. Contrast

but • however • nevertheless • yet • although • (even) though
in spite of / despite • whereas • on (the) one hand / on (the) other hand

► Numerical methods are used more and more; **nevertheless**, they are not suitable in every case.

3. Contrast to what people think

actually • in fact • in actual fact • in reality

► Most people think Newton alone invented calculus, but **actually** Leibniz developed it independently.

4. Introducing something obvious / generally accepted

obviously • naturally • of course
clearly • doubtless

► **Obviously**, unless the sequence is bounded, it cannot converge.

5. Making a general statement

in general • generally speaking
on the whole • as a rule

► **As a rule**, a polynomial of degree n has n complex roots (counted with multiplicity).

6. Clarification

for example • for instance • e.g. • such as
that is to say • in other words • i.e. • namely

► There is exactly one even prime, **namely** two.

III – VOCABULARY (2): LINKING YOUR IDEAS (CAUSE & CONSEQUENCE)^{xxv}

Expressing why something happens and what follows

1. Adverbs and conjunctions

■ Cause

Followed by a noun phrase:

because of • owing to • due to • on account of • thanks to
- = = = +

► **Owing to** rounding errors, the algorithm failed to converge.

Followed by a clause:

because • as • since

► **As** the matrix is singular, it has no inverse.

■ Consequence

consequently • therefore • thus • hence • as a result • thereby

► The series is absolutely convergent; **therefore**, it converges.

► We divided by the common factor, **thereby** reducing the fraction to lowest terms.

2. Verbs

■ Cause

to cause • result in • lead to • be responsible for
bring about • give rise to • trigger (off) • spark (off)

► A division by zero **led to** an undefined result.

■ Consequence

to come from • result from • arise from • stem from

► The contradiction **stemmed from** a false assumption.

3. Nouns

■ Cause

cause • reason • origin • source

► The real **cause** of the error was an off-by-one index.

■ Consequence

- consequence • result • outcome • effect
- by-product • spin-off

► The final **outcome** of the proof was a sharper error bound.

IV – VOCABULARY: LINK WORDS IN CONTEXT

Link words (also called connectors, discourse markers or conjunctions) are cohesive devices, used mainly to relate one sentence to another or to signal the speaker's attitude or style. They can be used in conversation to indicate agreement, contradiction, surprise, etc. Intonation is important as it helps to convey meaning.

1. SIMILARITY

AND

- *I study and I work.*

NOT ONLY/BUT ALSO

- *Not only do I study, but I also work.*
- *He is not only funny, but also intelligent.*

AS WELL AS

- *Please proofread for spelling mistakes as well as grammatical errors.*

IN ADDITION

- *In addition, dogs can be taught to feed themselves if you allow them access to your pantry.*

FURTHERMORE

- *Furthermore, they make wonderful walking buddies.*

MOREOVER

- *Moreover, they'll show you the way home when you're drunk.*

BESIDES

- *Besides not eating much, they won't ever chew on your electric cords.*
- *I'm not ready to get married yet. Besides, I enjoy living alone.*

APART FROM

- *Apart from their house in London, they also have a villa in Spain.*

2. CONTRAST

BUT

- *His mother won't be there, but his father might.*

HOWEVER

- *This restaurant has the best food in town. However, their staff are quite rude.*

NEVERTHELESS

- *I was in so much pain I didn't want to get up in the morning. Nevertheless, I went to football practice as usual.*

YET

- *I've asked you not to leave your dirty socks on the floor. Yet, you keep doing it.*
- *It was raining, yet we went out.*

ALTHOUGH

- *Although the sun was shining, it wasn't very warm.*

(EVEN) THOUGH

- *Anne was fond of Tim, though he often annoyed her.*

IN SPITE OF THIS/DESPITE

- *In spite of his age, he still leads an active life.*
- *In spite of feeling tired, we decided to go out.*
- *Despite his age, he leads an active life.*
- *Despite missing the train, he arrived on time for his English lesson.*

INSTEAD OF

- *Tickets will cost only £5, instead of the usual £6.50.*
- *Can't we deal with this now instead of waiting until tomorrow?*

WHEREAS

- *Some of the studies show positive results, whereas others do not.*

ON THE ONE HAND/ON THE OTHER HAND

- *On the one hand, England has the best language schools. On the other hand, it has the worst weather.*

3. CONTRAST TO WHAT PEOPLE THINK

ACTUALLY

- *Most people think that Fleming was the father of penicillin, but actually most of the work was done by Florey.*

IN FACT

- *I used to live in France; in fact, not far from where you're going.*

IN ACTUAL FACT

- *I thought the work would be difficult. In actual fact, it's very easy.*

IN REALITY

- *In reality, human beings and dinosaurs never lived on Earth at the same time.*

4. INTRODUCING SOMETHING OBVIOUS / GENERALLY ACCEPTED

OBVIOUSLY

- *Obviously, unless the birth rate decreases, there will be a major food crisis.*

NATURALLY

- *Naturally, I get upset when things go wrong.*

OF COURSE

- *Of course, there are other ways of doing this.*

CLEARLY

- *Clearly, this will cost a lot more than we realised.*

DOUBTLESS

- *You will doubtless find a solution.*

5. MAKING A GENERAL STATEMENT

IN GENERAL

- *In general, Japanese cars are very reliable and breakdowns are rare.*

GENERALLY SPEAKING

- *Generally speaking, the more you pay, the more you get.*

ON THE WHOLE

- *On the whole, I'm in favour of the idea.*

AS A (GENERAL) RULE

- *As a general rule, vegetable oils are better for you than animal fats.*

6. CLARIFICATION

FOR EXAMPLE / FOR INSTANCE

- *Wikis vary in how open they are. Some wikis, for example/for instance, allow anybody to edit content, while others only allow registered users to do this.*

E.G.

- *The website has a variety of interactive exercises (e.g. matching games, crosswords and quizzes).*

THAT IS TO SAY

- *Three days from now, that is to say, on Friday.*

IN OTHER WORDS

- *They asked him to leave — in other words, he was fired.*

I.E.

- *Service charge is included in all prices; i.e. you don't have to leave a tip.*

SUCH AS

- *Web 2.0 technologies, such as wikis, blogs and social networking sites, have changed the way that people use the internet.*

NAMELY

- *The quantity of heat is expressed in the same units as energy and work, namely joules.*

7. INTRODUCING A CAUSE/CONSEQUENCE

BECAUSE

- I did it because he told me to.

AS

- As you were out, I left a message.

SINCE

- We thought that, since we were in the area, we'd stop by and see them.

CONSEQUENTLY / AS A CONSEQUENCE

- Many parents today do not have time to cook healthy meals for their children. Consequently/As a consequence, many children grow up eating too much junk food.

THEREFORE

- We're going to experience some meteor showers in the next few days. Therefore, the number of miraculous self-healings will rise.

THUS

- You didn't tell me you wanted to come. Thus, we won't be taking you with us.

HENCE

- We suspect they are trying to hide something, hence the need for an independent inquiry.

AS A RESULT

- I've done a pranic healing course. As a result, I've been able to cure my neighbour's sick cat.

THEREBY (+ -ing verb)

- The arteries became gradually blocked, thereby reducing the oxygen supply.

Link words tend to vary according to the formality of the context.

Meaning	SCIENTIFIC PRESENTATIONS (ORAL)		ACADEMIC WRITING
	Basic form & formal use	More formal	Most formal
Addition	also first, second etc. ... too.	additionally besides further furthermore in addition next last but not least	equally important moreover similarly
Cause-effect	then therefore	as a result for this reason thus	as a consequence consequently hence
Comparison	also like too	as well as both... and... compared to in the same way as likewise neither... nor...	by comparison in common with similarly
Contrast	however	neither...nor... instead nevertheless	conversely in contrast to in opposition to on the contrary otherwise still ...whereas...
Time	after a while after that also at last currently earlier eventually finally first, second, etc... in the future in the past last next now	afterwards at the same time formerly immediately in the meantime later	concurrently previously simultaneously subsequently
Example	for example	for instance in other words	as an example as an illustration to exemplify
Summary - conclusion	finally therefore	after all all in all briefly last on the whole thus to sum up in conclusion	accordingly as a consequence in brief in closing in short in summary to conclude to summarise

1. Identify what helps to link ideas in the text below^{xxvii} (part of the transcript of a presentation)

"Good morning, everyone. I'm going to talk to you this morning about mathematics in society, and in particular, two very different issues in which mathematics plays a central role. I'm going to talk first about the topic of data privacy, and I will outline some of the major issues related to this current phenomenon. Then, I will go on and do the same for artificial intelligence and its impact on society. I'll also give you a summary of the points we have covered.

But before we begin, I have a little story to tell you. Before becoming a lecturer, I worked for Teachers Without Borders. For those of you that haven't heard of it, it's an international humanitarian organization that provides mathematics education both in France and in the developing world. I was based in Haiti, and it was a fantastic experience, at least from a personal point of view. I was working with children and young teachers, and the team was just great. But as you can imagine, the situation was far from ideal: textbooks were in short supply, and the conditions were just appalling. Consequently, there was a lot of frustration amongst the teaching staff, as well as for the Haitians themselves. One of the biggest problems was corruption, the dishonesty of some of those in power. Of course, the fact that there wasn't any real education system in place didn't help. It meant that supplies often didn't get through, or they were hijacked on the way. And the ones who actually suffered the most were the kids... Of course, the point of that story is to illustrate the fact that the role of education authorities in ensuring the numeracy of a nation is crucial. So - to get back to the main part of my lecture..."

2. Rewrite the sentences using the transitions in brackets. You may need to modify the structure of the sentence. ^{xxviii}

a. The mathematician didn't publish the proof. She presented it at a seminar. (instead of)

b. The formula looked complicated. It simplified to a single term. (however)

c. The calculation was delayed. There was an error in the first line. (because of)

d. You must check the denominator. You might divide by zero. (unless)

e. The student continued the proof. She could not justify the third step. (although)

f. The matrix was singular. The system had no unique solution. (consequently)

g. The data were noisy. The model produced an excellent fit. (in spite of)

h. The result was wrong. He had used the wrong formula. (as)

i. The team rescaled the variables. The algorithm could converge faster. (so that)

j. The proof was checked thoroughly. A gap was found two years later. (despite the fact that)

3. Choose the correct word from the list. ^{xxix}

1) The device enables temperatures to be monitored, improving the safety margin.

- a. thereby
- b. nevertheless
- c. namely
- d. e.g.

2) A new technique, the infra-red camera, means that dust surrounding new stars can be penetrated.

- a. whereas
- b. namely
- c. besides
- d. for instance

3) Superconductivity, when materials lose all resistance, will boost computer performance.

- a. thus
- b. obviously
- c. besides
- d. that is to say

4) Programmable electronic systems are more reliable. , they can be used to handle radioactive material.

- a. whereas
- b. such as
- c. moreover
- d. in other words

5) Computers can process data extremely fast. this, they have several serious drawbacks.

- a. in spite of
- b. whereas
- c. however
- d. moreover

6) The new engine is far more efficient., more work is required to reduce noise levels.

- a. nevertheless
- b. whereas
- c. i.e.
- d. besides

7) The presence of high concentrations of elements that are rarely found on Earth, iridium, suggests there was a meteorite impact at that spot.

- a. on the whole
- b. moreover
- c. such as
- d. that is to say

8) The data are stored on hard disk, they are easily accessible.

- a. actually
- b. whereas
- c. e.g.
- d. hence

9) Applicants for the job should speak at least one other European language French

- a. obviously
- b. besides
- c. moreover
- d. actually

10), in the initial stages there were minor problems to be overcome, but from then on, the prototypes were perfectly reliable.

- a. besides
- b. as a rule
- c. namely
- d. despite

4. Write chains of linked sentences, using different links words, each time based on one of the sentences below.^{xxx}

Example:

Artificial intelligence can now solve some competition-level mathematics problems.

1. **Clearly**, this means that the way we assess students may have to change.
2. **However**, these systems still cannot explain why their answers are correct.
3. The value of a rigorous, human-checked proof is **therefore** increasing.

1. Prime numbers lie at the heart of modern cryptography.

2. Quantum computers will be able to break today's encryption within a few decades.

3. The four-colour theorem was the first major result proved with the help of a computer, in 1976.

4. Within fifty years, most routine calculations will be carried out by machines.

5. Statistics is now used in almost every branch of science.

6. More and more of everyday life is governed by algorithms.

- in addition, furthermore, besides
- however, nevertheless, yet, in spite of, on the other hand
- but actually, but in reality, in fact
- obviously, clearly, doubtless
- in general, on the whole, as a rule
- in other words, that is to say, namely, consequently, therefore, thus

Chain 1:

1.



2.



3.

Chain 2:

1.



2.



3.

VII – HOMEWORK FOR NEXT CLASS

1. PRACTISE FOR YOUR PROJECT: CONNECTING THE DOTS

- a. Take at least 2 documents from your project, or that could be used in your project.
- b. Identify **8 linking words** (or expressions) in context between paragraphs or between ideas from 1-2 article(s), find synonyms and give a short explanation of their use.

e.g. *Plants need sunlight to grow. However, some can survive in low light.*

- *Linking word: however*
- *synonyms: nevertheless, yet*
- *explanation: used at beginning of a sentence / to express contrast*

/	Linking word (or expression)	Synonyms	Explanation of its use in context
1.			
2.			
3.			
4.			
5.			
6.			
7.			
8.			

c. Now, look at your project documents again: What is the link between them? Check the vocabulary from 'Linking your ideas (general)' and think of an interesting **research angle** (addition, contrast, cause, consequence, concession, ...) between documents. Fill in at least **two** tables. You should come back to this exercise later during the year and fill in more tables as you draw more **logical links between your documents**.

Link between book chapter n°....	and book chapter n°....

Link between the book chapters	and the popular science article

Link between the book chapters	and the scientific paper

Link between the book chapters	and the audiovisual resource

Link between.....	and additional documentation

VIII – PRACTISE FOR YOUR PROJECT: CONNECTING THE DOTS & PRESENTING ARTICLES

1. In groups of 5-6 students, focus on:

- the linking words/expressions within one article
- the links between the articles

Do not read a text. Use a few notes only (keywords on a small piece of paper) and a PowerPoint slide per article if you wish.

2. Listeners: Take notes of the information (choose one document per student) in the table below.

3. Be ready to report back on other students' documents.

<u>Document 1:</u>	<u>Document 2:</u>
<u>Document 3:</u>	<u>Document 4:</u>
<u>Document 5:</u>	<u>Document 6:</u>

4. Discussion: Feedback about the presentations.

A. LISTENING

➤ *1st listening*

1. Listen to the recording. Write down as many linking words from the audio as you can.

- Take notes of anything you understand.

➤ **2nd & 3rd listenings**

Answer the following questions.

1. What is Stephen Hawking's profession and what degrees did he have?

2. What are his greatest scientific contributions?

3. What impact did the disease (amyotrophic lateral sclerosis) have on his life?

4. What did a Californian computer expert create specifically for him?

5. In which countries did he use to give lectures/talks?

B. WRITING

Fill in the transcript with the missing connectors.

as a result of - as a rule - besides – despite - doubtless

- however - nevertheless - that is to say - thereby

PROFESSOR STEPHEN HAWKING

Stephen Hawking was _____ the most well-known theoretical physicist and cosmologist of our time.

At the age of 21, while he was still a young student, Stephen Hawking was diagnosed as having amyotrophic lateral sclerosis, _____, a degenerative motor neurone disease that prevents voluntary control of the muscles. He was informed that he did not have long to live.

_____ the disease, or as he himself says, perhaps _____ it, he continued with his studies and was awarded his PhD in 1966 and so started a brilliant scientific career. He has made important contributions to the theory of black holes and the origin of the universe.

_____ losing the ability to control his arm and leg muscles, he gradually lost the ability to speak clearly. This meant that, _____, at scientific conferences or seminars, he was obliged to speak via an interpreter.

In 1985, he caught pneumonia and was obliged to undergo a tracheotomy, **i.e.** an incision of the trachea.

The tracheotomy operation, _____, removed his ability to speak altogether. For a time, the only way he could communicate was by signalling with his eye muscles.

A computer expert in California designed him a programme linked to a voice synthesiser that could be activated with one finger, _____ enabling Hawking to move the cursor. That program allowed him to produce 15 words a minute.

Hawking was _____ very active. He travelled a great deal and in his late years gave lectures at different conferences in the US, India, South Korea and the UK.

X – HOMEWORK FOR NEXT CLASS

Each student should:

- be ready to explain their project (topic choice, ...) for 2 minutes.
- be ready to explain how and why they selected their documents for 1 minute
- fill in the two analysis tables about the scientific paper and popular science article (see next two pages)
- be ready to present your scientific paper and popular science article with PowerPoint for 4 minutes.



Use connectors.

ANALYSIS TABLE: SCIENTIFIC PAPER

Who and Where - Author - Publication Date - Researchers and institution named	
General topic - Context and previous work - Statement of the problem - Main results announced	
Methods - Definitions - Notation - Strategy of proof	
Results - Theorems - Lemmas - Propositions	
Discussion - Remarks - Corollaries - Extensions	

ANALYSIS TABLE: POPULAR SCIENCE ARTICLE

Who and Where - Author - Publication Date - Researchers and institution named	
The hook - How the article opens - Why the reader is told this matters)	
The idea - The mathematical concept, in the article's words - The same concept, in your words	
The evidence - What the article cites (a paper, an interview, a preprint) · - Is the original source identified?	
The telling - Analogies, images, metaphors used - What is left out - Anything that seems overstated - Remarks	

PROJECT: 1st POWERPOINT CHECKPOINT

In groups of 3.

Each student should:

- explain their project (topic choice, ...) for 2 minutes.
- explain how and why they selected their documents for 1 minute
- present your scientific paper and popular science article with PowerPoint for 4 minutes including the elements from the analysis tables.



Make links between your articles and use connectors.

The listeners act as the teacher. They should:

- ask why the student has chosen the topic.
- ask the student how and why they selected their articles
- ask questions about the presentation (e.g. What have you learnt while preparing this project?).
- give feedback about the presentation.
- give feedback about the PowerPoint .

See checklist next page.

CHECKLIST: STUDENT 1: ...

Tick as appropriate.

1. PRESENTATION

Topic choice	☐ Clearly explains the topic choice ☐ 2 mins
Article selection	☐ Explains <u>how</u> the articles were selected ☐ Explains <u>why</u> these articles were selected ☐ All selected articles are relevant ☐ 1 min
Presentation of research articles	☐ Presents the scientific paper clearly ☐ Presents the popular science article clearly ☐ Uses connectors to link ideas smoothly ☐ Makes links between the 2 documents ☐ 4 mins
Overall Delivery	☐ Speaks clearly ☐ Engages the audience (eye contact, posture, ...)

2. POWERPOINT (to be examined after the presentation)

Slides	☐ Front slide ☐ Overview ☐ General information ☐ Current research: 1 slide per article ☐ Bibliography ☐ Slide numbers
Front slide	☐ Student's name (first name – SURNAME) ☐ Studies ☐ Academic year ☐ Title ☐ Teacher's name ☐ Course title
Overview	☐ Title: Overview (NOT Summary/Table of Contents) ☐ No slide number next to each title
Content	☐ Key words (NO text) ☐ Bullet points ☐ Relevant pictures only ☐ Current research part: for each article: title of article, name of researchers, date of publication, and useful info (e.g. general topic, procedure, conclusions, remaining questions...)

3. SPELLING/GRAMMAR:

- ☐ spell check used
- ☐ grammar check used

4. CONSTRUCTIVE FEEDBACK ABOUT THE PRESENTATION

CHECKLIST: STUDENT 2: ...

Tick as appropriate.

1. PRESENTATION

Topic choice	☐ Clearly explains the topic choice ☐ 2 mins
Article selection	☐ Explains <u>how</u> the articles were selected ☐ Explains <u>why</u> these articles were selected ☐ All selected articles are relevant ☐ 1 min
Presentation of research articles	☐ Presents the scientific paper clearly ☐ Presents the popular science article clearly ☐ Uses connectors to link ideas smoothly ☐ Makes links between the 2 documents ☐ 4 mins
Overall Delivery	☐ Speaks clearly ☐ Engages the audience (eye contact, posture, ...)

2. POWERPOINT (to be examined after the presentation)

Slides	☐ Front slide ☐ Overview ☐ General information ☐ Current research: 1 slide per article ☐ Bibliography ☐ Slide numbers
Front slide	☐ Student's name (first name – SURNAME) ☐ Studies ☐ Academic year ☐ Title ☐ Teacher's name ☐ Course title
Overview	☐ Title: Overview (NOT Summary/Table of Contents) ☐ No slide number next to each title
Content	☐ Key words (NO text) ☐ Bullet points ☐ Relevant pictures only ☐ Current research part: for each article: title of article, name of researchers, date of publication, and useful info (e.g. general topic, procedure, conclusions, remaining questions...)

3. SPELLING/GRAMMAR:

- ☐ spell check used
- ☐ grammar check used

4. CONSTRUCTIVE FEEDBACK ABOUT THE PRESENTATION

CHECKLIST: STUDENT 3: ...

Tick as appropriate.

1. PRESENTATION

Topic choice	☐ Clearly explains the topic choice ☐ 2 mins
Article selection	☐ Explains <u>how</u> the articles were selected ☐ Explains <u>why</u> these articles were selected ☐ All selected articles are relevant ☐ 1 min
Presentation of research articles	☐ Presents the scientific paper clearly ☐ Presents the popular science article clearly ☐ Uses connectors to link ideas smoothly ☐ Makes links between the 2 documents ☐ 4 mins
Overall Delivery	☐ Speaks clearly ☐ Engages the audience (eye contact, posture, ...)

2. POWERPOINT (to be examined after the presentation)

Slides	☐ Front slide ☐ Overview ☐ General information ☐ Current research: 1 slide per article ☐ Bibliography ☐ Slide numbers
Front slide	☐ Student's name (first name – SURNAME) ☐ Studies ☐ Academic year ☐ Title ☐ Teacher's name ☐ Course title
Overview	☐ Title: Overview (NOT Summary/Table of Contents) ☐ No slide number next to each title
Content	☐ Key words (NO text) ☐ Bullet points ☐ Relevant pictures only ☐ Current research part: for each article: title of article, name of researchers, date of publication, and useful info (e.g. general topic, procedure, conclusions, remaining questions...)

3. SPELLING/GRAMMAR:

- ☐ spell check used
- ☐ grammar check used

4. CONSTRUCTIVE FEEDBACK ABOUT THE PRESENTATION

CHECKLIST: STUDENT 4: ...

Tick as appropriate.

1. PRESENTATION

Topic choice	☐ Clearly explains the topic choice ☐ 2 mins
Article selection	☐ Explains <u>how</u> the articles were selected ☐ Explains <u>why</u> these articles were selected ☐ All selected articles are relevant ☐ 1 min
Presentation of research articles	☐ Presents the scientific paper clearly ☐ Presents the popular science article clearly ☐ Uses connectors to link ideas smoothly ☐ Makes links between the 2 documents ☐ 4 mins
Overall Delivery	☐ Speaks clearly ☐ Engages the audience (eye contact, posture, ...)

2. POWERPOINT (to be examined after the presentation)

Slides	☐ Front slide ☐ Overview ☐ General information ☐ Current research: 1 slide per article ☐ Bibliography ☐ Slide numbers
Front slide	☐ Student's name (first name – SURNAME) ☐ Studies ☐ Academic year ☐ Title ☐ Teacher's name ☐ Course title
Overview	☐ Title: Overview (NOT Summary/Table of Contents) ☐ No slide number next to each title
Content	☐ Key words (NO text) ☐ Bullet points ☐ Relevant pictures only ☐ Current research part: for each article: title of article, name of researchers, date of publication, and useful info (e.g. general topic, procedure, conclusions, remaining questions...)

3. SPELLING/GRAMMAR:

- ☐ spell check used
- ☐ grammar check used

4. CONSTRUCTIVE FEEDBACK ABOUT THE PRESENTATION

PROJECT: CURRENT RESEARCH ANALYSIS

RESEARCHERS & TOPIC

- This research study was carried out by X from Y University in (country). It investigates (the topic).
conducted

-	Scientists Researchers X	from Y University	(have)	carried out conducted	a study	on about	(the topic).
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PUBLICATION

- It was published in (name of the journal) in (January) 2026.
- They published their work in “
- The work appeared in “

PROCEDURE / WHAT WAS EXAMINED

➤ **Sequence**

First,	First of all,	The first step	was
Second,	Then,	The second step	“
Third,	Finally,	The third step	“
...		...	

➤ **Verbs**

The researchers scientists team	tested ...
	carried out an experiment about/on ...
	analysed...
	observed...
	examined...
	claimed that...
	reported that...
	said that...

CONCLUSIONS

- The	researchers	(have)	found (evidence)	that ...
	scientists		demonstrated	
	team		discovered	

- This suggests that...

- The	results	showed	that...
	findings	suggested	

- The	researchers	(now) believe	that...
	scientists	concluded	
	team	found out	
		discovered	

- The	researchers'	discovery	implies that...
	scientists'		
	team's		

REMAINING QUESTIONS

- Several challenges remain including how...

- Further studies will focus on + noun / -ing verb

- It is still unclear how...

APPENDIX 1: MOCK EXAMS – GRAMMAR & VOCABULARY

I - Choose either the present simple or the present continuous verb tense and write the correct form.

A Trekking Journal

November 12, 2013

Today (be) _____1 the second day of my trek around Mount Annapurna. I am exhausted and my legs (shake) _____2; I just hope I am able to complete the trek. My feet (kill, really) _____3 me and my toes (bleed) _____4, but I (want, still) _____5 to continue.

Nepal is a fascinating country, but I have a great deal to learn. Everything (be) _____6 so different, and I (try) _____7 to adapt to the new way of life here. I (learn) _____8 a little bit of the language to make communication easier; unfortunately, I (learn, not) _____9 foreign languages quickly. Although I (understand, not) _____10 much yet, I believe that I (improve, gradually) _____11.

I (travel, currently) _____12 with Liam, a student from Leeds University in England. He (be) _____13 a nice guy, but impatient. He (walk, always) _____14 ahead of me and (complain) _____15 that I am too slow. I (do) _____16 my best to keep up with him, but he is younger and stronger than I am. Maybe, I am just feeling sorry for myself because I am getting old.

Right now, Liam (sit) _____17 with the owner of the inn. They (discuss) _____18 the differences between life in England and life in Nepal. I (know, not) _____19 the real name of the owner, but everybody (call, just) _____20 him Tam. Tam (speak) _____21 English very well and he (try) _____22 to teach Liam some words in Nepali. Every time Tam (say) _____23 a new word, Liam (try) _____24 to repeat it. Unfortunately, Liam (seem, also) _____25 to have difficulty learning foreign languages. I just hope we don't get lost and have to ask for directions.

II - Choose either the *past simple* or the *present perfect* verb tense and enter the correct form.

1. A: Did you enjoy the lecture on number theory?
B: I'm not sure. I (study, never) _____ that topic in depth.
2. Professor Nguyen (arrive) _____ at the university last semester.
3. My study partner and I (solve) _____ problems together for over two years.
We still meet weekly to practise.
4. Euler was a brilliant mathematician. He (publish) _____ many influential papers. One day, you might write something just as impactful.
5. I (feel, not) _____ this confident with proofs since I (take) _____ that logic course.
6. Mathematical tools (evolve) _____ significantly. When we first (start) _____ using software, the department (have, only) _____ a few licenses. Since then, we (gain) _____ access to a full suite of tools.
7. I (tell) _____ her to double-check the theorem, but she (skip) _____ the final step and (make) _____ a mistake.
8. Listen, Jordan, I don't care if you (miss) _____ the tutorial this morning. You (be) _____ late to class too many times. You're on academic probation.
9. Martha is from a school that didn't offer advanced maths, so she (see, never) _____ a real mathematical proof. She should attend the next workshop.
10. How sad! Mr Allen (dream) _____ of solving the Riemann Hypothesis before retiring, but he didn't succeed. He (prove, never) _____ it.
11. In the last century, mathematical research (become) _____ more collaborative. In the past, it (take) _____ years to verify a complex proof. The process (be) _____ slow and meticulous. Things (change) _____ a lot with modern tools.
12. Wow, I can't believe how much your problem-solving skills (improve) _____ since the last time I (see) _____ your work.

III - Use the verbs in brackets in the correct future tenses – will-future, going to-future, Simple Present or Present Continuous.

1. The seminar on quantum computing (start)_____ at 11:45.
2. We (have)_____ lunch with the visiting professor after the colloquium.
3. It (get)_____ quite cold during the field experiment in the mountains tomorrow evening.
4. Look at the radar – it (rain)_____ soon. We should postpone the telescope session.
5. On Sunday at 8 o'clock I (meet)_____ my study group to prepare for the exam.
6. They (fly)_____ to the international conference in Geneva on Friday evening.
7. Wait! I (drive)_____ you to the lab so you don't miss the experiment.
8. I (see)_____ my thesis advisor in April to discuss my research progress.
9. Listen! Someone's at the lab door. I (open)_____ it for you.

IV - The following sentences are answers to questions. If the words in bold are the missing information, write the questions.

1. He goes to the lab **twice a week**.

2. I have tried to solve the **equation** five times.

3. The observatory is **200 kilometres** from the university.

4. She bought **a new graphing calculator**.

5. There were **more than 1000 people** at the conference.

6. Kate had to wait **an hour** for the simulation to finish.

7. Tom likes **linear algebra**.

8. The textbook cost **€10**.

9. The lecture series starts **this week**.

10. Mary joined the research team **in 2011**.

11. John met his supervisor **at the summer school**.

12. They are staying **at the university residence**.

13. He paid for the conference registration with **his credit card**.

14. Kim measured a metal rod that was **160 centimetres** long.

On Moodle:

- Extra grammar tests pdf

Online: additional grammar tests

- <http://www.englisch-hilfen.de/en/exercises/tenses/mix.htm>
- <http://www.englisch-hilfen.de/en/exercises/tenses/mix2.htm>
- <http://www.englishpage.com/verbpage/verbs29practicetest.htm>
- <http://www.englishpage.com/verbpage/verbs30finaltest.htm>
- <http://www.agendaweb.org/verbs/mixed-tenses.html>
- [http://www.english-grammar.at/online_exercises/tenses/tenses_index.htm_\(all_tenses\)](http://www.english-grammar.at/online_exercises/tenses/tenses_index.htm_(all_tenses))
- <https://engelsklaslokaal.nl/en/english-grammar-exercises/two-english-verb-forms/mix-of-tenses/>
- <https://www.english-4u.de/en/tenses-exercises/tenses-mixed.htm>

GRAMMAR

What to study for the exam recap:

Grammar topic	Grammar book: section & page numbers
Questions	Section 8 (Questions), pp. 103–107
Past simple - past continuous	Section 4 (Past tenses), pp. 39–46
Present perfect	Section 5 (Perfect tenses), pp. 47–53, 56–57, 60
Past perfect	Section 5 (Perfect tenses), pp. 61–63
Future	Section 3 (Future), pp. 25–32
Conjunctions	Section 17 (Conjunctions), pp. 239–243

Do not forget to study [irregular verbs](#). You are also expected to know the names of the various tenses.

VOCABULARY (

Find the missing word(s) in context. The first letter as well as a synonym, antonym or an explanation of the word is given to help you.

1. Between 2005 and 2025, the average world death rate is expected to decline only s..... (a little, by a small amount)
2. The biosphere refers to the t..... layer surrounding the Earth where living organisms are found. (small transversal dimension, >< thick)
3. The original estimation of the cost of building the Three Gorges Dam in China was i..... (imprecise)
4. A chain reaction is a series of reactions in which the result of each s..... causes the next. (step, part of the process)
5. If you add two odd numbers together, you get an e..... number. (>< odd)
6. The Arctic Ocean is joined to the Pacific Ocean by the Bering Strait, a narrow and s..... channel with a depth of no more than 55 m. (>< deep)

What to study for the exam recap: coursebook key vocabulary (~ Quizlet Term 1)

APPENDIX 2: GRAMMAR — HOW TO ASK QUESTIONS

I - DISCOVER

1. CLOSED QUESTIONS: YES-NO QUESTIONS

It is here → _____
He studies Pharmacy. → _____
He has got a new laptop. → _____
They went to UMONS. → _____

RULE : yes-no questions

➤ **Single verb (present simple or past simple)**

- 'to be' :
- other verbs:

➤ **Compound verb (more than one part)**

2. OPEN QUESTIONS: WHO - WHAT - WHY - WHEN - WHERE - HOW- HOW MUCH - HOW MANY - WHOSE

Newton is <u>an English scientist</u> .	→	_____
Newton discovered gravity <u>in England</u> .	→	_____
William loves <u>Kate</u> .	→	_____
<u>William</u> loves Kate.	→	_____
She has requested <u>an appointment</u>	→	_____
It is <u>Mark's</u> car.	→	_____

RULE : open questions

➤ **Single verb (present simple or past simple)**

- 'to be' : WH-word + ...

- other verbs: WH-word + ...



Exception: If the subject is WHAT or WHO - e.g. William loves Kate.

WHO/WHAT + ...

➤ **Compound verb (more than one part)**

WH-word + ...

Note: Whose + ...

II - THEORY

WORD ORDER³

In questions, the word order *subject-verbs-object* is the same as in [affirmative sentences](#). The only thing that's different is that you usually have to put the auxiliary verb (or the main verb "be") before the subject. Interrogatives are put at the beginning of the sentences:

interrogative	auxiliary verb	subject	other verb(s)	indirect object	direct object	place	time
What	would	you	like to tell	me			
	Did	you	have		a party	in your flat	yesterday?
When	were	you				here?	

You don't use an auxiliary verb if you ask for the subject. In this case the interrogative simply takes the place of the subject.

interrogative	verb(s)	object
Who	asked	you?

Another useful website: <http://www.englisch-hilfen.de/en/grammar/frage.htm>



³ Source : <http://www.ego4u.com/en/cram-up/grammar/word-order/questions>

QUESTION WORDS⁴

We use question words to ask certain types of questions. We often refer to them as *WH* words because they include the letters *WH* (for example *WHY*, *HoW*).

question word	function	example sentence
what	asking for information about something	What is your name?
	asking for repetition or confirmation	What? I can't hear you. You did what?
what...for	asking for a reason, asking why	What did you do that for?
when	asking about time	When did he leave?
where	asking in or at what place or position	Where do they live?
which	asking about choice	Which colour do you want?
who	asking what or which person or people (subject)	Who opened the door?
whom	asking what or which person or people (object)	Whom did you see?
whose	asking about ownership	Whose are these keys? Whose turn is it?
why	asking for reason, asking what...for	Why do you say that?
why don't	making a suggestion	Why don't I help you?

⁴ Source: <https://www.englishclub.com/vocabulary/wh-question-words.htm>

how	asking about manner	How does this work?
	asking about condition or quality	How was your exam?
how + adj/adv	asking about extent or degree	see examples below
how far	distance	How far is Pattaya from Bangkok?
how long	length (time or space)	How long will it take?
how many	quantity (countable)	How many cars are there?
how much	quantity (uncountable)	How much money do you have?
how old	age	How old are you?
how come (informal)	asking for reason, asking why	How come I can't see her?

Another useful website: <http://www.anglaiscours.fr/poser-une-question-en-anglais-what-where-who-why-how-whose-when-which-etc.html>



III – EXERCISES

1. Yes-no questions: Transform the following sentences into questions.

- | | | |
|-----------------------------|---|-------|
| a. She is a student | → | _____ |
| b. It's snowing | → | _____ |
| c. I speak Spanish | → | _____ |
| d. I went to Gent yesterday | → | _____ |
| e. I'm going to the party | → | _____ |
| f. I should study more | → | _____ |
| g. I will try to come | → | _____ |
| h. She has gone to NY | → | _____ |

2. Open questions: Ask questions with *what, why, when, where, how, whose...* The answers are underlined.

- | | | |
|--|---|-------|
| a. She is at university <u>during the week</u> | → | _____ |
| b. The students are <u>in building 6</u> | → | _____ |
| c. I am leaving <u>at 6</u> | → | _____ |
| d. I am at university <u>to study science</u> | → | _____ |
| e. She is feeling <u>terrible</u> | → | _____ |
| f. I am studying <u>my physics lesson</u> | → | _____ |
| g. He starts to work <u>at 8 a.m.</u> | → | _____ |
| h. He studies <u>in Mons</u> | → | _____ |
| i. He lived <u>in Brussels</u> for 3 years | → | _____ |
| j. He gave a presentation <u>yesterday</u> | → | _____ |
| k. She has gone <u>to London</u> | → | _____ |
| l. <u>He should study more</u> | → | _____ |
| m. Paul knows <u>Virginia</u> | → | _____ |
| n. <u>Paul</u> knows Virginia | → | _____ |
| o. It is <u>Steve's</u> house | → | _____ |

3. Ask questions. The answers are the words in bold. Please keep all relevant information in the question.

1. The students have prepared **a presentation about a black hole**.

_____?

2. He went **to the hill**.

_____?

3. They regularly cut **the grass**.

_____?

4. **He** has given Harry a nice present.

_____?

5. The guests made **an awful mess**.

_____?

6. This cave is around **26 metres** in depth.

_____?

(What is the depth....)

7. They went home **because they were tired**.

_____?

8. We call Mary **every weekend**.

_____?

9. He stays **in bed** all day long.

_____?

10. My suitcase weighs **24 kilograms**.

_____?

11. He has stayed in London **for a few hours**.

_____?

12. She wore black jeans **last Tuesday**.

_____?

13. Tina has lived in a small village in Wales **since she was a little girl**.

_____?

14. They went up **1860** steps to reach the top.

_____?

15. We took **the second elevator**, because the first one was broken.

_____?

APPENDIX 3: GRAMMAR — LIST OF TENSES

Tense	Active	Passive
Present simple	<i>I watch.</i>	<i>I am watched.</i>
Present continuous	<i>I am watching.</i>	<i>I am being watched.</i>
Present perfect simple	<i>I have watched.</i>	<i>I have been watched.</i>
Present perfect continuous	<i>I have been watching.</i>	-
Past simple	<i>I watched.</i>	<i>I was watched.</i>
Past continuous	<i>I was watching.</i>	<i>I was being watched.</i>
Past perfect simple	<i>I had watched.</i>	<i>I had been watched.</i>
Past perfect continuous	<i>I had been watching.</i>	-
Be going to	<i>I am going to watch him.</i>	<i>He is going to be watched (by me).</i>
Future simple	<i>I will watch.</i>	<i>I will be watched.</i>
Future continuous	<i>I will be watching.</i>	-
Future perfect simple	<i>I will have watched.</i>	<i>I will have been watched.</i>
Future perfect continuous	<i>I will have been watching.</i>	-
Imperative	<i>Watch !</i>	<i>Be reassured !</i>
Conditional present	<i>I would watch.</i>	<i>I would be watched.</i>

Match the tense and its form.

Present simple	•	•	I am studying
Present continuous	•	•	I have been studying
Present perfect simple	•	•	I studied
Present perfect continuous	•	•	I had studied
Past simple	•	•	I had been studying
Past continuous	•	•	I study
Past perfect simple	•	•	I have studied
Past perfect continuous	•	•	I was studying

APPENDIX 4: GRAMMAR — PAST AND PERFECT TENSES

	PAST SIMPLE	PAST CONTINUOUS
Use	Action <u>completely in the past</u> , dated by a time marker or through context	Action happening in the past and <u>interrupted by another action</u>
Affirmative form	➤ Regular verbs: infinitive + -ED e.g. Yesterday, I played tennis. ➤ Irregular verbs: to be studied by heart e.g. In 2015, I went to Scotland.	WAS/WERE + ING-verb e.g. I was watching TV when the phone rang.
Interrogative form	DID + subject + infinitive e.g. Did you visit Scotland last year?	WAS/WERE + subject + ING-verb e.g. Were you watching TV when the phone rang?
Negative form	DIDN'T + infinitive e.g. I didn't visit Scotland last year.	WASN'T / WEREN'T + ING-verb e.g. I wasn't watching TV when the phone rang.

	PRESENT PERFECT SIMPLE	PAST PERFECT SIMPLE
Use	➤ Action that <u>started in the past and goes up to now</u> ➤ Finished actions that have <u>consequences now</u> ➤ <u>News items</u>	Action that happened <u>before another past action</u>
Affirmative form	HAVE/HAS + past participle e.g. . I have lived in Mons since 2015. . I have eaten all the cakes (there are none left). . A plane has crashed in Bangkok.	HAD + past participle e.g. . I bought a train ticket and went to the platform. But the train had already left. . It was a song I hadn't heard before. . I went into the office. It was empty. Natalie had gone home.
Interrogative form	HAVE/HAS + subject + past participle e.g. Have you done it?	HAD + subject + past participle e.g. Had he finished his exam when the bell rang?
Negative form	HAVEN'T/HASN'T + past participle e.g. I haven't been to Paris for a long time.	HADN'T+ past participle e.g. I hadn't finished my exam when the bell rang.

Identify the tense of the verbs and justify its use.

1. I have studied for 6 hours.
2. I have been studying for 6 hours.
3. Yesterday, I studied for 6 hours.

4. I've had two cups of tea this morning.
5. I had two cups of tea this morning.

6. I was having a bath when the bell rang.
7. The bell rang while I was having a bath.

8. When I arrived, everybody applauded.
9. When I arrived, everybody was applauding.

10. After he finished watching the film, he went to bed.
11. After he had finished watching the film, he went to bed.

12. I was walking the dog when I saw my neighbour.
13. I had already walked the dog when I saw my neighbour.

APPENDIX 5: GRAMMAR — LIST OF IRREGULAR VERBS

Irregular verbs

Verb	Past simple	Past participle
arise	arose	arisen
be	was / were	been
beat	beat	beaten
become	became	become
begin	began	begun
bend	bent	bent
bite	bit	bitten
bleed	bled	bled
blow	blew	blown
break	broke	broken
bring	brought	brought
broadcast	broadcast	broadcast
build	built	built
burn	burnt	burnt
burst	burst	burst
buy	bought	bought
catch	caught	caught
choose	chose	chosen
come	came	come
cost	cost	cost
creep	crept	crept
cut	cut	cut
deal	dealt	dealt
dig	dug	dug
do	did	done
draw	drew	drawn
dream	dreamt	dreamt
drink	drank	drunk
drive	drove	driven
eat	ate	eaten
fall	fell	fallen
feed	fed	fed
feel	felt	felt
fight	fought	fought
find	found	found
fly	flew	flown
forbid	forbade	forbidden
forget	forgot	forgotten

Verb	Past simple	Past participle
forgive	forgave	forgiven
freeze	froze	frozen
get	got	got
give	gave	given
go	went	gone
grow	grew	grown
hang	hung	hung
have	had	had
hear	heard	heard
hide	hid	hidden
hit	hit	hit
hold	held	held
hurt	hurt	hurt
keep	kept	kept
kneel	knelt	knelt
know	knew	known
lay	laid	laid
lead	led	led
lean	leant	leant
learn	learnt	learnt
leave	left	left
lend	lent	lent
let	let	let
lie	lay	lain
light	lit	lit
lose	lost	lost
make	made	made
mean	meant	meant
meet	met	met
pay	paid	paid
put	put	put
read	read	read
ride	rode	ridden
ring	rang	rung
rise	rose	risen
run	ran	run
say	said	said
see	saw	seen

Verb	Past simple	Past participle
sell	sold	sold
send	sent	sent
set	set	set
sew	sewed	sewn
shake	shook	shaken
shine	shone	shone
shoot	shot	shot
show	showed	shown
shrink	shrank	shrunk
shut	shut	shut
sing	sang	sung
sink	sank	sunk
sit	sat	sat
sleep	slept	slept
slide	slid	slid
smell	smelt	smelt
sow	sowed	sown
speak	spoke	spoken
spell	spelt / spelled	spelt / spelled
spend	spent	spent
spill	spilt	spilt
spit	spat	spat
split	split	split
spoil	spoilt	spoilt
spread	spread	spread
spring	sprang	sprung
stand	stood	stood
steal	stole	stolen
stick	stuck	stuck
sting	stung	stung
strike	struck	struck
swear	swore	sworn
sweep	swept	swept
swell	swelled	swollen
swim	swam	swum
swing	swung	swung
take	took	taken
teach	taught	taught

Verb	Past simple	Past participle
tear	tore	torn
tell	told	told
think	thought	thought
throw	threw	thrown
understand	understood	understood
wake	woke	woken
wear	wore	worn
weep	wept	wept
win	won	won
write	wrote	written

APPENDIX 6: GRAMMAR – FUTURE TENSES

PREDICTION	▪ Calculating e.g. <i>I think it will rain this evening.</i> (calculating, weather forecast) ▪ We think, we believe e.g. <i>Don't take that glass. You'll break it.</i> (I know you, I think you are very clumsy – Reformulation: If you take that glass, you will break it)	▪ Future starting now e.g. <i>I think it is going to rain this evening.</i> (I can see clouds in the sky) - e.g. <i>Look out! You are going to break that glass!</i> (There is an obstacle, you are going to fall or to drop it... I can see it starting to happen now)	
PLAN, INTENTION	Social arrangement e.g. <i>I'll see you tomorrow</i>	Plans (arrangements may or may not have been fixed) e.g. <i>. I am going to speak at a conference.</i> <i>(no fixed arrangement)</i> <i>. I am going to speak at a conference on 15 Dec.</i> <i>(fixed arrangement)</i> <i>. We are going to go to Paris next year.</i>	Plans with already <u>fixed</u> arrangements – the time / place is given e.g. <i>. I am speaking at a conference on 15 Dec.</i> <i>. We are going to Paris next year.</i>
DECISION	Making the (quick) decision now e.g. <i>. We have got a lot of mail to answer. - OK, I'll take care of it</i> <i>(I am making the quick decision now)</i> <i>. Which one do you want? I'll take this one.</i>	The decision was made earlier e.g. <i>We have got a lot of mail to answer. - I know. I'm going to answer them on Tuesday.</i> <i>(I decided a while ago that I would answer the mail on Tuesday)</i>	
PROMISE	e.g. <i>Please write every week – Of course, I will</i> <i>(I promise)</i>		
THREAT	e.g. <i>Do your homework or I'll take away your PlayStation from you.</i>		
REFUSAL	e.g. <i>The car won't start.</i>		
OFFER	e.g. <i>I'll carry your bag.</i>		

PRESENT SIMPLE	Event always happens a certain time e.g. <i>Her plane arrives at 15.30.</i> Or Event is in the timetable e.g. <i>The lesson starts at 10 am.</i>
-----------------------	--

Choose the best form.

1. It *will rain/is going to rain* – Look at those clouds.
2. If it gets any colder, it *will snow/is going to snow*.
3. Alice *will have / is going to have* a baby.
4. If the baby takes after his mum, he *ll have/is going to have* blue eyes.
5. I *am going to play / am playing* with Stan this Saturday.
6. He *ll win/he's winning* tomorrow. He always does.
7. Don't tell her. She *ll tell/is going to tell* everybody else.
8. She *ll get married/she's getting married* on Friday at the local church.

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These references are included for illustrative and educational purposes.